

ICRA-C

International Conference on Research Advancements in Chemistry-2024

21-23 AUGUST

ICRA-C 2024

BOOK OF ABSTRACTS

Organized by
Department of Chemistry, School of Natural Sciences NUST, Islamabad Pakistan.

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BACKGROUND OF ICRA-C 2024

ICRA-C 2024 is the 3rd International Conference in the series, organized by the Department of Chemistry, School of Natural Sciences at the National University of Sciences and Technology, Islamabad, Pakistan. The conference is being held on 21-23rd August 2024, with active participation from prominent researchers and scholars from industry and academia. The conference's main objective is to provide a forum for exploring novel and innovative research advancements in chemistry and material sciences. The chemistry department has arranged 02 plenary Lectures, 03 international keynote talks, 10 invited talks, 73 oral and 96 poster presentations to achieve the goal. ICRA-C 2024 has attracted more than 360 researchers and scholars from Pakistan and other countries, including UK, KSA, Finland, Australia, China, Malaysia, UAE, South Korea, Cameron, Nigeria, Turkey, Barkindo Faso, etc. This abstract book includes the abstract selected for oral presentations and poster presentations.

It is a matter of great pleasure for the chemistry department that scholars and researchers from Gilgit Baltistan to Karachi take an active interest through their participation. Other than the designated talks and poster presentation, almost 360 people registered for the conference as attendees. ICRA-C 2024 will be a powerful platform for researchers, academicians, and students to engage in knowledge and technology sharing. It is believed that the event will encourage participants to exchange experiences, discuss current challenges independently, to promote future collaborations and knowledge transfer between participants. It will also identify fruitful avenues for future research in chemistry. This conference will also encourage participants to reciprocate experiences and debate the current research challenges.

ABOUT NUST

The National University of Sciences and Technology (NUST) is a public sector university established in 1991. NUST is a multi-campus university with the main campus located in H-12, Islamabad. The university offers undergraduate and graduate degrees, including doctoral and professional degrees in more than 35 disciplines with 60+ streams.

NUST facilitates its students and faculty with male/female student's hostels and faculty residential apartments. Moreover, it provides facilities for sports and extracurricular activities. Currently there are more than 20 active clubs and societies indulged in academic, adventurous, and welfare activities.

NUST is proudly positioned at 353 as per recent QS world university rankings. Also, NUST is the only Pakistani university that stands among top 100 Asian Universities as per academic reputation indicator. Thus, NUST holds the singular honor of being #1 university in Pakistan both in the world and Asian rankings.

NUST

Defining Futures

ABOUT DEPARTMENT OF CHEMISTRY

The Department of Chemistry is a constituent part of School of Natural Sciences and situated at new SNS building, NUST, H-12, Islamabad. The Department of chemistry has started its voyage in year 2014 with its first batch of post graduate program. After a short period, BS Chemistry program was also launched in year 2018.

Within a short span, the Chemistry department has achieved its global standing among top 450 chemistry departments in the world according to recent QS ranking. We have the vision to raise chemistry department as a premier academic, research, and teaching institute, which would be internationally recognized for excellence in training of scholars, and for research in frontier fields of chemical science.

Entire Chemistry FMs are PhD qualified, graduated from world renowned universities and have won research projects like HEC-NRPU, TTSF, Local Challenge Funds and PSF. Currently, the Chemistry department has 168 UG and 81 PG students. In BS Chemistry Program, three specializations are offered named Organic chemistry, Physical chemistry, and Inorganic chemistry. We are intended to commence Industrial chemistry specialization in our BS chemistry program in near future. Chemistry department has 07 teaching and research labs equipped with modern research instruments including FTIR Spectrophotometer, High Performance Liquid Chromatograph, Potentiostat, UV-visible Spectrophotometer, Fluorescence Spectrometer, Gas Chromatograph Mass Spectrometer, Raman Spectroscope, CHNS Elemental Analyzer and Powder X-Ray diffraction.

GREETINGS FROM THE CONVENOR

On behalf of the organizing team of 'International Conference on Research Advancements in Chemistry' (ICRA-C 2024), it is with great pleasure that I welcome you to the International Conference on Research Advancements in Chemistry (Chemistry for a Green Future). With a sense of pride and satisfaction, I would like to say that with the blessings of Almighty ALLAH and active support of the national and international scientific community, the dream of ICRA-C has come alive. I am reminded of the pivotal role that chemistry plays in shaping a sustainable and environmentally responsible future. This conference brings together leading scientists, researchers, and innovators from around the globe, united by a shared commitment to advancing green chemistry and its applications.

The Department of Chemistry is a constituent part of School of Natural Sciences, National University of Sciences & Technology. Department had started its voyage in year 2014 with its first batch of post graduate program. After a short period, BS Chemistry program was also launched in year 2018. It is a matter of pride that within a short span, the chemistry department has emerged as a highly dynamic and energetic academic & research entity. It has achieved its global standing among the top 450 chemistry departments in the world according to recent QS ranking. We have the vision to raise the chemistry department as a premier academic, research, and teaching institute, which would be internationally recognized for excellence in training of scholars, and for research in frontier fields of chemical sciences.

Being the Conference convenor, it's gratifying to release abstract book of ICRA-C 2024. Aim of this abstract book is to provide a scientific platform for chemistry researchers to get their intellectual work recognized in literature. Over the course of this conference, we will explore cutting-edge research, share groundbreaking ideas, and engage in thought-provoking discussions that will pave the way for a more sustainable world. The diverse range of topics covered, from renewable energy sources to eco-friendly materials, reflects the multifaceted nature of green chemistry and its potential to address some of the most pressing challenges of our time. The research abstract accepted in ICRAC 2024 has been evaluated and peer

reviewed by highly qualified researchers from the relevant fields. I am confident that this book will provide captivating reading overview to the readers. Subject book contains works that are rich in content and truly mirror the diverse and dynamic research titles. This abstract book encapsulates the Panorama of advancements in Chemistry and exhibits the recent developments in the chemical scientific community.

I would like to extend my heartfelt gratitude to our keynote speakers, presenters, and participants for contributing their knowledge and expertise. Your dedication to advancing the field of green chemistry is truly inspiring and vital to our collective mission.

I also wish to acknowledge the hard work and dedication of the organizing committee and our sponsors, whose efforts have been instrumental in making this conference a reality.

As we embark on this journey of discovery and collaboration, I encourage you to engage fully in the discussions, share your insights, and forge new partnerships that will continue long after the conference has concluded.

I admire the concerted efforts of the abstract committee team to bring out this abstract book of ICRAC 2024. My commendations to all contributors for making this book innovative and inspiring. I truly hope that the pages that follow will make some interesting reading for the readers. I am confident that ‘ICRA-C 2024’ would also continue to bring a similar advanced, diverse, and comprehensive repertoire of research work to its readers in future.

With best regards!

Prof. Dr. Manzar Sohail

HoD Chemistry/ Conference Convenor

GREETINGS FROM THE CHIEF ORGANIZER

Dear Readers!

On behalf of organizing team of 'International Conference on Research Advancements in Chemistry' (ICRA-C), we extend our passionate welcome to all readers. ICRA-C meant to provide a forum for both veteran and novice researchers in various genres of chemical sciences to showcase their research prowess, report their creative work, pen down their results and satiate their scientific pursuits. With a sense of pride and satisfaction, I would like to say that with the blessings of Almighty ALLAH and active support of national and international scientific community, dream of ICRA-C has come alive. I anticipate ICRA-C to keep pace with emerging chemical advancements and disseminate scientific research that meet national and international aspirations in the domain of 'Green Future'.

Being chief organizer, it's gratifying to release abstract book of ICRA-C. Aim of this abstract book is to provide a scientific platform for chemistry researchers to get their intellectual work recognized in literature. There is no denying the fact that such scientific literary exertions are obligatory for grooming young scientists and budding researchers. These readings broaden mental horizons and more significantly enhance their research comprehension. The research abstract accepted in ICRA-C 2024 have been evaluated and peer reviewed by highly qualified researchers from the relevant fields. I am confident that this book will provide captivating reading overview to the readers. Subject book contains works that are rich in content and truly mirror the diverse and dynamic research titles. This abstract book encapsulated the Panorama of advancements in Chemistry and exhibits the recent developments in chemical scientific community.

I applaud the concerted efforts of abstract committee team to bring out this abstract book of ICRA-C 2024. My commendations to all contributors for making this book, innovative and inspiring. I truly hope that the pages that follow will make some interesting reading for the readers. I am confident that 'ICRA-C' would also continue to bring similar advanced, diverse, and comprehensive repertoire of research work to its readers in future.

Happy Reading!

Prof. Dr. Azhar Mahmood

Chief Organizer ICRA-C 2024

ORAL PRESENTATION

Sustainable and Inclusive Drug Discovery from Nature – *Examples of Our Recent Work*

M. Iqbal Choudhary, Mustafa (PBUH) Prize Laureate, H. I., S. I., T. I.

International Center for Chemical and Biological Sciences

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Subject Category: Organic Chemistry

Abstract: Sustainable drug discovery and development is based on the principles of the use of renewable resources, and green chemistry methods. Nature-based discovery of drugs against prevailing and emerging diseases is sustainable and inclusive. Our work in the past two decades has been focused on the discovery of new drug leads from medicinal plants, as well as synthesis of bioactive analogs by biocatalytic transformations. We have been working largely at the interface of chemistry and biology for the discovery of chemical constituents from plants used in traditional medicines, as well as for designing of new bio-transformed products with therapeutic potential. This has resulted in the identification of several novel drug leads against various therapeutic targets. Emphasis has been on the discovery of natural products, and their analogs against chronic disorders, including cardiovascular, cancer, diabetes, inflammatory diseases, and Alzheimer's and Parkinson's diseases. Novel analogues of existing drugs such as tibolones (postmenopausal osteoporosis), exemestane (anti-cancer drug), medrysone (anti-inflammatory agent) and many others were synthesized using biotransformation tools. Thus, potent anti-inflammatory, and anti-cancer agents were identified as lead molecules. Furthermore, inhibitors of key enzymes related to several diseases were also identified.

The Green and Sustainability aspects behind Thermogalvanic Waste Heat Harvesting

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Email: leigh.aldous@kcl.ac.uk

Subject Category: Materials Science

Abstract: Waste heat is a hugely abundant but largely under-utilised energy source, from human industry to solar irradiation through to body heat. However, there is a lack of economically viable methods of valorising this (*e.g.* converting it to a more widely applicable form of energy). Electrochemistry can be applied here, with one example being thermogalvanic waste heat harvesting. A simplified description of a thermogalvanic device is two electrodes sharing an electrolyte containing two redox states; a temperature difference between the two electrodes can be thermoelectrochemically converted to electricity, via the entropy difference between the redox states.

This presentation will cover aspects such as modifying solvation to enhance performance,[1,2] fundamental aspects such as efficiency,[3] and moving towards devices and commercialisation.[4] In keeping with the theme of this Conference, this presentation will focus upon my recent research into the green and sustainability aspects behind these devices,[5-8] and in particular how to publish in related journals. While any system that can harvest waste heat is in theory green and/or sustainable, more is required in order to publish in top-tier journals dedicated to this topic.

References:

[1] K. Laws, M. A. Buckingham, M. Farleigh, M. Ma and L. Aldous, *Chemical Communications* **2023**, *59*, 2323. Link

[2] K. Laws, M. A. Buckingham and L. Aldous, *Chemical Science* **2024**, *15*, 6958. Link

[3] M. A. Trosheva, M. A. Buckingham and L. Aldous, *Chemical Science* **2022**, *13*, 4984. Link

[4] P. Phelan, L. Aldous, R. Taylor and B Obeng, *US Patent 11,346,103*, **2022**. Link

[5] M. A. Buckingham, F. Marken and L. Aldous, *Sustainable Energy and Fuels* **2018**, *2*, 2717. Link

[6] M. A. Buckingham, S. Hammoud, H. Li, C. J. Beale, J. T. Sengel and L. Aldous, *Sustainable Energy and Fuels* **2020**, *4*, 3388. Link

[7] M. A. Buckingham, K. Laws, J. T. Sengel and L. Aldous, *Green Chemistry* **2020**, *22*, 6062. Link

[8] M. A. Buckingham, K. Laws, E. Cross, A. Surman and L. Aldous, *Green Chemistry* **2021**, *23*, 8901. Link

Leveraging Electrochromic Supercapacitor Smart Windows for Futuristic Buildings

Numan Arshid

Sunway Center for Electrochemical Energy and Sustainable Technology (SCEEST), School of Engineering and Technology, Sunway University, Malaysia

Subject Category: Materials Science

Abstract: As the world is shifting toward sustainable and energy-efficient building practices, the use of smart windows is becoming prevalent in modern construction, especially in the areas of Passivhaus. These windows can dynamically regulate the access of light and heat in buildings, efficiently control indoor temperatures, and minimize reliance on artificial heating and cooling systems, resulting in a more sustainable and comfortable living environment. This talk will provide the fundamentals of electrochemical energy storage systems and highlight the current confusion between battery-grade and capacitive-grade electrode materials. Further, a detailed summary of current developments of our electrochromic electrode materials for smart windows will be presented. Current challenges and future considerations of the integrated energy storage and conversion devices will be covered.

Streamlining drug discovery for non-communicable diseases with potent and selective therapeutic compounds

Sumera Zaib

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Subject Category: Organic Chemistry/ Analytical Chemistry

Abstract: Non-communicable diseases (NCDs), such as cardiovascular diseases, cancers, chronic respiratory diseases, and diabetes mellitus, pose a critical threat to global health and sustainable development. These conditions are responsible for over 70% of all deaths worldwide, disproportionately affecting low- and middle-income countries, and placing immense strain on healthcare systems. The rapid increase in NCDs is driven by factors such as aging populations, urbanization, and lifestyle changes, including poor diet, lack of physical activity, and tobacco use. Therefore, the development of novel, drug-like molecules that are potent, selective and exhibit improved efficacy is mandatory to mitigate these fatal ailments. Our recent research efforts have focused on designing, synthesizing, and thoroughly analyzing a variety of novel inhibitors targeting key enzymes involved in neurological disorders, metabolic dysregulations, and other pathophysiological conditions. These enzymes include acetylcholinesterase (AChE), butyrylcholinesterase (BuChE), urease, glucosidase, glucoamylase, maltase, monoamine oxidase A/B (MAO A/B), carbonic anhydrase, beta-secretase, dipeptidyl peptidase-IV (DPP-IV), angiotensin converting enzyme (ACE), aldose reductase, and aldehyde reductase. In vitro studies have demonstrated the efficacy of these inhibitors, while in vivo experiments have provided their detail pharmacokinetic and pharmacodynamics profiles. In silico analyses were employed to elucidate binding mechanisms and predict bioactivity, thereby accelerating the drug discovery process and ensuring rational optimization for enhanced specificity and reduced off-target effects. Among the synthesized compounds, notable inhibitory activity and selectivity were observed with coumarin-iminothiazolidinone hybrids, keto esters, isobenzofuranone derivatives, piperidinyl-quinoline N-acylhydrazones, hydrazine compounds, quinoline-thiosemicarbazone hybrids, and (piperidin-1-yl)quinolin-3-yl)methylene)hydrazinecarbothioamide derivatives. Consequently, our findings represent a significant advancement in the development of therapeutics for NCDs and provides potential future therapeutics to address the challenging landscape of NCDs.

Functional Nanomaterials – Tuning the Size and Surface Chemistry for Applications in Biomedical Science, Renewable Energy Technologies and Environmental Remediation

Irshad Hussain

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Subject Category: Materials Science

Abstract. The unique chemical and physical properties of nanoscale materials have triggered great scientific interest to explore their potential applications in biomedical sciences, reenable energy technologies, environment, catalysis, and industry etc. The chemical and physical properties of metal/ metal oxide nanoparticles can generally be tuned by controlling their size, shape, and surface chemistry. In this regard, we have developed several reproducible protocols based on chemical reduction and precipitation approach to prepare functionalized metal/metal oxide nanoparticles from subnanometer to over 100 nm in aqueous/organic media with a decent control over their size, shape, and surface chemistry. Many of these metal nanoparticles have been used as building blocks to design/synthesize new nanostructured materials with tunable nanoscale features using template-based and template-less strategies. The functionalized metal/metal oxide nanoparticles/nanoclusters possess interesting optical, recognition and catalytic/bio-catalytic properties and currently we are focusing on the applications of these nanoparticles and nanocomposites in biomedical sciences (i.e., bio-sensing especially bacterial detection, bio-imaging, drug delivery, multidrug resistance), environmental remediation (detection and removal of organic/inorganic pollutants from water, CO oxidation, and CO_2 capture and conversion) and renewable energy technologies (mainly H_2 production & storage and electrode materials for batteries) based on electrocatalytic and photocatalytic approaches. This talk would, therefore, be an overview of interdisciplinary research activities of Functional Nanomaterials Group at LUMS to synthesize customized inorganic/organic nanoparticles with tunable size and surface chemistry, and their composites having unique chemical and physical properties, and subsequent applications in biomedical sciences, environment, catalysis and renewable energy technologies.

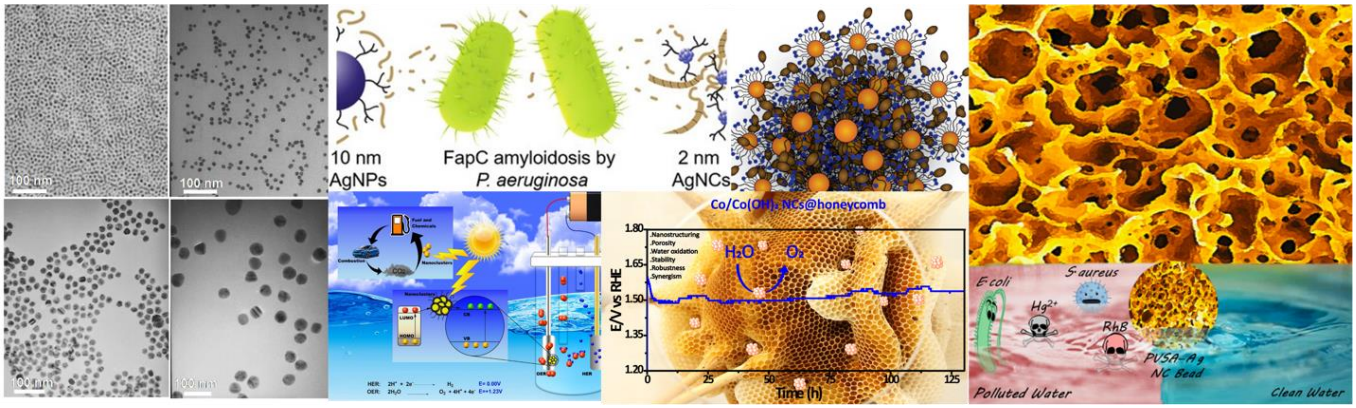


Figure showing electron microscopic images/cartoons of different functional nanomaterials prepared in my group at LUMS.

Investigation of novel pyrazolo[4,3-*e*]tetrazolo[1,5-*b*][1,2,4]triazine sulfonamide derivatives as potential candidates against colon cancer

Mariusz Mojzych

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Subject Category: Organic Chemistry

Abstract:

Background and aims: Cancer therapy is one of the most important challenges of modern medical and chemical sciences. Among the many methods of combating cancer, chemotherapy plays a special role. Imperfect modern chemotherapy justifies continuing the search for new, more effective, and safe drugs *i.e.* without an impact on normal cells. Sulfonamides are the classic group of chemotherapeutic drugs with a broad spectrum of pharmacological activity. Recent literature reports show that sulfonamide derivatives have antitumor activity *in vitro* and *in vivo*. Thus, the objective of this study was to synthesize novel pyrazolo[4,3-*e*]tetrazolo[1,5-*b*][1,2,4]triazine sulfonamide derivatives with anticancer potential against colon cancer cells [1,2].

Results: The research carried out included MTT assay, DNA biosynthesis, cell cycle analysis, Annexin V binding assay, ethidium bromide/acridine orange staining, and caspase-8, -9, and -3/7 activity. Moreover, the concentrations of important molecules (sICAM-1, mTOR, Beclin-1, cathepsin B) involved in the pathogenesis and poor prognosis of colorectal cancer were also evaluated by ELISA. Obtained results demonstrated that the novel compounds were able to induce apoptosis through intrinsic and extrinsic pathways and was capable of decreasing sICAM-1, mTOR, cathepsin B concentrations, whereas increased Beclin-1 concentration was detected in both colon cancer cell lines (DLD-1 and HT-29).

Conclusions: The novel compounds represents promising multi-targeted potential in colorectal cancer, but further *in vivo* examinations are needed to confirm the claim. The most active derivative turned out to be tricyclic sulfonamide **MM131** that possesses promising anticancer properties. It inhibits the viability and proliferation of DLD-1 and HT-29 cells. The molecular mechanism of action is associated with the induction of both extrinsic and intrinsic apoptotic pathways. This is related to increased caspase-8 and caspase-9 activity. **MM131** leads to the inhibition of important proteins engaged in the progression and metastasis of colorectal cancer such as sICAM-1, cathepsin B, and mTOR.

Interfacial Electrochemistry: from fundamentals to applications

Angel Cuesta Ciscar

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Subject Category: Electrochemistry

Abstract: Interfacial electrochemistry aims at understanding the structure of the electrode-electrolyte interface and the interfacial charge transfer processes happening therein with the greatest possible detail, i.e., at the atomic level. Such information is crucial for the development of electrochemical devices with applications, amongst other, in energy conversion and storage.

In my lecture, I will first give an overview of the fundamental problems dealt with in my group and of the techniques we use to tackle those problems. Then, I will show an example of applied research based on the knowledge acquired from that fundamental work, describing an acid-base redox-flow battery that uses hydrogen as the charge carrier without consuming it and can both generate power and desalinate water during discharge.

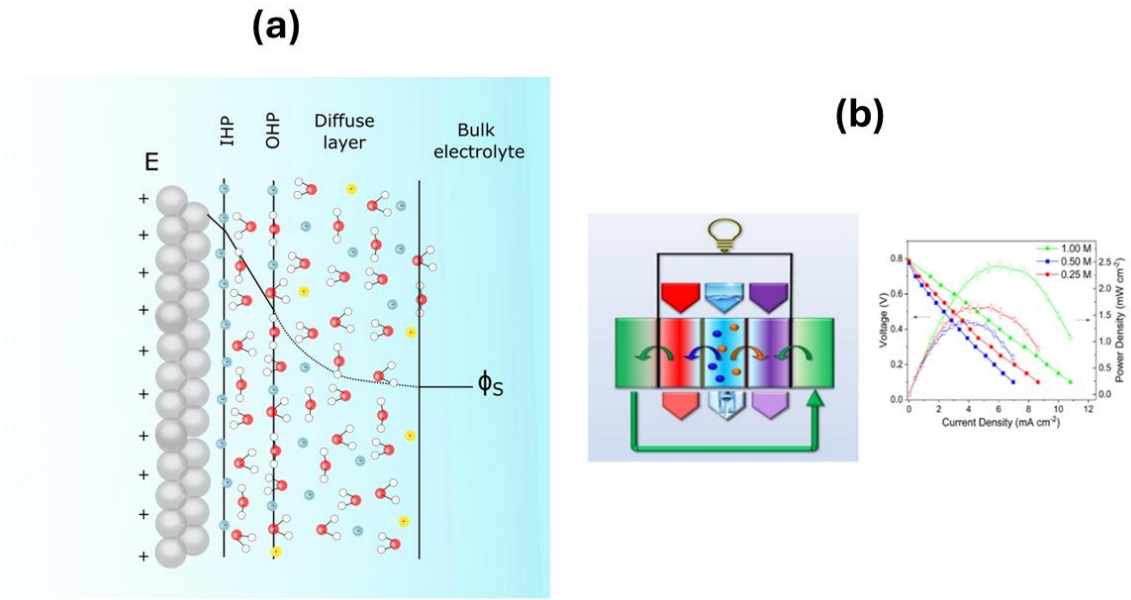


Figure 1. (a) Scheme depicting Stern's description of the electrochemical interface in the presence of specifically adsorbing anions for the particular case of a positively charged surface; **(b)** Scheme of the acid-base redox-flow battery (left) and polarization curve during discharge at different concentrations of acid and base.

Employing electrochemistry for improved asset management

Geoffrey Will

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Subject Category: Materials Science

Abstract: The study of electrochemistry holds profound significance in understanding material degradation rates and their broader implications for technological progress. Over centuries, corrosion has been meticulously investigated, leading to a deeper grasp of mechanisms and the development of advanced materials. A fundamental comprehension of corrosion processes and effective management strategies is indispensable for optimizing material usage and implementing preventative measures. Factors such as material type, environmental conditions, and historical context must be holistically considered to predict degradation mechanisms accurately. Electrochemistry has traditionally served as a cornerstone in unravelling corrosion mechanisms and rates. Recent advances in electronics and data analysis further enhance our ability to precisely quantify these phenomena and educate future generations. At an industrial level, assessing asset corrosion often relies on sophisticated inspection techniques utilizing imaging and other tools to assess material conditions. Often overlooked are the electrochemical techniques available for the evaluation of corrosion rates, the integrity of coating and evaluation of service life. The integration of these into predictive models capable of providing insights into performance, maintenance/inspection and the cost are key efficient use of an asset. With expanding data capabilities and analytical techniques, leveraging electrochemistry becomes more accessible, ensuring longevity and reliability in material applications.

Advanced Porous Membranes with Tuneable Morphology for High Performance Lithium Batteries

Arshad Hussain

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Subject Category: Materials Science

Abstract: Lithium batteries (LBs) are considered as one of the most potential promising power batteries due to their superior potential energy density. Unfortunately, uncontrolled Li dendrite growth during cycling leads to an unstable interface, a limited cycling life, and a low Coulombic efficiency, further results in severe capacity decay and safety issues. To address these problems, an effective approach is proposed to realize uniform Li nucleation. Herein, we demonstrate a thermally stable polymer for making porous membranes with a tuneable morphology via a versatile non-solvent treatment method for LBs applications. The final morphology of polymer membranes is controlled by selecting the appropriate non-solvent and an optimized performance could also be obtained. Notably, our functional porous polymer membrane with abundant polar amine functional groups could enable dendrite-free Li deposition, showing stable and low overpotential cycling durability at a current density and capacity. With such a membrane, the batteries cell shows superior cycle stability and rate performance, which are higher than those of commercial separators. Remarkably, this work offers a simple and facile method to prepare advance porous membranes for LBs with improved dendrite resistance.

Keywords: Lithium metal batteries, Porous membrane, Tuneable morphology, Dendrite

Low cost and Sustainable Hybrid Materials for Energy Conversion and Storage Applications

Prof. Dr. Zafar Hussain Ibupoto

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Subject Category: Material Science

Abstract: Hybrid materials with low cost and sustainable growth process are highly desirable for the clean energy, clean environment. The sustainable synthesis is one of most wanted UN mission to be emphasised for the minimum or zero release of toxic chemicals into environment. Hybrid materials have proven as efficient materials for the diverse applications such as green energy, and environmental applications owing to their high specific surface area, large exposure of catalytic sites, and fast electron communication. In this talk, we will focus on the fundamental understanding of nanostructured materials, the present research results on the green energy, and energy storage systems. We will also focus on the scientific challenges in these fields and their future aspects.

Keywords: Nanostructured materials, Green energy, Environment, Biomedical applications

The Relationship Between Oxide-Ion Conductivity and Cation Vacancy Order In the Hybrid Hexagonal Perovskite $Ba_3VWO_{8.5}$

Asma Gilanea,b, Sacha Fopb,Falak Shera and Abbie C. Mclaughlinb*

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Subject Category: Material Science

Abstract: The study of oxide ion conductors has recently gained great attention owing to their use in electrolytes for solid oxide fuel cells (SOFC) and solid oxide electrolyzer cells (SOEC), oxygen separation membranes, oxygen sensors, and oxygen pumps. Currently, high working temperatures (> 700 °C) for SOFCs is a major hurdle that leads to poor durability of components, increased costs, slow start-up times, and issues with selection of compatible materials. Significant research efforts are being devoted to the development of oxide ion conductors that can operate at intermediate temperatures (400-600 °C). Substantial oxide ionic conductivity has recently been reported in cation-deficient hexagonal perovskite $Ba_3M'M''O_{8.5}$ derivatives, with disordered hybrid 9R-palmierite average structures, which has opened a new avenue for researchers to search novel oxide ionic conductors within this family.

We have synthesized the phase pure $Ba_3VWO_{8.5}$ hexagonal perovskite derivative and the crystal structure and electrical properties investigated using various analytical techniques. The electrical characterizations demonstrate that $Ba_3VWO_{8.5}$ is a good oxide ion conductor with total conductivity of 6.2×10^{-4} S cm^{-1} in air at 900 °C. While previously reported $Ba_3NbMoO_{8.5}$ and $Ba_3NbWO_{8.5}$ present random distribution of cationic vacancies, X-ray and neutron diffraction experiments demonstrate that the cationic vacancies are ordered on the M2 sites in $Ba_3VWO_{8.5}$, resulting in a structure where $M1O_x$ palmierite-like layers are separated by empty octahedral cavities. Bond-valence site energy (BVSE) analysis on different phases demonstrates that ordering of the cationic vacancies hinders long-range oxygen diffusivity parallel to the *c*-axis in $Ba_3VWO_{8.5}$, thus explaining the reduced ionic conductivity of this compound.

New synthetic non-toxic mono-azo acid dyes with prominent antibacterial properties for potential application on polyamide fabrics

*Muhammad Sohail Khan Rafiq[a], Abdul Jabbara[b], *, Salma Farooq[c], Muhammad Salman[a], Sajida Munsif[d], Zaheer Ul-Haq[d]*

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bTextile Institute of Pakistan, Karachi, Pakistan

cNED University of Engineering and Technology, Karachi, Pakistan

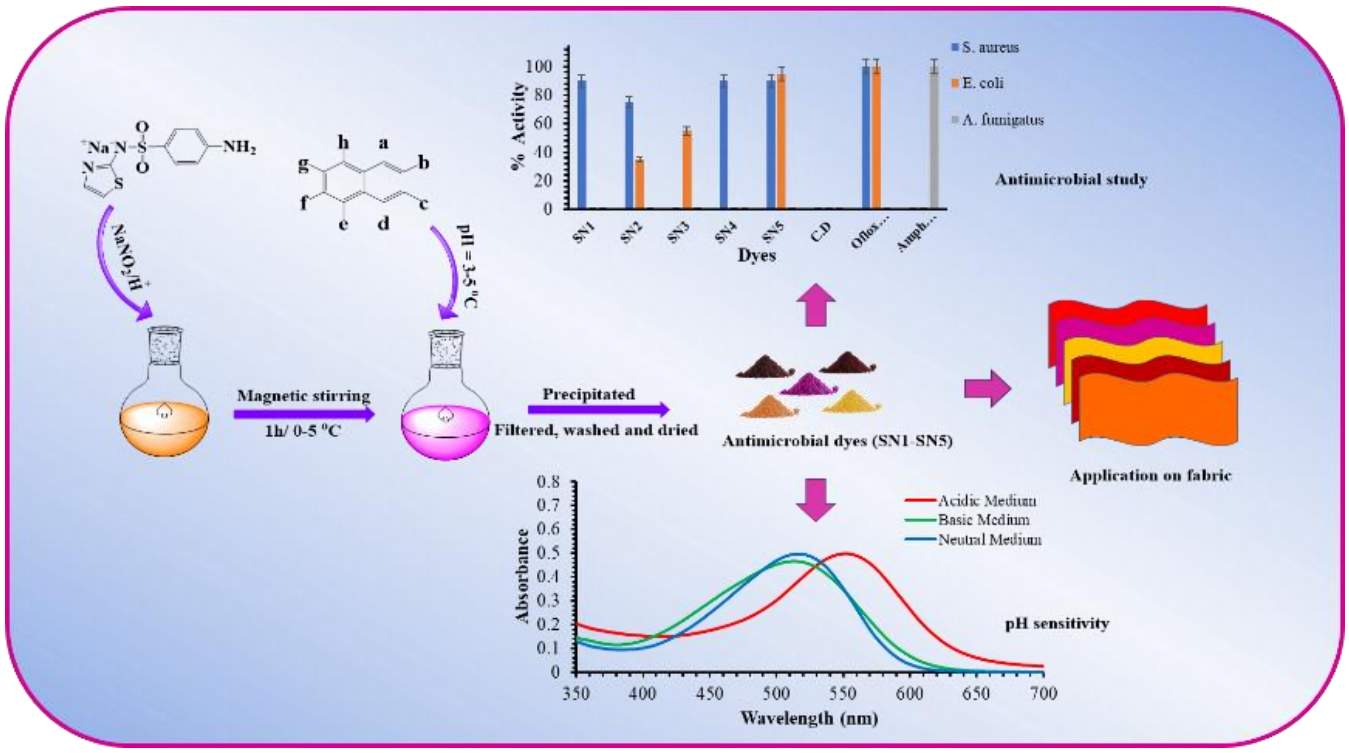
dDr. Panjwani Center for Molecular Medicine and Drug Research, International Center for Chemical and Biological Sciences, University of Karachi, Karachi, Pakistan

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Subject Category: Organic Chemistry

Abstract: In pursuit of improving dyeability and incorporating antimicrobial attributes into polyamide fabric, a series of mono-azo acid dyes (SN1-SN5) was synthesized. Sulfathiazole was employed as an intermediate in the synthesis process of the dyestuff. The resultant dyes were characterized by employing various analytical techniques. The antimicrobial efficiency of these colorant was established through testing against Aspergillus fumigatus, Staphylococcus aureus, and Escherichia coli, displaying a spectrum of mild to strong antibacterial activity. Dye SN5 exhibited the highest potency, showing growth inhibition of 90% and 95% against both bacterial strains respectively. Since the acid dyes are the prime contenders for utilization in food, pharmaceuticals, and cosmetics, an assessment was conducted on BJ Human Fibroblast Cells to evaluate any plausible adverse impacts. The results revealed that all the dyes exhibited no cytotoxicity. The dyeing properties of these colorants, including color representation, build-up, exhaustion, and fixation, were thoroughly examined following their application on the fabrics. The fastness properties were found to be considerably good to excellent. The influence of acidic and alkaline conditions on the absorption maxima was also explored. The dye SN3 showed a substantial bathochromic shift with increase in the solution's acidity in comparison to other dyes.

Graphical Abstract:



An Updated Overview (2003–2023): Investigating Marine Life as a Significant Breakthrough in the Fight Against Prostate Cancer

Amjad Hussain [a], Ammara Sohail [a]*

[a] Institute of Chemistry, University of Okara, Okara-56300, Pakistan

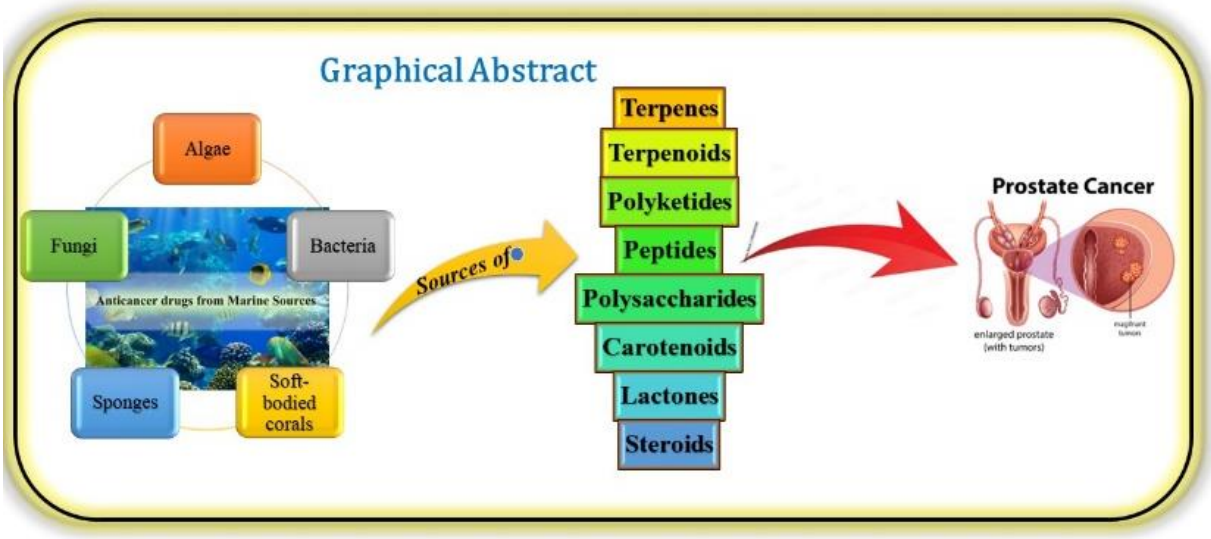
** E-mail: amjadhussain@uo.edu.pk*

Subject Category: Organic Chemistry

Abstract:

Currently, cancer is the biggest threat that the mankind is facing. Prostate cancer ranks as the second most common cancer among men. Environmental or geographical variables, age, and family history are risk factors for prostate cancer. Chemotherapy failures have led to a higher death rate among cancer patients. In the past ten years, researchers have made significant progress in using naturally occurring medications to counteract this worst threat. Compared to terrestrial creatures, marine organisms have been proved as a promising source of novel bioactive agents. Marine natural products are preferred over other natural products due to their signaling pathways for carcinogenesis. The anti-prostate cancer drugs derived from fish, algae, fungi, bacteria, sponges, soft-bodied corals, and other diverse marine species are the subject of this study. Terpenoids, carotenoids, polyketides, peptides, lactones, quinones, polysaccharides, and steroids are some of the classes to which these natural compounds belong. This research survey reports the data published during the entire duration between 2003 and 2023 that includes anti-prostate agents derived from marine sources.

Graphical Abstract:



Sacrificial Template Synthesis of Tailorable Conductive MOFs for Electrocatalytic CO_2 Reduction to Ethanol

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Subject Category: Materials Science

Abstract: In an effort to combat climate change, the electrochemical CO_2 reduction reaction (CO_2RR) has drawn a lot of interest as a potentially effective method for turning CO_2 into valuable chemicals and fuels [1]. Since copper and copper-based compounds have the appropriate intermediates binding energetics that permit serial reaction cascades, they are known to be highly effective in producing deeply reduced multi-electron products. Even with their great industrial and economic qualities, ethanol and methanol—among the many reaction products that include C_1 and C_{2+} hydrocarbons and oxygenates—have proven difficult to obtain in high yield and selectivity. One explanation might be the very reductive environment at the cathode surface, which makes it challenging for the oxygenic intermediates to remain, particularly when they go beyond the C–C coupling potentials. To generate these multi-carbon alcohols with high turnover rates and selectivity, exquisite catalyst design and manufacturing are therefore required. These efforts attempt to overcome the linear scaling relation and optimize individual intermediate binding [2].

In order to enhance the oxygenate generation in CO_2RR, several strategies have been implemented. First of all, it has usually been believed that high local CO concentrations, whether from CO supplied to the feedstock [3], or CO generated by tandem catalysts [4], are advantageous for the production of alcohols. Secondly, the alcohol production has been effectively increased by using the synergies that arise from the interaction between the catalyst and substrate, such as Cu clusters supported on metals and oxidized carbons. Third, there has also been a significant synthesis of oxygenates from nitrogen-doped graphitic carbon and its complexation with Cu moieties[5]. Not to mention, the use of crystalline defects—particularly those that are near together—has lately been employed as a powerful tactic to promote the formation of alcohol [6]. These various strategies generally aim to reduce the overpotential for C–C interaction in order to lessen the very reductive electrolytic environment and to stabilize the oxygenated intermediates by using a companion active motif. Therefore, to target alcohol generation in CO_2RR, it is essential to create effective electrocatalysts based on fundamental understandings of the C–C coupling and oxygenation process.

Herein, we demonstrate a general sacrificial template strategy for the synthesis of layer-stacked 2D *c*-MOFs via MOFs-to-*c*-MOFs transformations, in which the insulating 3D MOFs are taken as sacrificial

templates. To demonstrate the generality of this sacrificial template strategy, we have considered the use of different metals (Cu, Zn, and bimetal Cu/Zn) with ligand benzene hexathiol (BHT). The as-prepared 2D *c*-MOFs exhibit intrinsic electrical conductivities and high surface areas, which can be utilized for the CO_2RR. The resultant Cu-BHT hollow nanocubes present an improved surface area of 94.8 m^2 g^{-1} (45 time higher than that of the bulk Cu-BHT by traditional synthesis). We establish a breakthrough sacrificial template strategy to realize a general transformation from insulating 3D MOFs to conductive 2D MOFs and are able to precisely tune the 2D *c*-MOFs for target device functions, which pushes the further development of conductive MOFs for energy and electronics. In this paper, we first investigate the suitability of a solventless synthesis approach for producing heterometallic MOFs by doping the benchmark HKUST-1(Cu), and then we analyse the performance of these materials in the electrocatalytic conversion of CO_2. Zn(II), have been chosen as dopant metal because of its compatibility with the metal site and their known catalytic activity and long-term stability (Zn(II)) . In order to enhance the conductivity and active site functionality, sacrificial template synthesis is employed with the aid of hexa-benzenethiol (BHT) monomer and convert the Bimetallic HKUST-1 (Cu, Zn) into 3D Cu-BHT MOF. To analyse the doping process, the powdered material was fully characterized, and each doped-MOF was treated as an electrode for CO_2RR. In electrochemical carbon dioxide reduction studies for ethanol, Zn-doped Cu-BHT MOFs show enhanced Faradaic efficiency, with Cu_8-Zn_2 exhibiting a remarkable total Faradaic efficiency of 95.5% for ethanol generation at -1.1 V vs RHE. The stability of BHT-Cu_8-Zn_2 is demonstrated over a 72-hour testing period, maintaining an ethanol Faradaic efficiency of 89% and a stable current density. In-situ ATR-SEIRAS provides insights into the reaction intermediates, confirming the proposed ethanol mechanistic pathway on BHT-Cu_8-Zn_2. Additionally, computational studies support the experimental findings, further elucidating the electrocatalytic process. This work showcases a breakthrough sacrificial template strategy and displaying enhanced performance, mechanistic insights, and stability of Zn-doped Cu-BHT MOFs underscore their potential as promising candidates for sustainable CO_2 electroreduction technologies.

Discovering the anti-diabetic potential of Pomegranate peel metabolites by examining molecular interplay with the TXNIP protein

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Subject Category: Organic Chemistry

Abstract: *Punica granatum* (pomegranate) is traditionally used against diabetes, inflammation, and other diseases. This study explores the interaction of *P. granatum*-derived compounds with the diabetes-associated human thioredoxin-interacting protein (TXNIP). High glucose levels in diabetes induce TXNIP production, leading to β-cell apoptosis. Inhibiting TXNIP may reduce diabetes incidence. The study used GC-MS to identify compounds in *P. granatum* peel and computational algorithms to assess their therapeutic potential against diabetes. Peel extracts were screened for antioxidant, anti-inflammatory, anti-diabetic, antimicrobial, and wound healing properties. Phytochemical and GC-MS analyses identified bioactive compounds, and molecular docking (Auto Dock Vina) predicted their binding tendencies with TXNIP. The peel exhibited notable antioxidant, anti-inflammatory, and anti-diabetic activities, attributed to phenols, tannins, and steroids. Key compounds identified included bis (2-ethylhexyl) phthalate and 9-octadecenoic acid, with high binding affinities to TXNIP (-4.5 and -5.0 kcal/mol, respectively), suggesting their potential as anti-diabetic agents. This study validates the traditional uses of *P. granatum* peel and demonstrates the efficacy of computational approaches in discovering pharmacologically active phytochemicals. Further lead optimization of identified ligands is recommended.

Molecularly Imprinted Polymers: Promising Tool for the Selective Detection of Dengue

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Subject Category: Analytical Chemistry and Material Science

Abstract: Dengue virus is affecting almost 50 million peoples annually and at present no vaccine is available for preventing the spread of this virus, therefore it is crucial to diagnose this at earliest. Dengue NS1 protein is produced by all dengue serotypes and is therefore chosen as target analyte and is present in blood in high quantities at early stages of infection as an excellent disease biomarker. Molecularly imprinted polymers (MIPs) can offer a highly selective recognition sites within polymeric matrices for the selective binding of NS1 protein molecules. Efficiency of imprinting technique for preparing imprinted cavities to bind selectively with target protein (NS1) was demonstrated herein. The process of using a mass sensitive transducer-QCM and Interdigital Electrodes (IDEs) is most ideal and accurate in-situ analytical technique for studying receptor-protein interactions. Apart from mass-sensitive based sensors, electrochemical based sensor system and photonic crystals were also developed for the detection of dengue NS1 protein. The fabricated sensor may have potential technological applications in the field medical diagnostics for dengue virus detection.

Graphical Abstract:

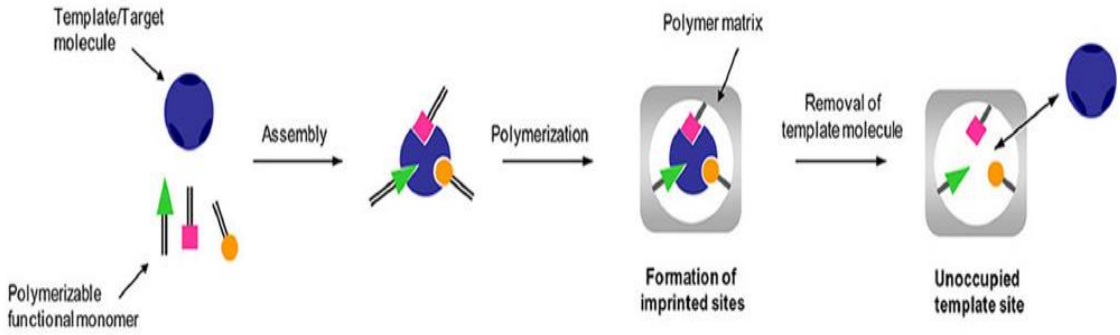


Figure. Molecular Imprinting

Study the effect of experimental parameters on N-doped carbon materials as an electrochemical supercapacitor's electrode

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Subject Category: Organic Chemistry

Abstract: The hierarchical porous nitrogen-doped carbon materials (NDCMs) were prepared from an organo-ionic dye precursor via high temperature carbonization methodology. The various experimental parameters during carbonization and the post-carbonization were investigated. The synthesized NDCMs were characterized by physicochemical techniques; FTIR, Raman spectroscopy, XRD Diffraction, XPS, BET/BJH, SEM, EDS, TEM, and further tested as an electrode material for supercapacitors in different aqueous electrolytes. The results suggested that multistep heating profile at relatively low annealing temperature under nitrogen atmosphere, 1.0 M $H_2SO_{4(aq)}$ electrolyte, and post acid etchings result in more capacitive electrode material attributed to various kind of doped nitrogen configurations, reasonable surface area and hierarchical porosity. The NDCM; $NCM_{(EBT)}$-700 exhibited specific capacitance of 203 F g^{-1} at 3 A g^{-1}, high rate capability and cyclic stability subjected to reasonably high specific surface area (614 m^2 g^{-1}) with hierarchical porosity and three distinct types of nitrogenous functionalities viz pridinic (N-6), pyrrolic (N-5) and quaternary (N-Q).

Cleaning the planet earth- A game changes in climate change

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Subject Category: Materials Science

Abstract: We survive with different environmental issues in this world but now increasing globalization and continuous population growth, everyone feels the stress whether familiar with climate change or not but directly or indirectly we are all at great risk. To overcome this major problem, we synthesized a novel nanocomposite containing magic material E which undergoes electron transfer mechanism exhibit electron-hole recombination in nanomaterial surface OM by fabricating the structures. Moreover, this dopped extended material is merged into the organic framework. In this way, a nanocomposite material synthesized which might be used for absorbing the multiple greenhouse gases and other hazardous pollutants providing a new way for not only minimizing the pollution in our environment but also provide a sustainable way for our planet earth. This nanocomposite after a careful examination of different instrumental techniques like Raman, Mossbauer, GC, Gas Detector and GC-MS deduced a comprehensive conclusion. The efficiency of the nanocomposites have shown a remarkable results and provides compatibility for our current and future needs.

Synthesis of Tri/Penta-Valent Antimony Complexes Using Phenolic Ligands: Evaluation of Their Antileishmanial Potential By In Vitro & In Silico Analysis

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Subject Category: Organic Chemistry/Medicinal Chemistry/Computational Chemistry

Abstract: Leishmaniasis, caused by diverse protozoan parasites of the genus *Leishmania*, is a negligent topical disease afflicting million globally, thus posing a significant public health challenge. In the present study four new antimony complexes were synthesized using phenolic ligands, and were characterized by FTIR and NMR spectroscopic techniques. Further structural parameters were evaluated *via* DFT studies. The antileishmanial activity of these complexes (**3a-3d**) was assessed *in vitro* against Leishmania strains, showing promising potential as antileishmanial agents. The compound **3a** was found most active among others, with inhibitory concentration value in the range of 6.6-9.4 µmol/L. The molecular docking analysis of these ligands against receptor protein from Leishmania major with PDB ID: **8FI6** revealed that these compounds have strong electrostatic interactions with receptor protein within the active pocket indicating that these complexes could be new therapeutic agents for leishmaniasis by leveraging the unique properties of tri/pentavalent antimony complexes.

Innovative structural modification of Glitazones *via* lead compound modification methodology

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Subject Category: Organic Chemistry

Abstract: Glitazones, a class of antidiabetic drugs that work by targeting PPAR-γ receptor, were banned in 2007 for associated adverse effects like Myocardial Infarction. In this study, safer alternatives of Glitazones were aimed through structural modification using lead compound modification methodology based on molecular hybridization. Employing both convergent and divergent synthesis, two series of 18 novel compounds were synthesized using reactions such as heterocyclization, nucleophilic substitution, addition elimination and condensation reactions. The synthesized compounds were characterized using NMR and IR spectroscopy. Molecular docking studies with PPAR-γ revealed that 14 out of 18 synthesized analogs exhibited better binding energies compared to the references, Pioglitazone and Rosiglitazone. These promising results set the stage for upcoming *in vitro* and *in vivo* evaluations, potentially leading to safer PPAR-γ targeting antidiabetic therapies. Our findings could pave the way for the next generation of Glitazone based treatments with better clinical outcomes.

Graphical Abstract:

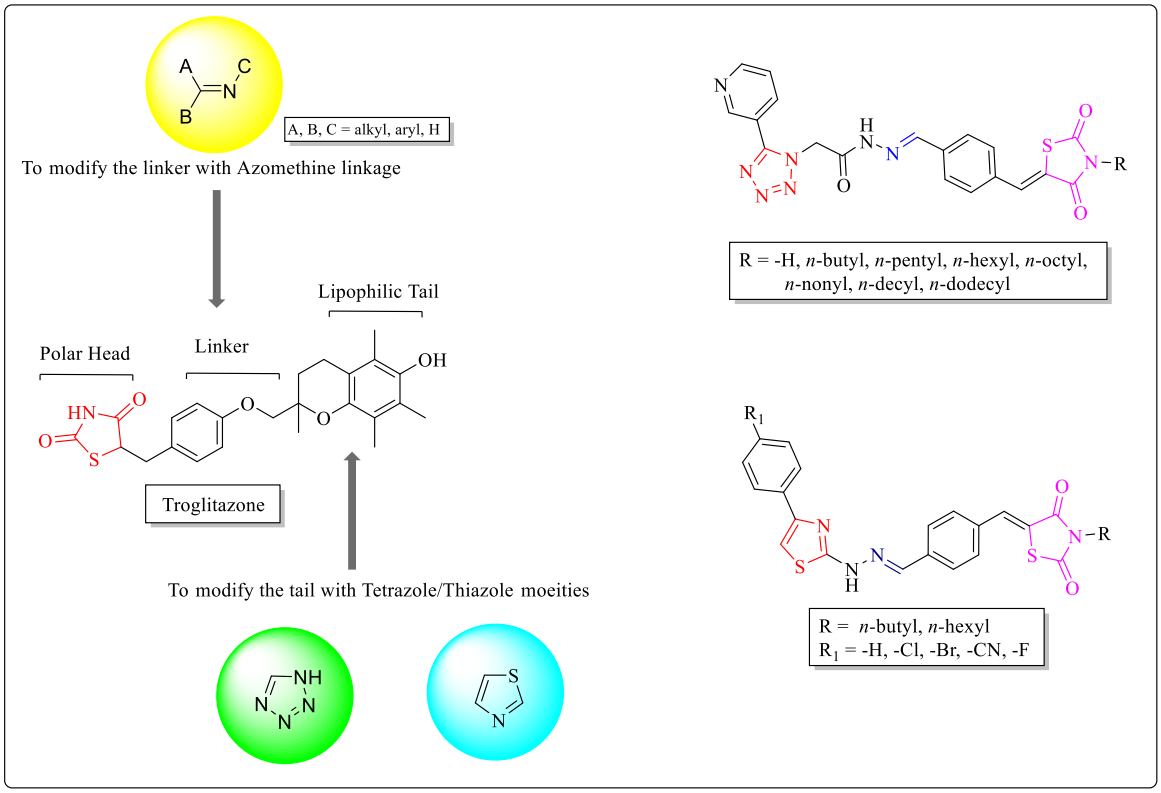


Figure: Structural Modification of Glitazones.

Amide Functionalized Mesoporous MOF LOCOM-1 as Stable Highly Active Basic Catalyst for Knoevenagel Condensation Reaction

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Subject Category: (Organic Chemistry, Organic Transformation, Heterogeneous Catalysis)

Abstract: Acyl-amide is extensively used as functional group and is a superior contender for designing of MOFs with guest accessible Functional Organic Sites. A novel acyl-amide-containing tetracarboxylate ligand, bis-(3,5-dicarboxy-pheny1)terephthalamide has been successfully synthesized. H_4L linker has some fascinating attributes as follows: (i) four carboxylate moities as coordination sites confirm affluent coordination approaches to figure a diversity of structures; (ii) two acyl-amide groups as guest interaction sites can engender guest molecules integrated into MOF networks through H-bonding interfaces and have prospective to act as Functional Organic Sites for condensation reaction. A mesoporous MOF ($[Cu_2(L)(H_2O)3]•4DMF•6H_2O$) have been prepared in order to produce amide FOS within MOF which will work as guest accessible sites. The prepared MOF was characterized by CHN analysis, PXRD, FTIR spectroscopy, and SEM analysis. MOF showed superior catalytic activity for knoevengal condensation. The catalytic system endures a broad variety of functional groups and presents high to modest yields of aldehyde containing electron withdrawing group (4-chloro, 4-fluoro, 4-nitro) offer yield > 98 in less reaction time as compared to aldehydes with electron donationg group (4-methyl). The amide decorated MOF (LOCOM-2) as heterogeneous catalyst can be simply recovered by centrifugation and recycled again without flagrant loss of its catalytic efficiency.

Opuntia monocantha: Validation of the anti-inflammatory and anti-arthritic activity of its polyphenolic rich extract in silico and in vivo via assessment of pro- and anti-inflammatory

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Subject Category: Pharmaceutical sciences

Abstract: Ethnopharmacological relevance: Opuntia monacantha belongs to the cactus family Cactaceae and is also known by cochineal prickly pear, Barbary fig or drooping prickly pear. It was traditionally used to treat pain and inflammation. O. monacantha cladodes showed pharmacological effects such as antioxidant potential owing to the presence of certain polysaccharides, flavonoids, and phenols. This research aimed to evaluate the anti-inflammatory as well as the anti-arthritic potential of ethanol extract of Opuntia monacantha (E-OM). Materials and methods: In vivo edema in rat paw was triggered by carrageenan and used to evaluate anti-inflammatory activity, while induction of arthritis by Complete Freund's Adjuvant (CFA) rat model was done to measure anti-arthritic potential. In silico studies of the previously High-performance liquid chromatography (HPLC) characterized metabolites of ethanol extract was performed by using Discovery Studio 4.5 (Accelrys Inc., San Diego, CA, USA) within active pocket of glutaminase 1 (GLS1) (PDB code: 3VP1; 2.30 Å). Results: EOM, particularly at 750 mg/kg, caused a reduction in the paw edema significantly and decreased arthritic score by 80.58% compared to the diseased group. It revealed significant results when histopathology of ankle joint was examined at 28th day as it reduced inflammation by 18.06%, bone erosion by 15.50%, and pannus formation by 24.65% with respect to the diseased group. It restored the altered blood parameters by 7.56%, 18.47%, and 3.37% for hemoglobin (Hb), white blood count (WBC), and platelets, respectively. It also reduced rheumatoid factor RF by 13.70% with concomitant amelioration in catalase (CAT) and superoxide dismutase (SOD) levels by 19%, and 34.16%, respectively, in comparison to the diseased group. It notably decreased mRNA expression levels of COX-2, IL-6, TNF-α, IL-1, NF-κβ and augmented the levels of IL-4 and IL-10 in real time PCR with respect to the diseased group and piroxicam. HPLC analysis previously performed showed that phenolic acids and flavonoids are present in E-OM. Molecular docking studies displayed pronounced inhibitory potential of these compounds towards glutaminase 1 (GLS1), approaching and even exceeding piroxicam. Thus, Opuntia monacantha could be a promising agent to manage inflammation and arthritis and could be incorporated into pharmaceuticals.

Potential Performance of ZnO/Zn_2TiO_4 Nanocomposites

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Subject Category: Material Science

Abstract: In this work, we report the synthesis of sub 70 nm nanocomposite of ZnO/ Zn2TiO4 by low temperature solution phase method. Titanium isopropoxide and zinc acetate were used as precursors and isopropyl alcohol and water as solvent. The formation of nanocomposites were studied by varying concentrations of precursors and surfactants at different pH values. All three ZnO/TiO_2/ Zn2TiO4 nanocomposites exhibited hexagonal wurtzite (ZnO) and anatase (TiO_2). ZnO, crystallite size in three ZnO/TiO_2 nanocomposites, was 62.67, 40.34 and 39.50 nm respectively. The natase phase TiO_2, crystallite size values in three ZnO/ Zn2TiO4 nanocomposite were 26.15, 25.36 and 21.24 nm, respectively. Morphological studies through FESEM showed formation of three ZnO/ Zn2TiO4 nanocomposites for entire samples. FTIR investigation depicted peaks in the range of 610-1250 cm^{-1} were attributed to Ti-O bond formation and peak located at 1350-1450 cm^{-1}, attributed to Zn-O bond formation. Also, EDX spectrum depicted the clear formation of ZnO/ Zn2TiO4 nanocomposite. Photocatalytic investigation carried out by analysing the UV-Visible absorption spectra of methylene blue dye on the surface of ZnO/TiO_2 nanocomposites, dye degradation at various irradiation time intervals, dye degradation at varied catalyst concentrations and the influence of pH on dye-degradation. Different sets of experiments were conducted under ambient conditions which showed an excellent photo catalytic degradtion efficiency.

First principal study on the adsorption of NO_2 gas on Al -doped WO_3 (001) and (100) surfaces

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Subject Category: Material Science

Abstract: A structural model of NO_2 adsorption on Al-doped WO_3 (001) and (100) surfaces has been developed. Utilizing the first-principles method based on density functional theory (DFT), we calculated the energy band, adsorption energy, and density of states. Our results indicate that Al doping significantly enhances both the energy and adsorption energy of the WO_3 (001) and (100) surfaces. This doping facilitates charge transfer from NO_2 to the WO_3 surfaces, causing the Fermi energy level to shift to a higher potential and reducing the band gap width. Consequently, the adsorption capacity for NO_2 gas molecules is substantially improved.

Impact of rGO and Nb2C-MXene Modified Electrodes on Performance Characteristics of Microbial Fuel Cell

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[b] Institute of Industrial Biotechnology, GC University Lahore, Lahore, Pakistan

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Subject Category: Analytical Chemistry / Material Science

Abstract: Water pollution and rapidly declining fossil based natural reserves are among the main global issues. Different attempts are being carried out worldwide to cope with these issues. Out of the available options, microbial fuel cells are getting acceptance as sustainable bio-electrochemical technology having capability of treating wastewater with simultaneous power generation. The current study was therefore aimed at investigating the impact of MFC's anode (carbon felt (CF)) modification with reduced graphene oxide (rGO) and Nb_2C-MXene on the operation performance regarding water treatment and power generation. Considerable improvement in the MFCs performance parameters was observed using modified anode i.e., rGO@CF and Nb_2C-MXene@CF comparative to the unmodified anode (CF). The power density values obtained for MFCs equipped with rGO@CF and Nb_2C-MXene@CF anodes were 294.91 mW/m^2 and 303.88 mW/m^2, respectively, which were significantly higher than 129.03 mW/m^2 observed for MFC using unmodified CF anode. Likewise, MFCs equipped with modified anode also displayed superior results regarding % COD reduction. The maximum % COD reduction observed was 78%, 79% and 64% for MFCs fabricated with rGO@CF, Nb_2C-MXene@CF and CF anode materials. It was revealed that, improvement of the MFC's anode with rGO and Nb_2C-MXene substantially improved the performance characteristics regarding the power production with simultaneous water treatment, most probably due to their higher surface area and conductivity. Hence, both rGO and Nb_2C-MXene exhibit potential to be employed as electrode material in MFCs for their better performance, however more studies are recommended for further optimization of other MFC operational parameters to make this technology more sustainable and cost effective.

The Electronic Structure of Uranium-Arene Complexes-Computational Studies

Riffat Parveen[a], Bess vlaisavljevich[b]

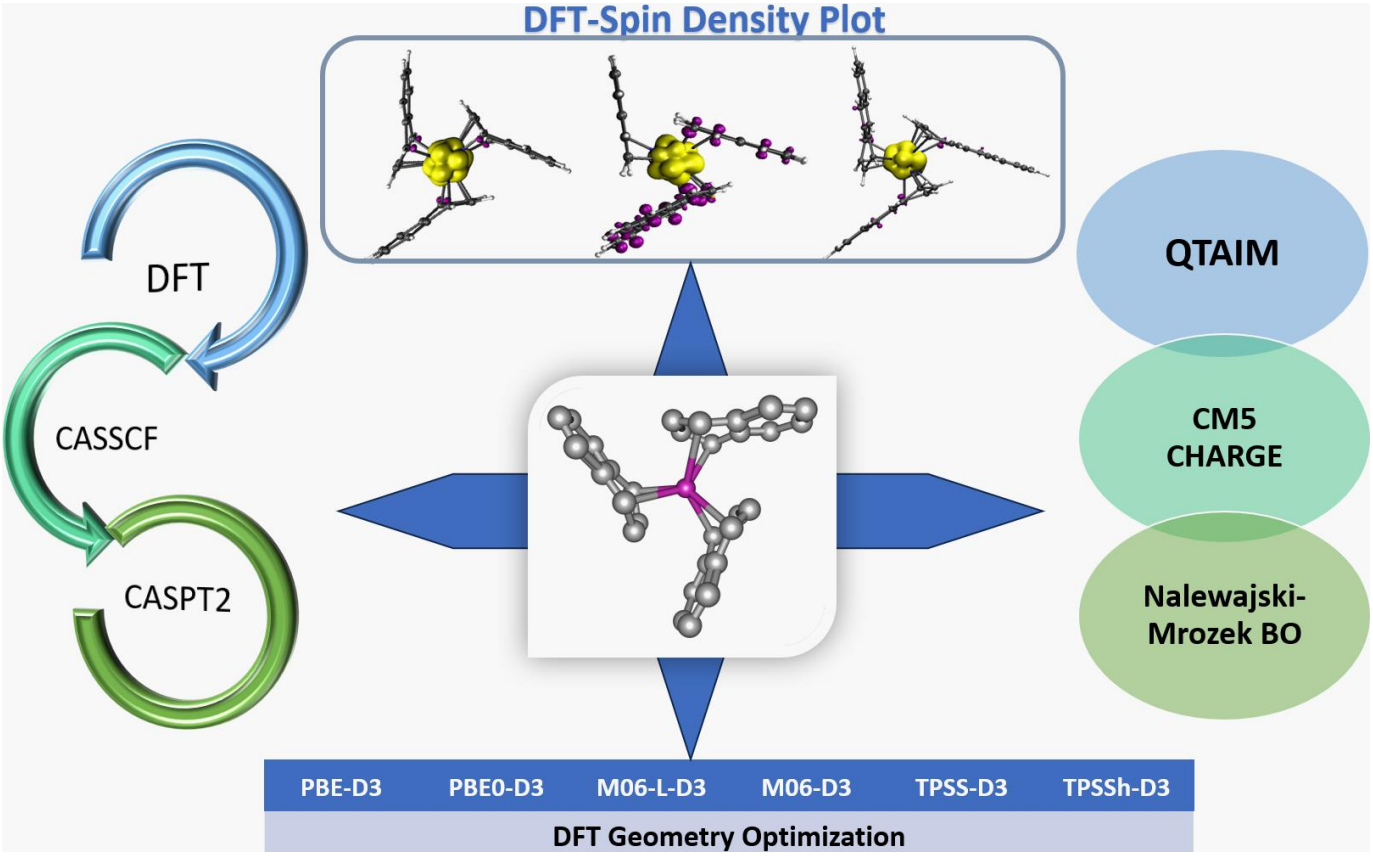
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Subject Category: Inorganic Chemistry

Abstract: The electronic structure of actinide complexes is interesting to explore due to the near degeneracies present in the valence orbitals and the ability of both the 5f and 6d orbitals to engage in bonding. In present work we have investigated the electronic structure of uranium-arene homoleptic complexes. This includes $[U(naphthenelide)_3]^{3-}$ and $[U(perylenide)_3]^{3-}$ and one heteroleptic complex of $[U(anthracenide)_3I]^{2-}$. Previous research has explored that the f orbitals on uranium and electrons from the arene can contribute in different types of bonding ranging from strong, highly covalent bonds to polarized donor-acceptor interactions between occupied orbitals on the ligand and empty orbitals on the metal. We have used a combine approach of DFT and multiconfigurational methods like CASSCF and complete active space second order perturbation theory (CASPT2), to study these complexes while ensuring that the electronic structure is properly described.

Graphical Abstract: (Optional)



Formulation of lip coloring cosmetics; A sustainable development of multifunctional products

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Subject Category: Material Science

Abstract: The synthetic colorants that contribute to the color of the cosmetics may cause harmful effects on continuous application as active ingredients in cosmetics. The present research work was designed to study the plant-derived colorants in skin coloring cosmetics. The work also involved the preparation of different formulations using various delivery vehicles to deliver these natural colorants. Colorants from different plants available in Pakistan were extracted, for example, red to purple shades colorants were extracted from the *Brassica oleracea* L., *Rosa indica* L., *Punica granatum* L., *Beta valgarus* L., etc. Three methods with three different polarity-based solvents ethanol, water, and acetone were used for the extraction of colorants. The anti-inflammatory activity of the extracts was done through the carrageenan inflammation model. Cytotoxicity of the extracts through hemolytic activity was calculated in terms of hemolytic activity. Characterization of ethanolic extracts of natural colorants was done by UV-visible spectroscopy, thin layer, and high-performance liquid chromatography. Formulations of different lip coloring cosmetics were prepared by incorporating natural colorants. The physical characteristics of prepared samples were carried out by using different physical tests after the application of prepared samples. Different physiochemical parameters were measured to characterize the formulated products, including pH, solubility, and skin irritation. All formulated cosmetics according to their melting point, breaking points, and perfume stability test. The development of multifunctional skin coloring cosmetics was accomplished using sustainable sources of coloring agents.

Characterization and SRIM Simulation of Ions Irradiated Nanowires

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Subject Category: Computational Chemistry

Abstract: Ni-NWs have emerged as a viable material due to their electrical, thermal, and magnetic properties. In present research work Ni-NWs are purchased from Plasma Chem (Germany) with diameter in the range 200 to 300 nm. Ni-NWs are irradiated by copper ions beam irradiation to study the changes in properties. Ni-NWs are exposed to copper ions beam of energies 30keV, 40keV and 50keV and fluence was 1 X 10^{15} ions/cm^2 at room temperature. Radiational effects of copper ions beam of various energies on the Ni-NWs structure and morphology are studied by characterizing the sample using X-Ray Diffraction and Transmission Electron Microscopy (TEM) techniques. XRD results show the characteristic peaks of Ni-NWs having cubic phase. Intensities of the peaks found to be reduced by changing the energies. Grain sizes were calculated by using Scherrer's equation. Calculations show that Crystallite size was increased up to 40 KeV and then started to be decreasing. Dislocation density and lattice strain is also calculated which showed opposite behavior compared to crystallite size. Dislocation density and lattice strain initially decreased and started increasing after 40 KeV. TEM images revealed that radius of Ni-NWs reduced by copper ions beams irradiation. Reduction in radius is observed due to lose of atom because of copper ions beam irradiations. Stopping Range of Ions in Matter (SRIM) was also performed to know about the presence of defects or ionization occurring due to irradiation in the un-irradiated and irradiated samples. This research highlights the changes in properties of Ni-NWs due to copper ions beam irradiations for its implementation as a transparent electrodes in optoelectronic devices.

Later, nano-welding technique for fabrication of X-, Y-, II- and T-shapes molecular junctions between silver nanowires (Ag-NWs) using irradiation by high energy proton beam is demonstrated [1, 2]. Thereafter, large scale networks of Ag-NWs is fabricated by high energy proton ions beam irradiation at different beam fluencies. Large scale welding is confirmed by transmission electron microscopy (TEM) and scanning electron microscopy (SEM) [3]. After those Ag-NWs networks have been irradiated with copper (Cu) ions

having MeV energy. The effect of ion fluencies on optical and electrical properties of Cu ion irradiated Ag-NWs are investigated. It has been observed that electrical conductivity and optical transmittance increased with the increase of Cu ion fluencies [1]. Similarly, nickel nanowires (Ni-NWs) meshes are fabricated by ions beam irradiation induced nanoscale welding of NWs on intersecting positions. Ni-NWs are exposed to beam of 50 KeV Argon (Ar^{+}) ions at various fluencies in the range ~10^{15} ions cm^{-2} to 10^{16} ions cm^{-2} to obtain welding of Ni-NWs and then exposed to 2.75 MeV hydrogen (H^{+}) ions to obtain strengthening in welds of Ni-NWs and modify electrical and optical properties [4].

Reduced Graphene Oxide-Zinc Sulfide Nanocomposite Decorated with Silver Nanoparticles for Wastewater Treatment by Adsorption, Photocatalysis and Antimicrobial Action

Hina Naeem[1]

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Subject Category: Materialsl Chemistry

Abstract: Reduced graphene oxide nanoflakes decorated with ZnS and ZnS-Ag nanoparticles are synthesized successfully through facile one-step chemical approach consisting of reducing metal precursors on rGO surface. Prepared rGO-ZnS nanocomposite is employed as an adsorbent material against two model dyes: malachite green (MG) and ethyl violet (EV). The adsorptive behavior of the nanocomposite was tuned by monitoring some parameters, such as the time of contact between the dye and the adsorbent, and the adsorbent dose. Experimental data were also simulated with kinetic models to evaluate the adsorption behavior, and the results confirmed that the adsorption of both dyes followed a pseudo 2nd order kinetic mode. Moreover, the adsorbent was also regenerated in a suitable media for both dyes (HCl for MG and ethanol for EV), without any significant loss in removal efficiency. Ag doped rGO-ZnS nanocomposite was also utilized as a photocatalyst for the degradation of the selected organic contaminant, resorcinol. The complete degradation of the phenolic compound was achieved after 60 min with 200 mg of rGO-ZnS-Ag nanocomposite under natural sunlight irradiation. The photocatalytic activity was studied considering some parameters, such as the initial phenol concentration, the photocatalyst loading, and the pH of the solution. The degradation kinetics of resorcinol was carefully studied and found to follow a linear Langmuir–Hinshelwood model. An additional advantage of rGO-ZnS and rGO-ZnS-Ag nanocomposites was antibacterial activity against Gram-negative bacterium, E. coli, and the results confirmed the significant performance of the nanocomposites in destroying harmful pathogens.

Investigations on shape-property relationship of magnetic and persistence luminescence nanomaterials coupled in a single assembly

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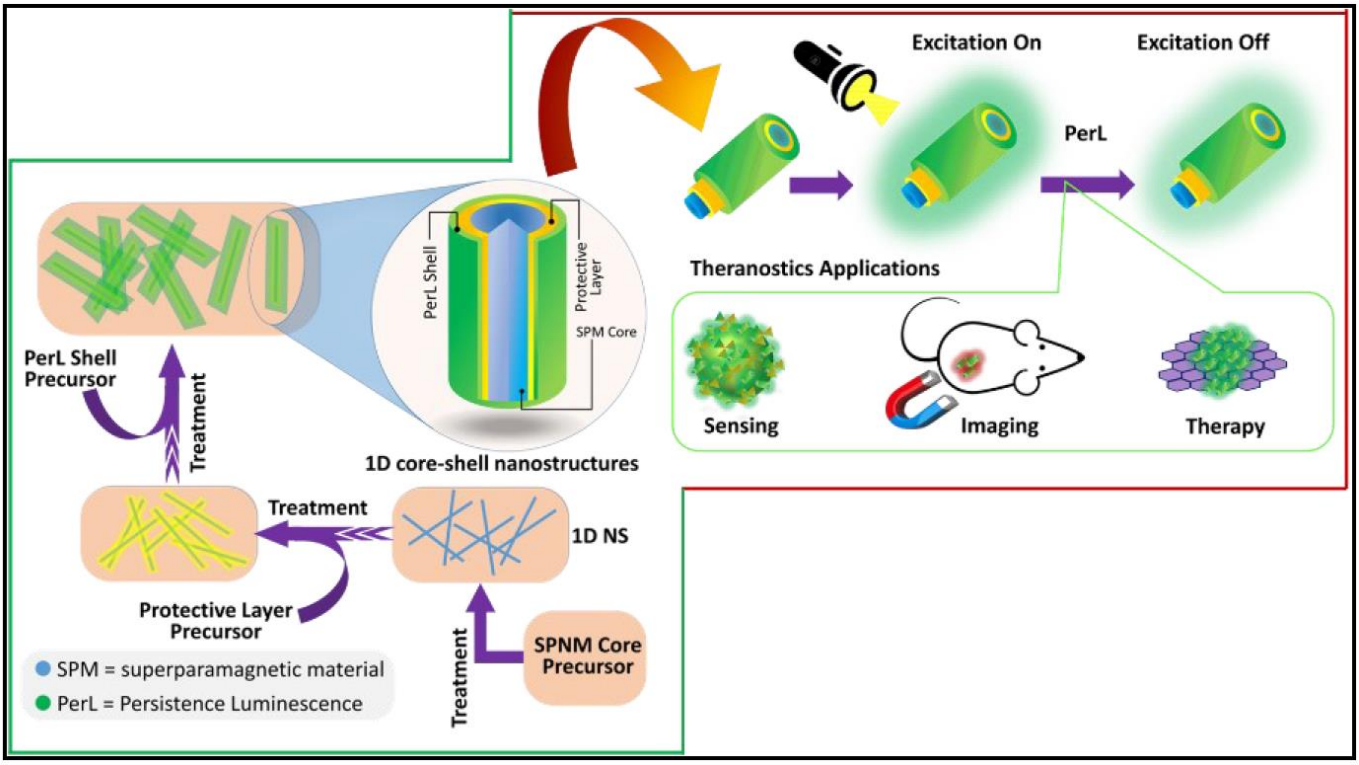
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Subject Category: Material Science

Abstract: Like size, shape of nanostructures has a profound impact over their functional properties. In context of the single assembly made up of magnetic and persistence luminescence (PersL) materials at a time, the importance of shape as property influencer further pronounces since magnetic and PersL improve with anisotropy of their particle shape. However, anisotropical single assembly of magnetic and PersL materials have not been explored so far. Therefore, we focused upon design and development of 1D single assembly having magnetic and PersL properties at a time. It is pertinent to mention here that such assemblies are of particular interest for bio-medical applications like bio-imaging, bio-tracking, and image-guided therapy. 1D single assembly of magnetic and PersL materials were constructed in a core-shell fashion by a three-step protocol. The magnetic and PersL materials were coupled in such a way that the former constituted the core while the latter formed shell of the resulting 1D single assembly. An interface layer made up of silica (SiO2) was incorporated in between the core and shell. The purpose of the passivating layer was to hinder quenching of emission radiations of PersL shell by the magnetic core. The resulting materials were characterized by powder XRD, FE-SEM, TEM, HR-TEM, AFM, XPS, Raman/FTIR spectroscopy, TG, DSC etc., and formation of the desired multi-shell nanoarchitectures, MxOy@SiO2@PersL (M=Fe or Mn, x=1 or 3, y=1 or 4, PersL=SrAl2O4:Eu2+, Dy3+) were confirmed. Magnetic and luminescence properties were investigated of the developed core-shell nanoarchitectures with the aim to establish their shape-property relationships. It has been found that the materials are paramagnetic (manganese oxide core) and superparamagnetic (iron oxide core). The magnetization varied clearly with shape and size of the nanostructures. Furthermore, the luminescence properties like emission wavelength (l) and delay time were impacted by the shape as well as size of the resulting nanostructures. We anticipate that the successfully developed 1D magnetic and PersL single assemblies will find bio-medical applications with improved performances. Therefore, our future prospects of the developed nanomaterials are to establish their effectiveness in theranostics.

Graphical Abstract:



Tuning Nano-structure and Gas Permeation Properties of Silica Membranes by Alkylsilane

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Subject Category: Materials Science

Abstract: Vinyl-functionalized sol-gel derived nano-structured hydrogen-selective hydrophobic silica membranes are deposited on a porous α–Al2O3 tubular support with γ–Al2O3 as an intermediate layer. The influence of the vinyltriethoxysilane (VTES) contents on the properties of silica membrane precursor such as microstructure, hydrophobicity property and membrane performance are examined. The membrane system exhibits proper three layers' adhesion or integration, and smoothness confirms by the scanning electron microscopy (SEM). The thickness of the membrane is about 782.6 nm as depicted by FE-SEM image. Fourier transform infrared spectroscopy (FTIR) confirms incorporation of highly hydrophobic groups in the modified membrane. The contact angle of water droplet increases from 36o for the original to 97o for the modified ones with 0.4 moles of VTES. Average pore size of unsupported original membrane is 1.98 nm, while for the unsupported modified silica material at pore size of 2.23 nm, respectively. The membrane derived from the mixture of tetraethyl orthosilicate (TEOS) and 0.4 VTES exhibits the highest H2 permeance with 41.18 × 10-7 mol m−2 s−1 Pa−1 at 250 oC, and ideal selectivity of H2 over CO2 is 14.54. The modified membrane is nine times more hydrophobic which makes it more desirable for the hydrogen separation from H2/CO2 mixture under humid environment for high-temperature separation.

Keywords: silica membrane, carbon dioxide, hydrogen, hydrophobic properties, sol–gel method, gas separation.

Novel Strategy for the Transformation of Graphene oxide paper into reduced Graphene oxide paper

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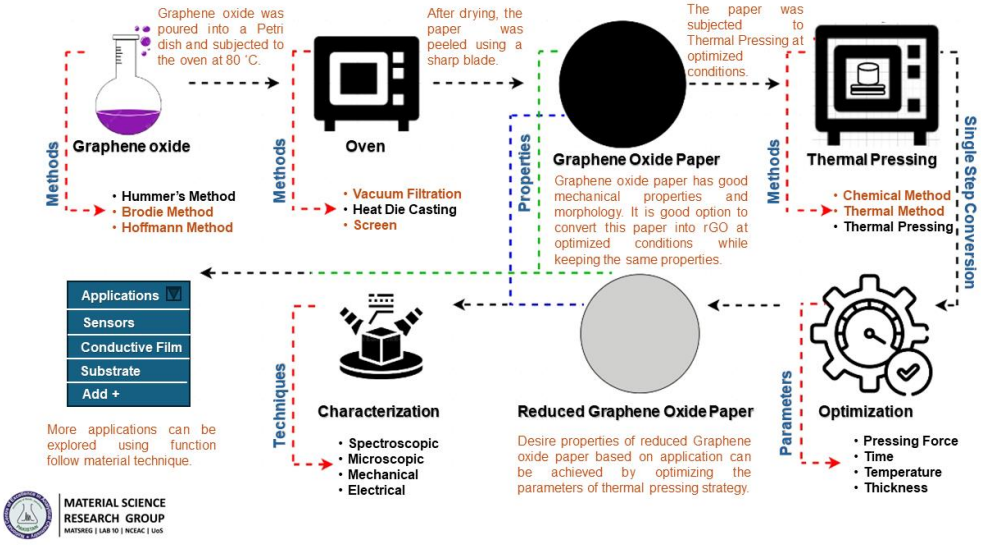
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Subject Category: Material Science

Abstract: There are various methods to synthesize reduced graphene oxide from graphene oxide but, the most frequent are chemical and thermal methods. These methods required additional steps and generated waste products. In this work, graphene oxide was fabricated using modified Hummer's method and shaped into paper using thermal die-casting method. Later, the GO paper was converted into rGO paper using a novel strategy named thermal pressing. The fabricated paper has been characterized by several spectroscopic, microscopic and mechanical techniques. The carbon and oxygen content in GO paper was 50% and 49% which was tailored to 67% and 32% after thermal reduction. The ATR spectra showed functional groups in GO paper while these peaks were in rGO paper confirming the transformation of GO into rGO. The SEM showed that the morphology of GO paper was so smooth at um scale while the rGO paper showed some wrinkles because of less pressing force. However, the mechanical strength of GO paper was 0.061 KN/m but rGO paper was not strong enough as compared to GO. The study presented enough evidence to support this novel reduction strategy for the conversion of GO into rGO paper while keeping the smoothness of the surface. This strategy is easily implemented on a commercial and industrial scale for the manufacturing of rGO paper.

Graphical Abstract:



Rational design and Synthesis of Energy Materials for Next-Generation Photovoltaic Solar Cells

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Subject Category: Materials Science

Abstract: Organic Solar Cells (OSCs) based on electron donor and electron acceptor building blocks are amongst the most promising renewable energy conversion solutions. Combining the features of low-cost and flexibility, OSCs are idealized for portable electronic devices, indoor light-harvesting applications and building-integrated photovoltaics. Recent breakthroughs in material design have resolved many of these issues by imparting a greater choice of synthetic protocols and precise optimization of optical and electronic properties,1, 2 thereby driving power conversion efficiency (PCE) over 20% for single junction solar cells with bulk heterojunction architecture.3, 4 Fused-ring non-fullerene electron acceptors have frequently exhibited champion PCEs owing to a well-connected π-conjugated framework which enables higher electron charge transfer and electron mobilities.5, 6 Herein, we will discuss new non-fullerene acceptors containing highly conjugated cores based on indacenodithiophene (IDT) and benzotrithienopyrrole derivatives. The nitrogen-rich groups impart both electron donating and electron withdrawing features into the molecules resulting in interesting balance of charges across the framework. Computational studies of the molecules reveal the possibility of band gap engineering by the introduction of halogenated substitutes on the periphery, thus allowing structure tuneability for optimum photovoltaic performance. The molecules are synthesized from low-cost precursors using mild reaction conditions and scalable reaction methodologies. Further, ongoing photophysical and electrochemical studies are used to elucidate optoelectronic properties and device performance.

Greener Formation of Glycine Max-Derived Nanoparticles For Evaluating Their Anti Proliferative And Apoptotic Impacts In Hepatocellular Carcinoma (Hcc)

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Subject Category: Material Science

Abstract: One of the biggest burdens in the field of oncology is hepatocellular carcinoma, which has a major detrimental impact on world health due to its rising incidence and death rates. Glycine max is a plant that has been used for ages in many forms of traditional medicine and has demonstrated promising anti-cancer properties. The current work aims to synthesize Glycine max-derived nanoparticles in an environmentally friendly manner and examine their anti-proliferative and apoptotic effects in epatocellular cancer. In this work, a silver nitrate water solution and the extract from Glycine max pods were used to create nanoparticles. FT-IR, SEM-EDX, X-ray diffraction, size distribution analysis, zeta potential determination, and UV-Vis spectroscopy were used to characterize the synthesized nanoparticles. Then, utilizing the HepG2 cell line, the in-vitro investigation was carried out using the trypan blue assay, crystal violet assay, enzyme linked immunosorbent assay (ELISA), and LDH. After 48 hours, all groups were assessed to determine the effectiveness of the Glycine max-derived nanoparticles. Using a phase contrast microscope, the morphology of the cells was examined. The findings showed that while the HepG2 cells in the treated groups showed morphological changes linked to cell death, such as decreased confluency, the untreated group's cells retained a fibroblastic shape. The Trypan blue assay indicated a rise in cell death in HepG2 cells after treatment. The treated malignant groups exhibited decreased viability, proliferation, and angiogenesis, as evidenced by crystal violet staining, MTT assay, and ELISA VEFG. However, at a dose of 30 µg/mL, GM-NPs demonstrated normal proliferation. This demonstrated that the treatment specifically targeted cancer cells by increasing apoptosis and decreasing proliferation, with no adverse effects observed. Lastly, in order to evaluate their anticancer efficacy against hepatocellular carcinoma, clinical human trials are also necessary.

Key Words: Hepatocellular Carcinoma, Bioinspired, nanoparticles, Glycine *max*

Chiral Recognition of Cysteine, Methionine, and Homocysteine Using Pillar[5]arene Macrocycle

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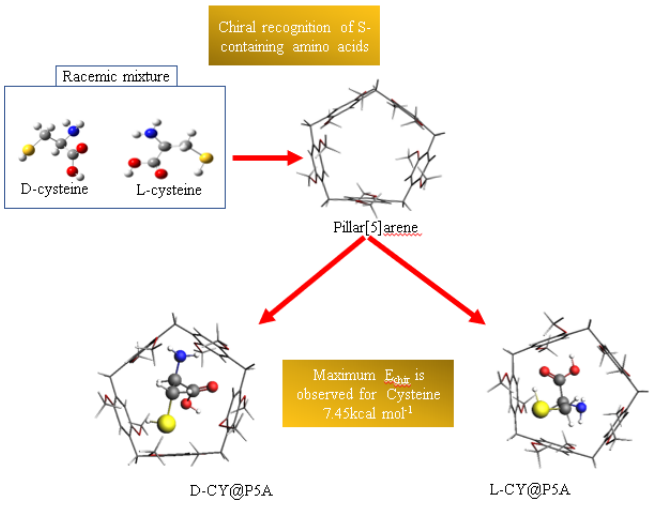
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Subject Category: Physical Chemistry and Material Science

Abstract: Enantiomers behave differently in chiral media due to their distinct spatial arrangements, making chiral recognition crucial for enantioselective separations. In this study, we used the chiral P5A macrocycle to recognize enantiomers of sulfur-containing amino acids, including cysteine (CY), methionine (MT), and homocysteine (HCY), through density functional theory (DFT) calculations. We used geometry optimization, thermodynamic stability analysis, noncovalent interaction analysis, and electron decomposition analysis (EDA) to investigate the chiral behavior of the complexes formed by P5A with the *L*- and *D*-isomers of the chosen amino acids. The interactive conformations of the complexes suggest that the amino acids are physiosorbed within the central cavity of the macrocycle. The most significant chiral differentiation occurs between *L*- and *D*-cysteine, with an energy disparity of 7.45 kcal/mol. The results indicate that the *L*-CY@P5A shows higher stability compared to its *D*-CY counterpart. In contrast, *D*-HCY and *D*-MT show higher stability than their *L*-isomer counterparts, probably due to the flexibility of these isomers. NCI analysis shows homocysteine complexes have the strongest attractive forces (electrostatic, dispersion) with minimal repulsive exchange, followed by cysteine and methionine complexes. The results indicate that the P5A macrocycle effectively differentiates between *L*- and *D*-amino acids, especially those with rigid and small structures.

Graphical Abstract:



Study of Iron Coated Soft Magnetic Composites (SMCs) for Applications in Transformer Core

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Tariq[a], Rub Nawaz Shahid and Hassan Waqasb

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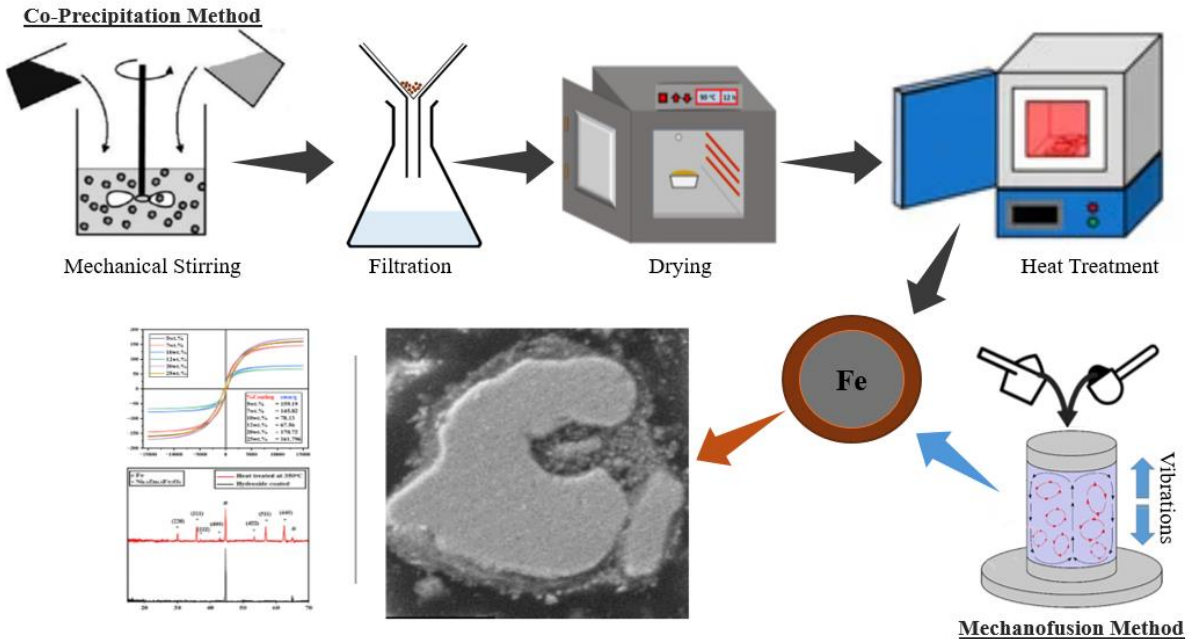
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Subject Category: Material Science

Abstract: This study investigates the synthesis of Ni-Zn ferrite coated iron cores for transformers operating in high frequencies. The research emphasis on achieving low coercivity, high magnetic saturation and minimal eddy current losses. Fe particles coated with $Ni_{0.5}Zn_{0.5}Fe_2O_4$ ferrites are prepared through coprecipitation and mechanofusion method. Various weight percentages for coating material are studied to optimized best coated material with superior magnetic properties. The structural properties and phases were identified using X-ray diffraction (XRD), scanning electron microscopy (SEM) and energy dispersive x-ray spectroscopy (EDS), to determine the morphology and particle size of synthesized powders. The magnetic properties were checked by vibrating sample magnetometry (VSM). VSM results showed that the magnetic saturation density of Fe particles coated through the co-precipitation route is 159 emu/gm for 5wt.% ferrite coating and 170 emu/gm for 20wt.% ferrite coating, synthesized through the mechanofusion method. Results confirmed the formation of a thin layer of ferrite on Fe particles through both techniques and showing superior magnetic properties while maintaining the stability across high frequency range.

Graphical Abstract: (Optional)



Improved hydrophilicity and antifouling performance of Silver embedded polyelectrolyte multilayers (Ag-PEM) based membranes

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Subject Category: Material Science

Abstract: The study demonstrates the improved hydrophilicity and antifouling performance of Silver embedded polyelectrolyte multilayers (Ag-PEM) based membranes. The process of membrane separation is the most efficient but suffers fouling problems for this reason polyelectrolyte multilayers based thin film composite (TFC) asymmetric membranes were prepared which are enhanced by silver nanoparticles for antifouling properties. Silver (Ag) nanoparticles provide antifouling and antimicrobial properties to the PEM membranes. Using the phase inversion process, the ultrafiltration (UF) support is prepared transforming the polymer solution from liquid to solid phase, using Polyether sulfone (PES) polymer. The PES support was coated with strong polyelectrolytes (PAH/PSS) which was followed by crosslinking to boost retention without compromising permeability. The membrane was then coated Ag-Np and immobilized to form stable Ag-Np. Performance evaluation of the membranes was carried out in terms of permeability, dye rejection and fouling tests with BSA and E.coli. The results shows us that enhancing the PEM membrane with Ag-Np decreases the permeability and increases the dye rejection up to 70% due to dense membrane structure. The fouling tests also indicates significant resistance against the foulant and complete killing of the bacteria on the Ag-Np membranes. The characterization of the membranes were done through FTIR, SEM and contact angle measurements.

Evaluation Haematological Parameters With Liver Enzymes And Renal Dysfunction Markers In Beta-Thalassemia Major Patients

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Subject Category: Analytical Chemistry

Abstract: Thalassemia is one of the most common genetic diseases, and beta-thalassemia major is its severe form. The present study deals with the analysis of liver function, estimation of important serum ions and hematological characteristics in beta-thalassemia patients and controls. The study included 54 patients with beta-thalassemia major each matched with 54 healthy individuals of corresponding sex and age. The levels of alanine transaminase (ALT), alkaline phosphatase (ALP) and aspartate transaminase (AST) were assessed in order to evaluate the liver function. Hemoglobin (Hb), ferritin, hematocrit (HCT), mean corpuscular volume (MCV), mean corpuscular hemoglobin (MCH), mean corpuscular hemoglobin concentration (MCHC), creatinine and uric acid level were also measured. Importantly, beta-thalassemia patients had higher ALT ($P < 0.001$), AST ($P < 0.05$), and ALP ($P < 0.001$) levels and lower creatinine levels ($P < 0.001$) compared to healthy controls. Higher iron levels ($P > 0.05$) were found in patients compared to healthy individuals. Hematological parameters such as Hb ($P < 0.001$), ferritin ($P < 0.05$), HCT ($P < 0.001$), MCV ($P < 0.05$), MCH ($P < 0.05$), except MCHC ($P > 0.05$) significantly measured in patients. A great difference was observed between the patients and the control group for uric acid ($P < 0.05$). Our study showed differences in liver function, renal function, serum iron content, and hematological characteristics between beta-thalassemia patients and controls.

Efficacious electrocatalysis of oxygen evolution reaction by hydrothermally fabricated nanoflowers-like Cr-MOF/g-C_3N_4

*Muzamal Javed[a]; Ayesha Saeed[a]; Zahoor Ahmad[a],**

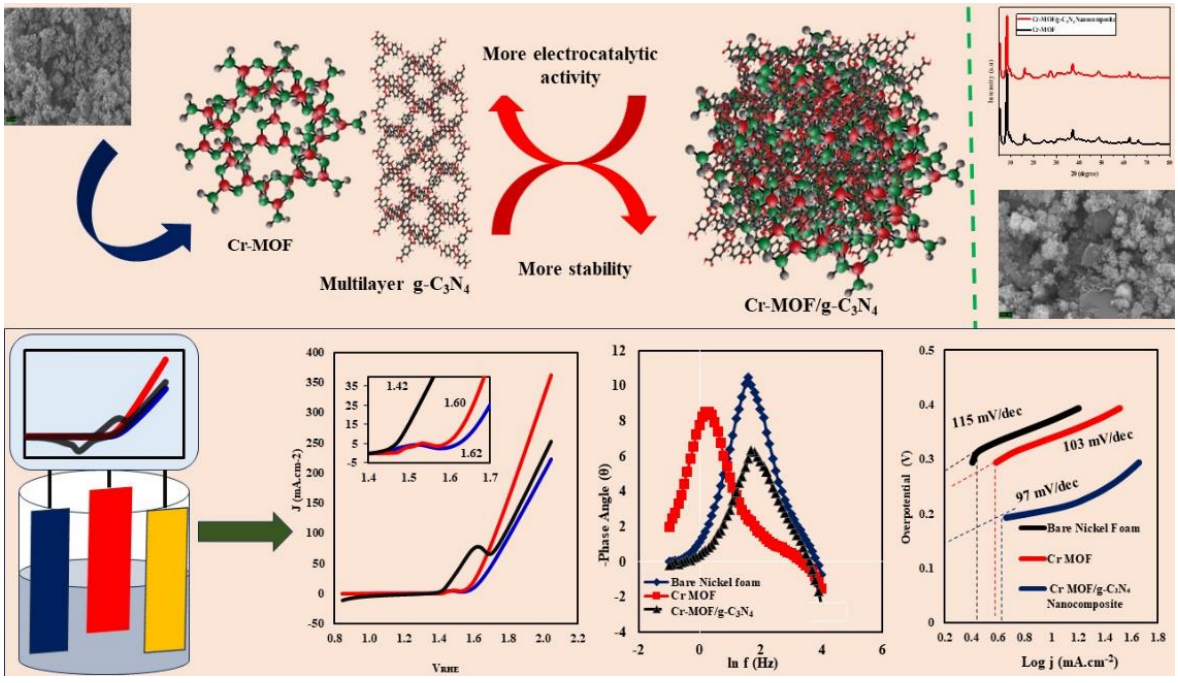
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Subject Category: Physical Chemistry

Abstract: Green hydrogen fuel, facilitated through water electrolysis, is a viable and ecofriendly alternative for the world's augmenting energy demands. The present study employed a facile hydrothermal method for Cr-MOF and Cr-MOF/g-C_3N_4 synthesis. Electrocatalytic performance and stability of electrocatalysts were scrutinized by utilizing cyclic voltammetry (CV), linear sweep voltammetry (LSV), electrochemical impedance spectroscopic (EIS) techniques and chronoamperometric potential cycling analysis (CPE). The nanocomposite exhibited remarkably lower onset potential of 1.4V (vs RHE), higher peak current with substantial exchange current density of 4.5mAcm^{-2}, reduced overpotential of 213mV and 97mV/dec Tafel slope value. Furthermore, analysis of critical metrics including double-layer capacitance (C_{dl}), electrochemically active surface area (ECSA), and impedance revealed an enhanced electroactive area in the Cr-MOF/g-C3N4 nanocomposite, evidenced by its increased double-layer capacitance (4.1x10^{-3} mF cm^{-2}) and ECSA (1.025 cm^2) and lessened value of R_{ct} (0.4059 Ω). However, reduction in surface roughness factor implies underlying structural alterations within the composite. CPE under optimal experimental conditions demonstrated significant long-term stability for both electrocatalysts.

Graphical Abstract



Deep eutectic solvents based green methods for extraction and analysis of Safranin O dye in aqueous samples using central composite design

Azmat Wali[a], Muhammad Naveed[b], Tanvir Khan[b]

[a]University of Malakand, Chakdara Dir Lower Khyber Pakhtunkhwa Pakistan

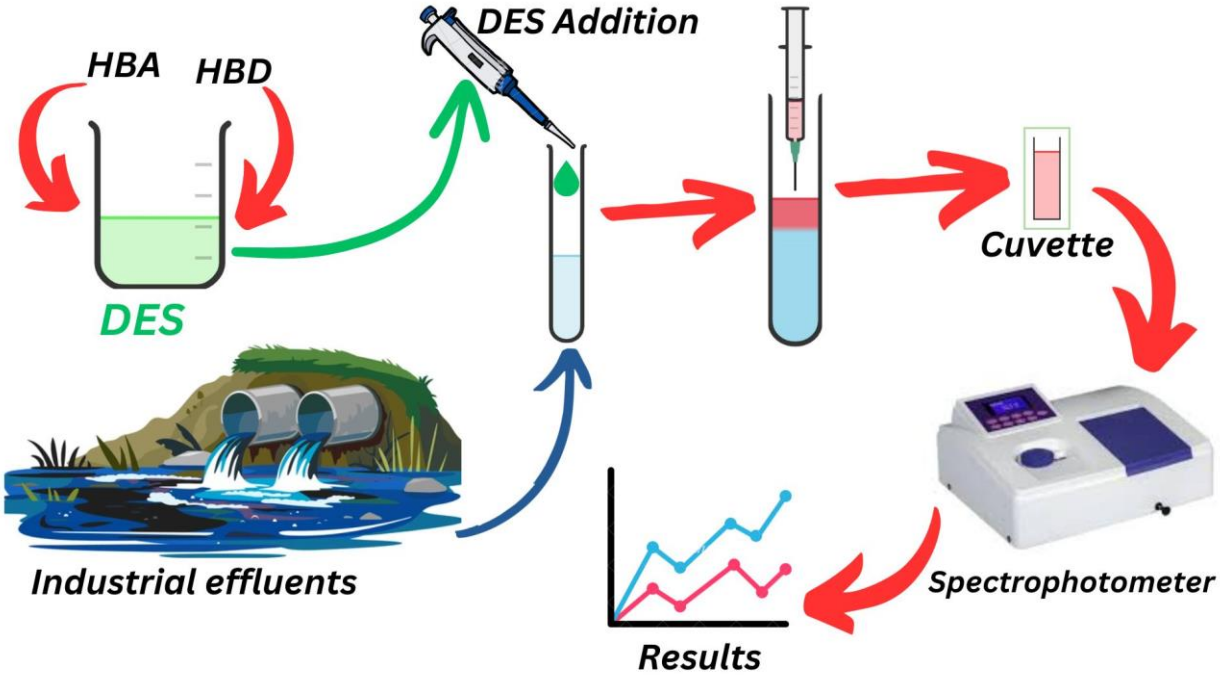
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Subject Category: Analytical Chemistry

Abstract: This study looks into the use of deep eutectic solvents (DES) to extract Safranin O dye from an aqueous media. Because of its toxicity and longevity in water bodies, safranin O, a cationic dye frequently employed in biological staining, presents a risk to the environment. The study focuses on using DES as an effective and environmentally friendly solvent for color removal. As DESs are made up of HBDs and HBAs, therefore it is biodegradable, less volatile solvent that can be adjusted, making it a sustainable substitute for traditional organic solvents. DES containing thymol (HBA) and ethylene glycol (HBD) in the molar ratio (1:1) were used, because of its highest percent recovery. Important variables were methodically examined, such as pH, extraction duration, the temperature, and molar ratio of the constituents of DESs. The results showed that Safranin O could be efficiently extracted from aqueous solutions using DES, with extraction efficiencies higher than the already existed methods. Significant hydrogen bonding and electrostatic interactions were revealed when the interaction mechanisms between DES and Safranin O were clarified through the use of spectroscopic and computational approaches. Limit of detection (LOD), and limit of quantification (LOQ) were 2.5 and 4.5 µg/L respectively with relative standard deviation RSD 3.2%. The results highlight DES's potential as an environmentally friendly dye removal solvent and how it might advance sustainable water treatment systems. This research study addresses a crucial issue in environmental water samples monitoring by presenting a novel and environmentally friendly method of extracting and analyzing dyes.

Graphical Abstract:


HBA
HBD
DES
DES Addition
Cuvette
Industrial effluents
Spectrophotometer
Results

A Novel Liquid-liquid Micro-extraction Based Method for the Extraction and Determination of Crystal violet from Aqueous Medium by Using DES and UV Visible Spectrophotometer Analysis

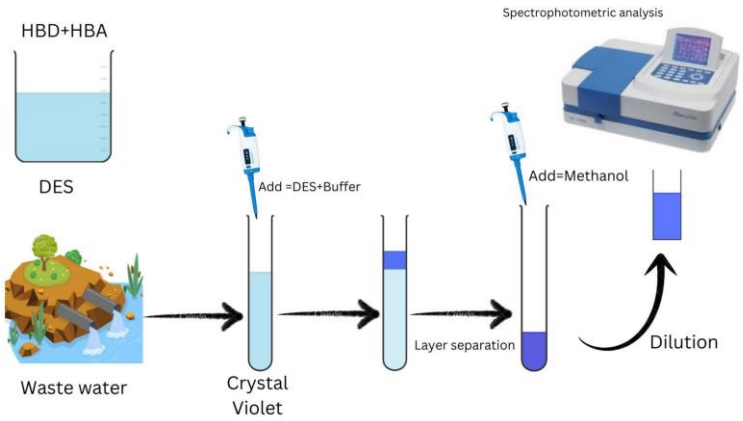
Muhammad Dawood, Azmat Wali, Salman Khan*
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Subject Category: Analytical Chemistry

Abstract: The study compares the efficiency of deep eutectic solvent as a green solvent to that of other conventional solvents, emphasising the advantages of its low toxicity, biodegradability, and tunability.In the current study, Dye crystal violet has been extracted from aqueous media utilising a deep eutectic solvent (DES) that contains thymol and decanoic acid in a 1:1 molar ratio. Dye crystal violet is a well-liked cationic dye in the textile sector, renowned for its vivid colour and water solubility.However, the environment and public health are seriously threatened by its presence in wastewater due to its potential toxicity and resistance to biodegradation. Optimising the extraction process involved looking at variables such as pH, temperature, DES volume, centrifugation, and extraction time.As a consequence, the DES of thymol and decanoic acid demonstrates a strong affinity for Dye crystal violet , achieving noteworthy extraction efficiency under ideal circumstances. It is hypothesised that the DES component and dye molecule will interact electrostatically and through hydrogen bonds as the extraction method.It was shown that the extraction process's limits of quantification (LOQ) and detection (LOQ) are 2.5 mg/L and 3.2 mg/L, respectively.This work examines how deep DES can be used for dye removal applications, offering a sustainable wastewater treatment option and advancing green chemistry. This study not only illustrates how a particular dye is extracted, but it also shows how synthetic dye affects the environment. Additionally, this research offers a platform for developing efficient, environmentally friendly extraction techniques for various industrial pollutants.

Graphical Abstract:



Design, Synthesis and In-silico Evaluation of New Coumarin Derivatives

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Subject Category: Organic Chemistry / Pharmaceutical Chemistry

Abstract: Alzheimer's disease (AD) is the most common cause of dementia all over the world, ranked as sixth leading cause of death in developed countries. Various drug targets are available for the treatment of AD, however, the available therapies target cholinesterase and N-methyl-D-aspartate. In this study, coumarin nucleus was combined with other heterocyclic moieties to produce compounds with potential neuroprotective potential. The designed compounds were evaluated through molecular docking and MM-GBSA calculations to predict binding potential with various AD targets including acetylcholinesterase, butyrylcholinestrase, β-site amyloid precursor protein cleaving enzyme along with some inflammatory markers; TNF-α and NFκB. Total of 47 compounds were designed. D-10 showed highest binding affinity against various targets ranging from -6.404 Kcal/mol to -10.268 Kcal/mol. It was further subjected to molecular dynamics simulation studies. Physicochemical properties were also evaluated. These compounds can serve as leads for synthesizing further derivatives with multi-target neuroprotective potential. Synthesis of these compounds is the second phase of this study.

Designing of Small Conjugated Organic Materials for Highly Efficient Blue Organic Light Emitting Diods

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Subject Category: Organic Chemistry

Abstract: Flexible, lightweight, and cost-effective organic light-emitting diodes hold significant potential for applications in displays and lighting. However, achieving highly efficient and stable deep-blue OLEDs remains challenging due to the scarcity of effective functional materials. Consequently, the creation of stable blue emitters is a critical focus in OLED development. Fluorene compounds have been extensively studied for blue-emitting materials due to their excellent light-emitting properties. The continuous demand for new small molecules that can serve as efficient red, green, or blue emitters is a crucial factor in the development of full-color OLEDs for displays and lighting applications. Low molecular weight compounds are comparatively easier to synthesize, isolate, purify, and spectroscopically characterize. These advantages make small molecules with intriguing photophysical properties highly appealing to researchers. A double bond conjugation strategy is proposed to design high-efficiency blue-emitting small molecules by modulating the conjugation degree with different spacers. In this study, a new rigid chromophore will be created through the Heck reaction. Examining variations in the core structure will be essential for understanding their effects on optical, thermal, and electronic properties. The resulting compounds are expected to be promising candidates for use in blue OLEDs as materials for charge transport and emission.

Exploration and Optimization of Hydrothermally Synthesized Ni-Co Oxides NPs as Superb OER Electrocatalysts

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Subject Category: Analytical Chemistry

Abstract: As a fuel, green hydrogen shows zero carbon emission and an elevated calorific value, making it an encouraging replacement for fossil fuels in meeting the world's energy needs. Herein, single metallic oxide (NiO) and Bi-mixed metallic oxide ($Ni_{0.8}Co_{0.2}O$ and $Ni_{0.6}Co_{0.4}O$) nanoparticles are synthesized using a facile hydrothermal approach from low-priced and abundant raw materials. These nanoparticles (NiO, $Ni_{0.8}Co_{0.2}O$, and $Ni_{0.6}Co_{0.4}O$) exhibit low onset potentials (1.40, 1.36, and 1.38 V_{RHE}) and overpotentials of 460, 400, and 390 mV at decade-current density, with high TOF values and long-time stability during constant potential electrolysis. These outcomes indicate that these nanoparticles are excellent electrocatalysts for OER and outperform pioneering electrocatalysts (RuO2 and IrO2) and other reported electrocatalytic materials such as $NiFe_2O_4$ and $CoFe_2O_4$. Furthermore, high current densities, reduced Tafel slopes, large exchange current densities, and big mass activities suggest that the Bi-mixed metal oxide nanoparticles are better OER electrocatalysts compared to single metal oxide (NiO) nanoparticles. This is achieved by tuning the concentration of Co in these nanoparticles, resulting in the maximum active sites, the highest redox potential, and lower charge transfer resistance. Due to their cost-effectiveness, simple synthesis, and exceptional electrocatalytic performance, these nanoparticles could be utilized in future solar water-splitting devices.

A novel liquid-liquid micro-extraction based method for the extraction and determination of vardenafil from aqueous medium by using DES and HPLC analysis

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Subject Category: Materials Science

Abstract: This work presents a liquid-liquid micro-extraction-based method for the extraction of vardenafil from aqueous media by using Deep Eutectic Solvents (DES), prior to the HPLC analysis. As DES is a green, cheap, easily synthesizable, and biodegradable solvent and therefore was considered as base for the developed method. The DES was synthesized by the combination of hydrogen bond acceptor (choline chloride) and acceptor (malonic acid) in 1:2 molar ratios. Verdenafil showed good interaction with the proposed DES as compared to water and therefore moved to the DES Phase from the aqueous phase. The quantitative analysis of the method was performed by using UV-visible spectrophotometry. The experimental conditions including DES volume, THF volume, pH, sonication and centrifugation time were optimized. The method is highly acceptable in terms of percent recovery (101-102%), LOD (2μg/mL), and percent RSD (1.29 to 1.48%) and therefore shows dormancy compared to other developed methods. The method was used for the analysis of vardenafil in real water samples and food supplements.

Synthesis and characterization of a new Archimedean rhombicuboctahedral polyoxometalate Rb2.5 [(Co(H2O)6)0.5(PW12O40)].16H2O: Probing magnetic characteristics and electrochemical water splitting catalysis

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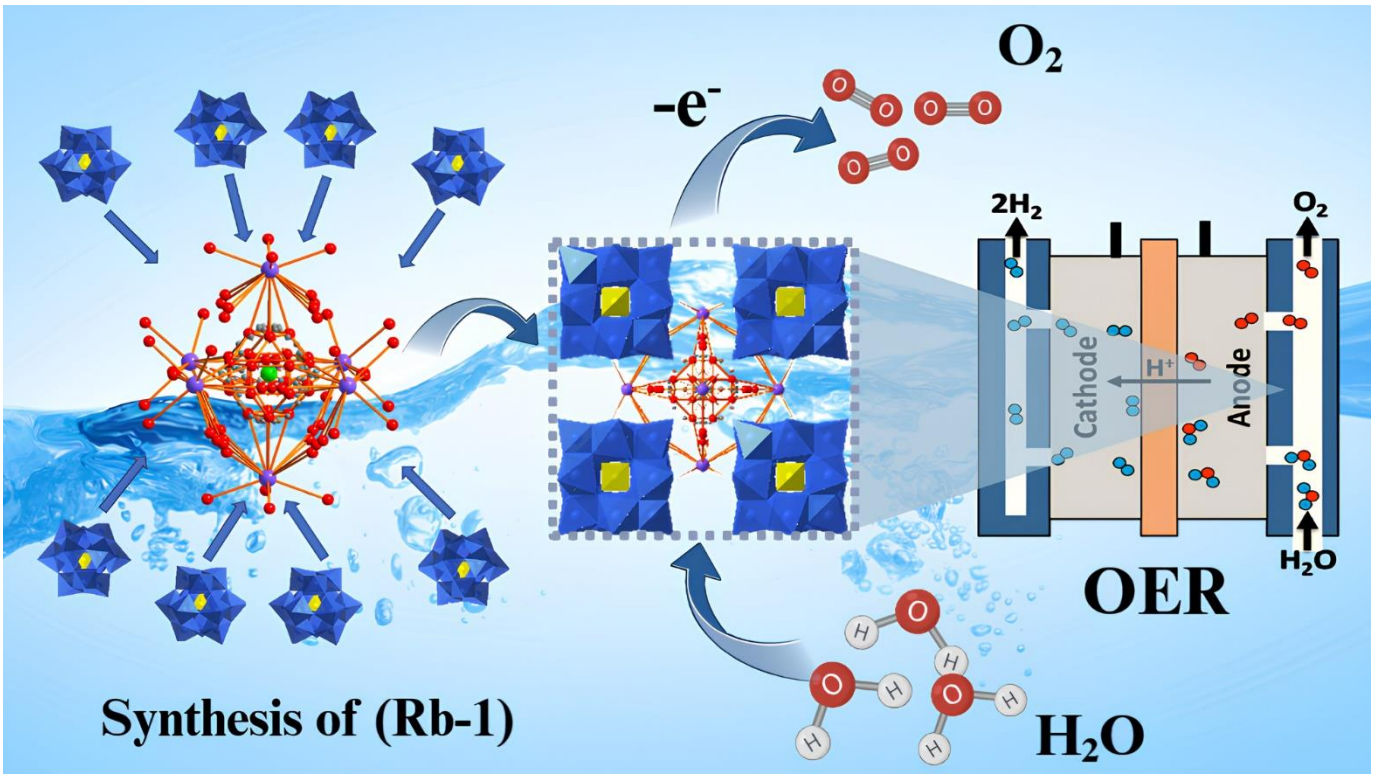
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Subject Category: Inorganic Chemistry / Material Science

Abstract: This study describes the synthesis, structure characterization, and magnetic/electrochemical properties of a new Rubidium salt of cobalt-based polyoxometalate (POM). A complex $Rb_{2.5}$ $[(Co(H_2O)_6)_{0.5}(PW_{12}O_{40})].16H_2O$ (Rb-1) was successfully synthesized in a single-pot reaction. The synthesis was done using pre-synthesized $[Co_2(\mu\text{-}OH_2)(O_2CCMe_3)_4(HO_2CCMe_3)_4]$ and tri-lacunary Keggin anion Na_9 $[A\text{-}\alpha\text{-}PW_9O_{34}].7H_2O$. The structure of Rb-1 was verified through Fourier Transform Infrared Spectroscopy (FTIR), Energy-Dispersive X-ray (EDX), and single-crystal X-ray crystallography. The crystal structure revealed a cubic crystal system (a, b, c = 22.5801(8) Å and α, β, γ = 90°) with a central Co (II) surrounded by six water molecules. The magnetic properties of Rb-1 were investigated via a superconducting quantum interference device (SQUID), which revealed antiferromagnetic behavior in Rb-1. Moreover, Rb-1 deposited on a Ni-foam electrode with a geometrical area of 1 × 1 cm^2 functioned as an anode for efficient oxygen evolution reaction (OER) in water splitting. The Rb-1/Ni-Foam electrode showed excellent overpotential values of 207 mV at 10 $mA.cm^{-2}$ and 390 mV at 100 $mA.cm^{-2}$ current densities. The electrode had fast kinetics, as indicated by a low Tafel Slope of 51.76 $mVdec^{-1}$ and a small charge transfer resistance (RCT) of 5.904 ohms. It exhibited remarkable stability over an 8 h test period, emphasizing its practical value.

Graphical Abstract:



Study of enhanced structural and photocatalytic properties of $Mn_2Fe_2O_4$ by Zn^{2+} substitution

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Subject Category: Analytical Chemistry

Abstract: The hydrothermal process was used to synthesize zinc-doped manganese ferrite ($Mn_xZn_{1-x}Fe_2O_4$ where x = 0.0, 0.1, 0.2, 0.3, 0.4, 0.5). This was done to study the impact of zinc on the photocatalytic activity of manganese ferrite. The photocatalytic activity of the synthesized ferrite was assessed by examining the degradation of Methylene Blue dye. FTIR, SEM, UV-Vis, XRD spectroscopic techniques have been used to analyze the manganese ferrites and zinc-encapsulated Mn ferrites produced hydrothermally as dye-degrading agents. X-ray diffraction confirmed the formation of a cubic hexagonal spinel structure. The FTIR analysis, conducted in the range of 500 to 4000 cm^{-1}, revealed absorption bands for the Fe-O bond at 442 cm^{-1}. The particle size, determined through X-ray studies, ranged from 19 to 34 nm. The photocatalytic activity of Zn-doped Mn ferrite nanoparticles was compared, and the photocatalytic activity of the nano-ferrite samples was assessed by exposing a solution of methylene blue organic dye to ultraviolet light with an intensity of 500W.The extensive use of colors in the textile industry causes 15% or so of unfixed dyes to contaminate water systems as effluent, which presents serious health risks. These dyes are resistant to the mechanisms that cause natural degradation because of their complex molecular structures. But the Advanced Oxidation Process, which uses the •OH hydroxyl radical—a reactive species produced from Fenton—offers a practical and effective substitute for traditional techniques in the degradation of organic contaminants. The experiments have shown that the efficiency of various Zn-doped Mn ferrites behaves irregularly when it comes to photodegradation. Under optimized conditions, it has been observed that 30mg of the photocatalyst demonstrates the highest efficiency in degrading MB at pH 2 within a time frame of 5 minutes. These nanoparticles have the potential to be used for treating water systems that are contaminated with organic dyes.. This work will pave the way for future advancements in environmental remediation.

Green and one Pot Synthesis of Biphenyl Sulfonamides Derivatives and their Antimicrobial, Antioxidant, and Docking Studies

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Subject Category: Organic Chemistry, Material Science

Abstract: The use of ultrasonic radiation for the synthesis of various biologically active compounds is a popular research area due to simple and affordable sonochemical method, which often results in amorphous material. The aim of this study was the synthesis of some biphenyl sulfonamide derivatives by using ultrasound irradiations (sound energy) to promote a greener methodology. The reaction of appropriate primary amines with biphenyl sulfonyl chloride gives seven new biphenyl sulfonamide derivatives in excellent yields. The structure of all these compounds were elucidated through various spectroscopic techniques such as IR, ^{1}H-NMR, EIMS, ^{13}C-NMR, and elemental analysis. All compounds showed significant antibacterial activity having higher to moderate MIC values. In the case of antifungal activity most of the compounds inactive against fungal strains. Among the antioxidant activity, all the compound possesses good activity, some possesses remarkably excellent antioxidant activity. All biphenyl sulfonamide derivatives were subjected to in silico studies against dihydropteroate synthase (DHPS) enzyme and human carbonic anhydrase II (hCA II) to predict binding mechanisms to find potential inhibitors to treat bacterial infection and glaucoma respectively.

Eco-Friendly PVA-Urea-based Plastic Bag Alternatives

Ayesha Wasim, Mahnoor Naeema and Mamoona Irum, Dr. Mudassir Iqbal,*

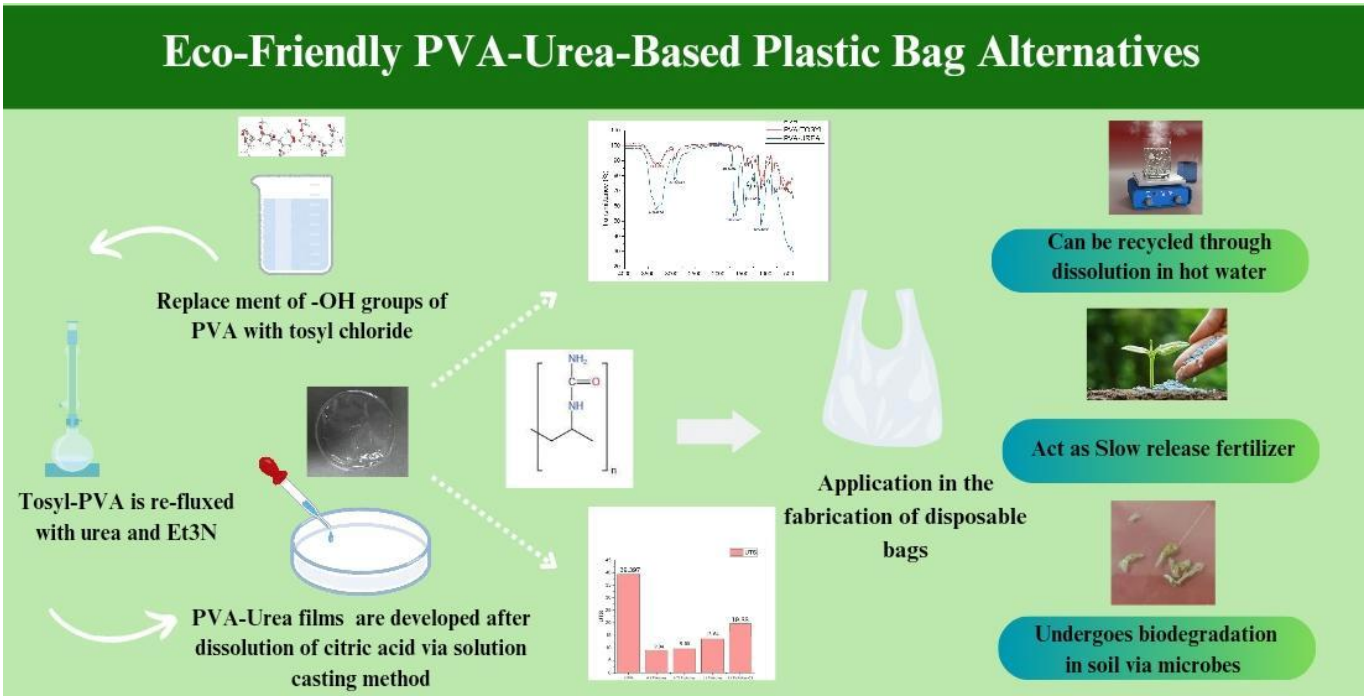
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Subject Category: Organic Chemistry/ Material Science

Abstract: The non-degradable nature of traditional petroleum plastics has posed serious threats to land and water ecosystems. They take many years to degrade and hence waste disposal is an issue. Therefore, the replacement of existing plastic with green biodegradable alternatives is the need of the hour. In our work, polyvinyl alcohol (PVA) is chemically functionalized with Urea and is fabricated into polymeric sheets. The films were synthesized in three distinct concentrations: a 1:1 ratio of PVA to urea, a 1:0.75 ratio, and a 1:0.5 ratio. Furthermore, PVA-Urea films have also been developed with and without the addition of plasticizers. FTIR has characterized the synthesized materials. Mechanical testing revealed that films 1:1 Poly Vinyl Alcohol-Urea with physical blending of citric acid exhibited the desired flexibility and strength for applications in plastic bags. Biodegradation studies were conducted to analyze the release of nitrogen content during the degradation process. The developed polymers have the attractive features of recyclability in a greener way i.e. dissolving in hot water and fabricating to the desired shapes and slow release of urea to the soil on biodegradation. The results of this research project demonstrate the successful synthesis of bio-degradable materials and highlight their potential for sustainable plastic bag application.

Graphical Abstract:



Asymmetric Nickel-Doped Chromium Molybdates As High-Performance Anode Material For Supercapacitors

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Subject Category: Physical Chemistry

Abstract: Nickel-doped Chromium Molybdates were synthesized by a convenient, cost-effective, and good-yield co-precipitation process for supercapacitor electrodes. Morphological and structural characterizations of materials were carried out by using scanning electron microscopy (SEM), energy dispersive spectroscopy (EDS), X-ray diffraction (XRD), and Raman spectroscopy. The electrochemical performance was evaluated by performing cyclic voltammetry, galvanostatic charge-discharge, and electrochemical impedance spectroscopy. Based on the synergistic effect of both Ni (II) and Cr (III) ions, specific capacitance increases with the doping of nickel until the parent lattice structure is maintained. The material with a nickel dopant concentration of 0.15 delivered the highest specific capacitance of 1572 F/g at 1 A/g among all other materials, good cyclic stability, and 100% coulombic efficiency. Moreover, the detailed kinetic analyses suggested that the dominance of the diffusion-controlled charge storage mechanism is responsible for the high specific capacitance of the materials. The present study demonstrated the traits of materials as potential anodes for designing high-performance asymmetric supercapacitors and also provided new insights for the development of high-performance materials for supercapacitors by using simple synthetic approaches.

Microwave assisted green methods for micro extraction and determination of selected dyes using green deep eutectic solvents

Muhammad Naveed[*]*, Azmat Wali , Tanvir Khan*

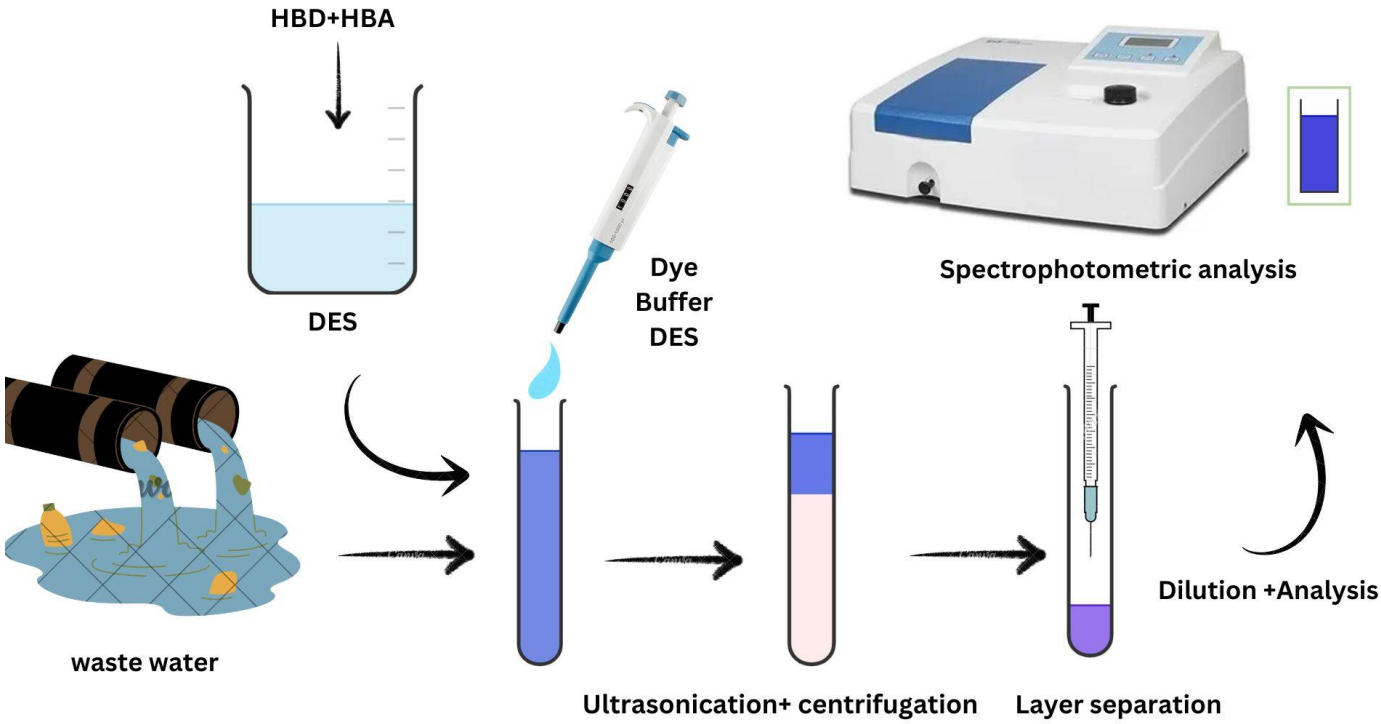
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Subject Category: Analytical Chemistry

Abstract: The study examines the effectiveness of DES as a green solvent compared to other traditional solvents, highlighting its benefits including low toxicity, biodegradability, and tunability. Present Research has been done on the extraction of Dye Blu 3 Basic from aqueous media using a deep eutectic solvent (DES) of choline chloride and ethylene glycol in a 1:1 molar ratio. A popular cationic dye in the textile industry, Dye Blu 3 Basic is known for its water solubility and vibrant colour. However, because of its potential toxicity and resistance to biodegradation, its presence in wastewater poses serious threats to the environment and public health. The extraction process was optimized by examining parameters like pH, temperature, DES volume, centrifugation, and extraction time. The result shows that the choline chloride and ethylene glycol DES exhibit high affinity for Dye Blue 3 basic, obtaining significant extraction efficiency in optimal conditions. The mechanism of extraction is postulated hydrogen bonding and electrostatic interaction between the DES component and dye molecule. it was observed that the limit of detection (LOQ) and limit of quantification (LOQ) for the extraction process are 2.5mg/L and 3.2mg/L respectively. This study reviews the potential of deep DES in dye removal applications, providing a long-term wastewater treatment solution and advancement in the field of green chemistry. This study not only highlights the extraction of specific dye but also indicates the environmental impact of synthetic dye. Furthermore, this work also provides a bench for creating effective, environmentally acceptable extraction methods for different industrial contaminants.

Graphical Abstract:


HBD+HBA
DES
Dye
Buffer
DES
Spectrophotometric analysis
waste water
Ultrasonication+ centrifugation
Layer separation
Dilution +Analysis

Synthesis, Characterization, pH Sensitivity, Solvatochromic Features, DFT Study, Biological Activity and Application of Monoazo Heterocyclic Disperse Dyes

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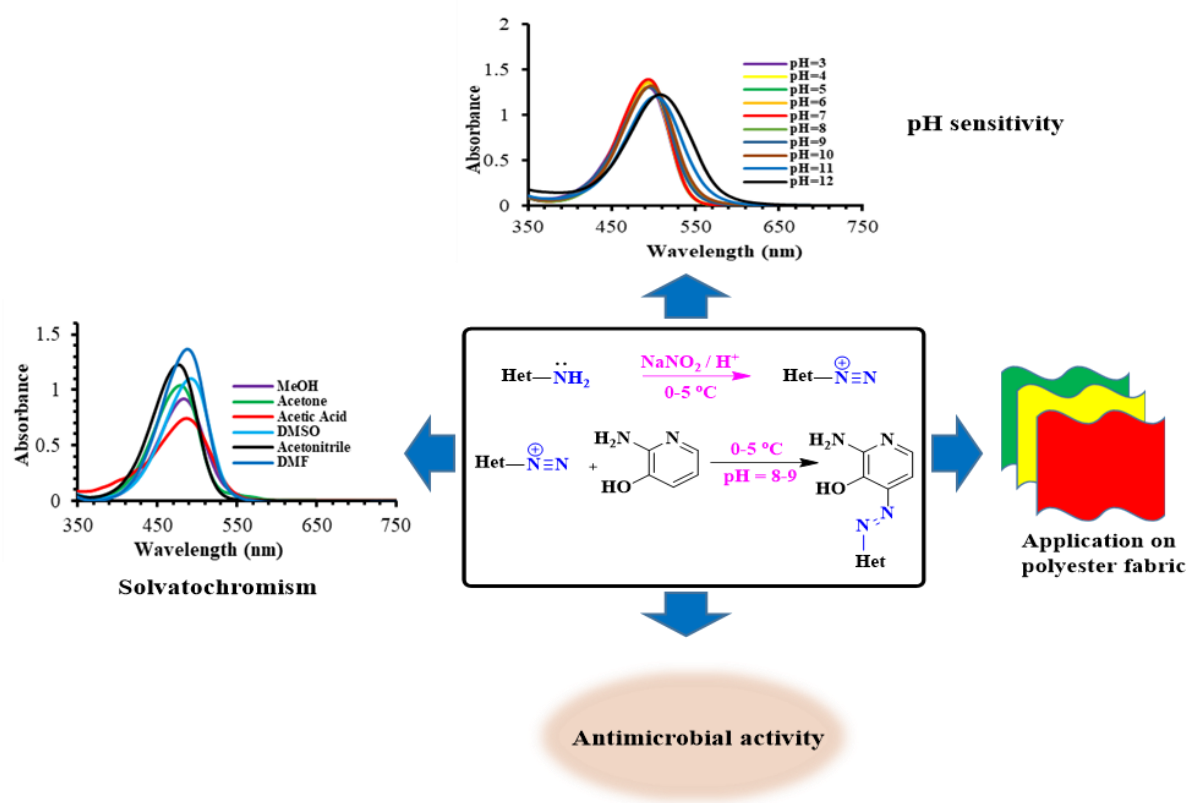
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Subject Category: Organic Chemistry

Abstract: In the present work monoazo heterocyclic disperse dyes were synthesized in good yield. 2-amino-3-hydroxypyridine was used as a coupling component with five distinct heterocyclic diazonium salts. All the dyes are new except the dye PY4. The obtained colorants were characterized by melting point, 1H-NMR, FT-IR, UV-Vis, FAB-LRMS and FAB-HRMS. Structure of the dye PY1 was further conformed by 13C-NMR (BB), 1H-1H COSY, 1H-13C HSQC, 1H-13C HMBC. The effect of pH variation on the λmax of these colorants were assessed in methanol and their solvatochromism analysis were performed in various organic solvents e.g. acetic acid, methanol, acetone, acetonitrile, dimethyl sulfoxide and dimethyl formamide. The geometries of the dyes were optimized at B3LYP/6-311G (d,p) level and their electronic excitation properties were determined using time dependent density functional theory. The computed data were in good agreement with the experimental one. These colorants were applied on polyester fabric as disperse dyes and evaluated for build-up study along with complete observation of their fastness to water, washing, sublimation, rubbing, perspiration and light. The pharmacological exploration of these dyes were performed in detail against human pathogenic fungi *Aspergillus fumigatus*, *Candida glabrata*, *Candida albicans* and *Trichophyton rubrum*. The dye PY5 was found to have the promising inhibition against all the used human pathogenic fungi.

Graphical Abstract:


pH sensitivity
Absorbance
Wavelength (nm)
pH=3
pH=4
pH=5
pH=6
pH=7
pH=8
pH=9
pH=10
pH=11
pH=12
Solvatochromism
MeOH
Acetone
Acetic Acid
DMSO
Acetonitrile
DMF
Het—NH2
NaNO2 / H+
0-5 °C
Het—N≡N
0-5 °C
pH = 8-9
Application on polyester fabric
Antimicrobial activity

Identifying new isoform-selective PI3Kα inhibitors for targeted cancer therapy: a machine learning-driven computational approach

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[b]Department of Pharmacognosy, College of Pharmacy, King Saud University, Riyadh, Saudi Arabia

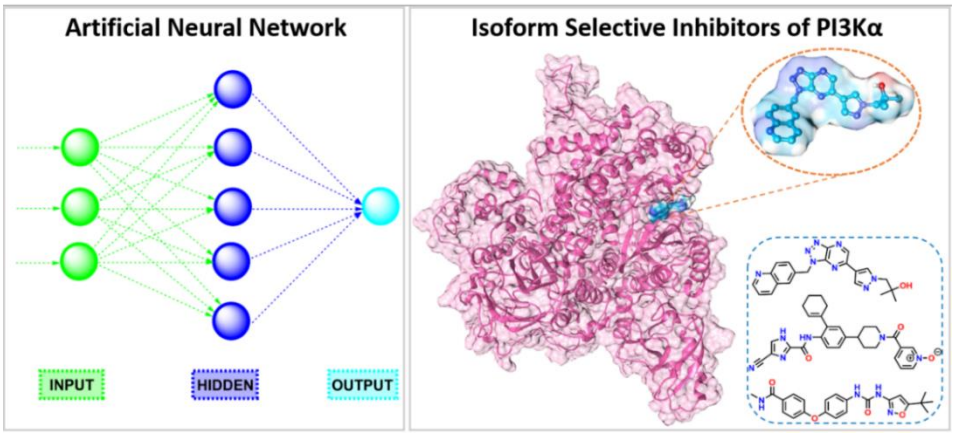
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Subject Category: Computational-Medicinal Chemistry

Abstract: Phosphoinositide 3-kinase alpha (PI3Kα), a frequently dysregulated kinase, plays a pivotal role in oncogenic diseases. The available drugs against PI3Kα-induced cancers are reported having side effects, prompting further research. However, the structural conservation in PI3K isoforms renders it challenging to design new isoform-selective PI3Kα inhibitors. We developed a hybrid protocol employing machine learning and *in silico* drug-discovery strategies to address this challenge. Using physicochemical data of known PI3Kα inhibitors, a machine learning classification model was developed to screen a library of compounds. This methodology predicted 662 compounds as PI3Kα inhibitors. The predicted hits were further characterized in the targeted cavity of PI3Kα using multiphase molecular docking, yielding twelve compounds interacting with the crucial residues of targeted site. Notably, one of these compounds was found to have potential activity in counteracting PI3Kα activity in a previous *in vitro* study. Subsequently, *in silico* drug-likeness and pharmacokinetic characterizations were performed to shortlist potential hits with promising ADME-T profiles. Further, mechanistic studies in dynamic mode endorsed the potential of identified hits in blocking the ATP-binding site of PI3Kα with higher binding affinities than the standard drug, alpelisib. These outcomes validate developed classification model and the devised computational strategy for identifying novel isoform-selective PI3Kα inhibitors.

Graphical Abstract:



An efficient Fe_2Se_3/Fe_2O_3 heterostructure electrocatalyst for oxygen evolution reaction

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Subject Category: Materials Chemistry

Abstract: In this study, novel iron-based electrocatalysts, including Fe2Se3, Fe2O3, and Fe2Se3/Fe2O3 heterostructures, have been developed to bring down overpotential in the oxygen evolution reaction (OER). Among these, the Fe2Se3/ Fe2O3 heterostructure emerges as a highly promising electrocatalyst for OER, offering exceptional efficiency, cost-effectiveness, and minimal overpotential. The Fe2Se3/Fe2O3 electrocatalyst exhibits outstanding electro-chemical performance, with a remarkable onset potential of 1.31 V (vs. RHE) and a minimum 160 mV over-potential at 20 mA/cm2 for OER. Notably, this prepared electrocatalyst demonstrates the lowest overpotential when compared to benchmark Fe-based OER electrocatalysts. Furthermore, during controlled potential elec-trolysis at 1.65 V (vs. RHE), the electrocatalyst maintains a stable current density of 65 mA/cm2, highlighting its impressive electrochemical robustness. The hydrothermal synthesis method used for the synthesis of Fe2Se3, Fe2O3, and their heterostructure is particularly very simple and straightforward. Which making them excellent candidates for water splitting applications, particularly in OER. These significant achievements accomplished through the utilization of non-precious metal electrocatalysts, open up new possibilities for practical OER implementations. Undoubtedly, these findings will greatly benefit the scientific community in the development of efficient electrocatalysts using non-noble metals for the electrochemical water splitting process.

Enhancing Organic Pollutant Degradation in Wastewater Using GQD/TiO2-Integrated Photocatalytic Concrete

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Subject Category: Material Science

Abstract: The integration of titanium dioxide (TiO_2) into concrete as photocatalytic concrete has gained considerable attention in addressing environmental concerns. This novel approach utilizes the photocatalytic properties of TiO_2 to effectively degrade organic pollutants, such as dyes and microbes, present in wastewater. By incorporating TiO_2 into concrete, the degradation of organic materials and airborne pollutants is achieved when the photocatalysts are exposed to light. Furthermore, the combination of TiO_2 with Graphene quantum dots (GQD) has shown promising results in enhancing pollutant degradation efficiency by shifting the wide band gap of TiO_2 from the UV range to the visible range. This research aimed to explore the effectiveness of different %mols (1, 5, 10, 15, 20 and 25mol%) of GQD/TiO_2 annealed at 3 different calcination temperatures (300°C, 400°C, 500°C). 15mo% loaded into concrete exhibit best degradation efficiency within 60 minutes for irradiation time to treat RB5 contaminated wastewater. Utilizing Response Surface Methodology (RSM), the study also focused on mathematical analysis to understand the probability and correlation of various parameters involved in the research. The ANOVA result showed significant model terms indicated by P-values less than 0.05 proving correlation between independent variables of input data. P-values greater than 0.050 indicated insignificant model terms showing that model reduction may improve this model. The findings of this study highlight the potential of photocatalytic concrete in effectively degrading organic pollutants and its suitability for widespread use in green infrastructure applications for the treatment of various wastes.

Synthesis and Characterization of Bimetallic Organic Framework (Zn-Mn BDC) MOF for Adsorption of Nitrobenzene

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Subject Category: Physical Chemistry

Abstract: Zinc manganese benzene dicarboxylic acid (Zn-Mn BDC) MOF was hydrothermally synthesized and then used as adsorbent for the removal of nitrobenzene (NB) from aqueous solution. The synthesized MOF was characterized by X-ray diffraction, Fourier transform infrared spectroscopy, thermogravimetric analysis, and BET surface area.. A number of experimental parameters for the batch adsorption studies were tuned, including temperature, agitation time, adsorbent dose, solution pH, and initial NB concentration.. It was found that the amount of NB removed increases with an increase in the temperature from 30 ^{o}C to 60 ^{o}C and pH from 2 to 6, whereas no notable change was observed in basic media. Different adsorption and kinetics models was applied on the experimental data. Pseudo-second order and Langmuir model with correlation coefficient (R2) values = 0.99 adequately suited the data. The adsorption capacity (qe) of Zn-Mn-BDC MOF for NB was found to be 287.73mg/g.

Synthesis of Porous Bimetallic Nanocatalyst for Selective Formate Production by CO_2 Electroreduction

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Subject Category: Material Science / Physical Chemistry

Abstract: The Ag_2SnO_3 nanoparticles were prepared by simple and cost-effective chemical reduction method. The morphology, optical properties, and composition of the catalyst were investigated. The as-synthesized nanocatalyst was employed to reduce CO_2 electrochemically in an aqueous electrolyte media at pH 6.8 using H-type cell. SEM images revealed that the catalyst looks like a sponge. The electrochemical results show higher current density (26 mA/cm^2), long term stability, and product selectivity on the electrode. The highest FE for formate was 89% at -1.3 V *vs.* Ag/AgCl. The FE was found to be higher as compared to monometallic (Ag, Sn) nanoparticles that indicate the synergistic effect between these metals.

Rational Design of Sulfur And Carbon Nanotube Networks Inside A Biomass-Derived Porous Carbon For High-Performance Li-S Batteries

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[1]*Institute of Chemical Sciences University of Swat Ala Abad, Tehsil, Charbagh, Swat, Pakistan*

Subject Category: Materials Chemistry

Abstract: Rechargeable lithium-sulfur batteries are widely recognized as a highly attractive candidate for future energy storage applications. This is primarily attributed to their exceptional energy density, substantial theoretical specific capacity, cost-effectiveness, and abundant availability. The limited adoption of sulfur as an electrode material can be attributed to various factors, including the shuttle effect, low conductivity properties, and volume expansion issues. The current study used a novel, straightforward carbonization process with a small amount of KOH as an activator to synthesize porous carbon from biomass-derived (peapod), showing an interconnected framework of thick nanosheets and blocky morphology can be achieved by adjusting the mass ratio of KOH and peapod. The porous carbon produced as a sulfur host exhibits several advantages in terms of enhanced conductivity and effective binding of polysulfides, achieved through physi-/chemisorption, endowing excellent electrochemical performance. The PC composite PC@S/MWCNTs demonstrates a notable initial capacity of 1330 mAhg−1, which decreases to 1115 mAhg−1 after undergoing 300 cycles of charging and discharging. Despite this decrease, the material retains approximately 83.8% of its original capacity. Additionally, the PC composite exhibits high discharge capacities at various current rates, specifically 1310, 1150, 940, 810, and 720 mAhg-1 when subjected to current rates of 0.1C, 0.5C, 1C, 2C, and 5C, respectively, as compared to PC@S.

A Novel Liquid-liquid Micro-extraction Based Method for the Extraction and Determination of congo red From Aqueous Medium by Using DES and UV Vis Spectrophotometer Analysis

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[a] *Abdul Wali Khan University Mardan Pakistan.*

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Subject Category: Analytical Chemistry.

Abstract: This study discuss a novel technique for extraction of Congo red dye from aqueous medium using a Deep Eutectic Solvent (DES) and analyze by UV Visible spectrophotometer. This DES composed from hydrogen bond donor (HBD) choline chloride and hydrogen bond accepter (HBD) thymol with 1:2 molar ratio. The DES offers a sustainable substitute for traditional organic solvents and due to low toxicity, environmental friendly and biodegradability, it classified as a green solvent. Congo red is non-biodegradable, carcinogenicity and synthetic dye and that is widely used in different areas such as paper and textiles industry, it poses serious problems to the environment and human health. The developed method for extraction has a high degree of precision and sensitivity, with a 3.2 µg/L Limit of Detection (LOD) and a 2.5 µg/L Limit of Quantification (LOQ). By using environmentally friendly solvents, this method not only successfully removes Congo red dye but also promotes sustainable industrial practices, which helps to reduce pollution and clean the environment.

Structural and Dynamical Properties of Human Aromatase *via* Molecular Dynamics Simulations

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Subject Category: Organic Chemistry

Abstract: Breast cancer has become one of the leading cause of death and a barrier to increasing life expectancy. It is a serious health issue among different cancers in women and approximately 2.3 million new cases were estimated to occur in 2020. Around 70% of breast cancer mainly depend on the overexpression of estrogen hormones. Estrogen is important for the development and progression of breast cancers by influencing the actions of genes involved in cell differentiation, multiplication and cell communication. In vertebrates, estrogens are produced via the aromatization of corresponding androgens by the rate-limiting enzyme aromatase, making it a promising therapeutic target for the development of specific inhibitors that reduce circulating estrogen levels. Aromatase, a microsomal membrane-bound cytochrome P450 monooxygenase complex, which promotes the aromatization of androgens via three oxygen-consuming steps and a specific NADPH cytochrome P450 reductase (CPR) acts as a source of electrons. However, the interactions between aromatase and CPR are also essential for its catalytic activity. We applied an integrated computational protocol consisting of protein-protein docking and atomistic molecular simulations to obtain a detailed understanding of structural and dynamical properties of the full-length aromatase embedded in realistic membrane environment, and the influence of the mutated CPR on the structure and activity profile of aromatase.

Probing the Oligomeric Possessions of IAV PB1-F2 Protein Via Molecular Dynamics Simulations

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Subject Category: Organic Chemistry / Physical Chemistry

Abstract:

In today's world, viruses present some of the most daunting challenges to humanity due to their unpredictable replication and rapid spread. Their mysterious nature makes it difficult for scientists to unravel their mechanisms of action. Viruses operate at various biological levels, hijacking cellular functions to replicate and evade the host's immune system. Among their tools are viroporins—specialized small hydrophobic proteins that increase membrane permeability by forming pores or self-oligomerizing into ion channels. We have studied one such protein, PB1-F2, encoded by Influenza A Viruses (IAV) which is known for its pro-apoptotic properties. The structural characteristics of PB1-F2 have long puzzled researchers. By using Molecular Modeling and Molecular Dynamics Simulation protocols, we explored the potential oligomeric structures of this protein. Our simulations, conducted in realistic mitochondrial membrane environments, revealed significant insights into the role of PB1-F2 in apoptosis. This study offers a practical approach to understanding PB1-F2 at the molecular level, providing valuable perspectives on the viral infection lifecycle and aiding in the development of effective therapeutics against the virus.

Salinity-Tolerant Oxygen Evolution using Corrosion-Resistant Ternary Metal Hydroxide Electrocatalysts

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Subject Category: Material Science

Abstract: Seawater electrolysis is a promising method for large-scale, cost-effective hydrogen production. However, catalyst performance can significantly deteriorate with increasing salt concentrations. In this study, a porous ternary metal hydroxide (NiCoFe@NF) is synthesized using a simple and scalable method, promoting homogeneous growth on a conductive substrate (NF) to create self-supported catalysts for the oxygen evolution reaction (OER) under alkaline conditions. As-obtained electrode repels chloride ions from the catalyst surface in the alkaline electrolyte. The NiCoFe@NF electrode demonstrates excellent catalytic activity and stability in alkaline electrolytes. Benefiting from the inherent chloride-repelling properties of the ternary metal hydroxide, the catalyst maintains high performance in natural seawater (1 M KOH) electrolytes, with an overpotential increase of less than 20 mV at 10 mA cm^{-2} when transitioning from freshwater to natural seawater. Additionally, the catalyst exhibits superior corrosion resistance with minimal degradation after 72 hours of stability testing. This research provides new insights into developing efficient OER catalysts with high salinity tolerance for seawater electrolysis.

Spectroscopic Studies of the Interactions of Bisphenol A with Cetyltrimethyl Ammonium Bromide

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Subject Category: Inorganic Chemistry / Physical Chemistry

Abstract: BPA is an endocrine-disrupting chemical used in different industries. BPA is released into the environment from different sectors and thus pollutes the environment. Surfactants are substances that can improve the solubility of the nonpolar compounds in water. BPA is less soluble in water, thus CTAB can improve the solubility of BPA in water. Studying the interactions is an important parameter to know about the best possible materials for removal. In the current literature, there is no study available about the effect of pH on the interactions of BPA with CTAB. Above pK_a BPA ionizes which affects the interaction behavior of BPA with CTAB. Here in this work, the interactions of BPA with CTAB are studied at different pH using UV-Vis spectroscopy. The binding constant (K_b), partitioning constant (K_c), and Gibbs free energies (ΔG_b and ΔG_p) are determined using UV-Vis spectroscopy. The K_b and K_c values increase with an increase in pH showing that the interactions become stronger with the increase in pH of the solution. At higher pH, BPA can interact with the head group as well as with the hydrophobic part of surfactants. Gibb's free energy shows that the process is spontaneous. This study can play an important role in the removal of BPA through surfactants.

Type High-Performance Tyrosine Nanosensor Based on Carbon Nanocoil and Manganese Tetraphenylporphyrin Composite

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Subject Category: Material Science

Abstract: Tyrosine is recognized as a key biomarker linked to various human diseases, necessitating the development of highly sensitive and selective sensors for its efficient and on-site detection in clinical samples. This research highlights the effective electrochemical sensing of tyrosine using a heterojunction of carbon nanocoils (CNC) and manganese tetraphenylporphyrin (MnTPP). The unique chiral and conductive properties of carbon nanocoils, combined with the catalytic activity of the MnTPP metal center, enable enhanced electron transfer kinetics in the CNC/MnTPP sensor compared to individual CNC and MnTPP components. The sensor was characterized, and its analytical performance assessed using cyclic voltammetry (CV), differential pulse voltammetry (DPV), electrochemical impedance spectroscopy, and chronocoulometry. Physical characterization was performed using FTIR, Raman, and UV/visible spectroscopy. The CNC/MnTPP/GC sensor demonstrated effective tyrosine detection over a wide concentration range, achieving a detection threshold as low as 0.021 μM at an optimal pH of 5. Additional evaluations revealed the sensor's excellent sensitivity, stability (both intra-day and inter-day), accuracy, and performance in human blood serum analysis. The CNC/MnTPP/GC sensor exhibited superior detection limits compared to previously reported sensors, underscoring its potential for clinical applications.

A Novel Liquid-liquid Micro-extraction Based Method for the Extraction and Determination of EBT From Aqueous Medium and by Using DES and uv visible spectrophotometer Analysis

Tanvir Khana,Azmat Walib ,Muhammad Naveedc.*

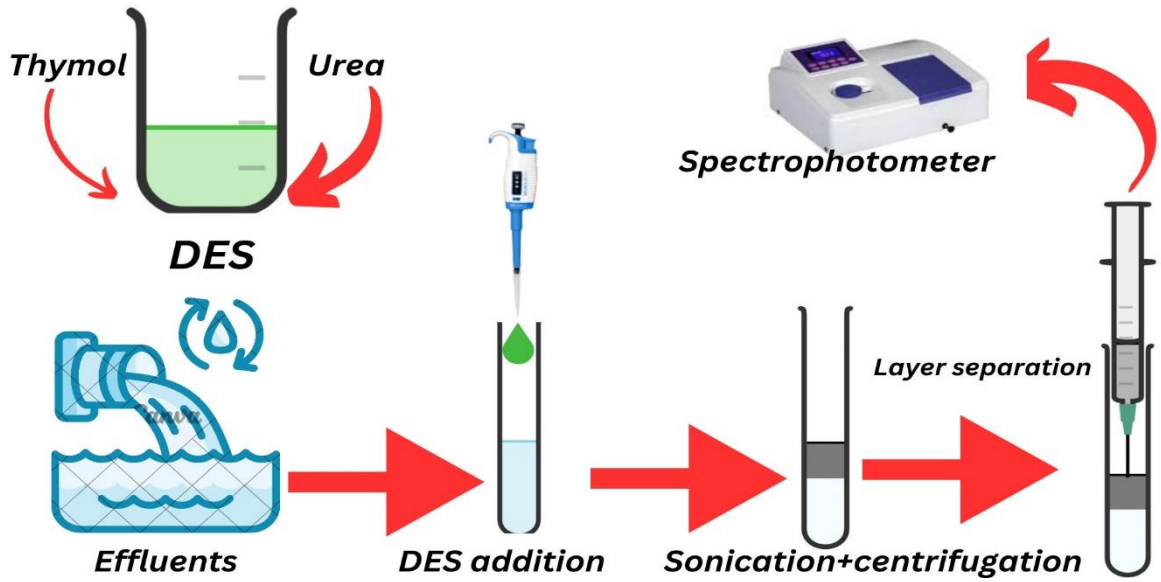
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Subject Category: Analytical Chemistry

Abstract: Deep eutectic solvents (DES) provide an effective and sustainable method for extracting EBT (Eriochrome Black T) dye. DES is a biodegradable, inexpensive, readily synthesized, and environmentally friendly solvent. Low toxicity and strong solubility power are the special characteristics of DES, which are created when thymol and urea were mixed in a 1:2 weight ratio for analysis. The toxicity of EBT dye and the use of DES in its extraction are examined in this work. Extraction efficiency is greatly impacted by the composition of DES such as thymol and urea. Variable optimization, including Ph and volume optimization, was studied as a means of increasing yield. In comparison to previous developed methods, the approach exhibits dormancy due to its high acceptability in terms of percent recovery (105 to 106%), LOD (2.1 mg/L), LOQ (3.0 mg/L), and percent RSD (1.41 to 1.12%). Subsequent developments in DES technology are expected to enhance extraction procedures and environmental sustainability, hence opening the door to more efficient and environmentally friendly analytical methods.

Graphical representation: (Optional)



Designing and synthesis of 2-Mercaptobenzimidazole derivatives to identify potential anti-hyperglycemic agents through computational approach

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Shifa College of Pharmaceutical Sciences, Shifa Tameer-e-Millat University Islamabad, Pakistan

Subject Category:

Abstract: One of the most widespread diseases recognized all over the world is diabetes, accounting for 1.5 million deaths each year. The main objective of the study was to identify anti-hyperglycemic agents acting through various mechanisms in the diabetes pathway. This study is focused on designing and synthesis of new derivatives of 2-mercaptobenzimidazole and their molecular dynamic studies to ascertain the binding interactions against α-glucosidase, PPaR-γ, DPP-IV and AMPK. One-pot synthesis of derivatives is also reported. Most of the designed compounds (A-10 = -10.5 Kcal/mol) showed significantly greater binding affinity with PPaR-γ as compared to the standard pioglitazone (-9.1 Kcal/mol). B-23 exhibited binding affinity of -9.6 Kcal/mol with PPaR-γ and 5.21 Kcal/mol with AMPK and was subjected to MD simulations which showed good stability profile. The study of binding interactions revealed the potential to further optimize the interactions by designing hybrids at different sites.

Electrochemical sensors for dopamine detection using nickel foam modified with 2-D rGO sheets decorated with transition metal oxides

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Subject Category: Physical Chemistry

Abstract: The development of sensitive and stable electrochemical sensors is highly desireable for dopamine detection. Herein, a facile method for craved nanocomposites synthesis as electrochemical sensors for dopamine detection with broad linear range and low detection limit has been reported. Various metal oxides and metal oxides/rGO nanocomposites have been synthesized and immobilized over nickel foam substrate as MO/rGO@NF sensors. The metal oxides (MO) nanoparticles and rGO were synthesized through coprecipitation method and modified hummer method, respectively. Scanning electron microscopy (SEM), FT-IR spectroscopy and X-ray diffractometer were engaged for the characterization of synthesized material. Electrochemical Impedance Spectroscopy (EIS) and Cyclic voltammetry (CV) were employed to evaluate the electrochemical performance of Fe_2O_3/rGO@NF, NiO/rGO@NF and CuO/rGO@NF electrodes as electrochemical sensors for dopamine detection. From different studies, it has been revealed that Fe_2O_3/rGO@NF shows the best results for dopamine detection which might be due to enhanced conductivity as compared to CuO/rGO@NF and NiO/rGO@NF. The detection limit (LOD) was calculated up to 50 µM with wide linear response over 60 to 100 µM (R^2= 0.99) for dopamine hydrochloride injection.

Outlook for organic solar cells with Non-Fullerene Future

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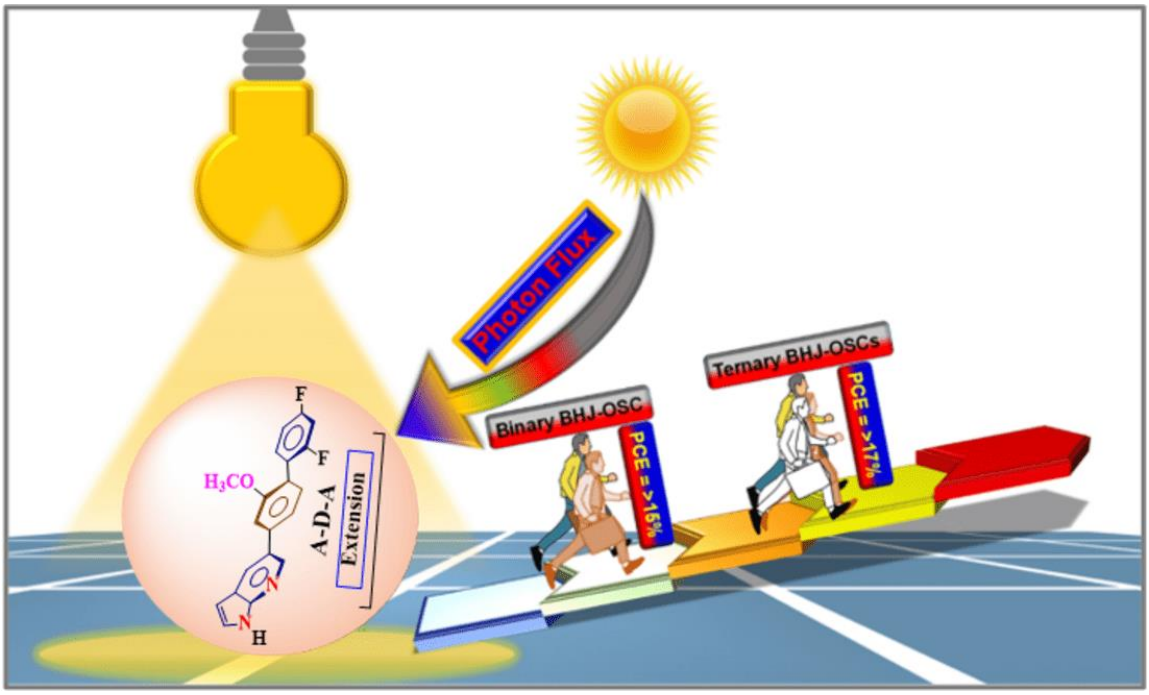
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Subject Category: Organic Chemistry (Organic synthesis/ Organic Solar Cells)

Abstract: The growing commercial and public interest in organic solar cells (OSCs) as a clean and renewable energy source is attributed to their low preparation costs, accessibility, sustainability, good adaptability, and non-toxicity. However, the primary obstacle currently limiting the development of OSCs is the selection of an acceptor material. High-performance acceptor materials play a crucial role in enabling OSCs to continuously enhance stability and raise power conversion efficiency (PCE) for greater performance. The fullerenes based acceptors in particular have taken over the environment. This talk describes the evolution of a new class of materials called non-fullerene acceptors (NFAs). These have resulted in a steady improvement in cell efficiency, and reaching the noteworthy 20% PCE is now possible. The most recent methods for improving the efficiency of organic solar cells as well as the many uses of non-fullerene acceptors (NFAs) on various OSCs are covered. Furthermore, the difficulties OSCs confront in enhancing their performance and their chances for advancement are disclosed, offering fresh concepts for creating the upcoming class of high-performance OSCs.

Graphical Abstract:



Effect of Triethanolamine (TEA) on the Optical Properties of CZTS Thin Films Deposited by Chemical Bath Deposition (CBD) Method

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Subject Category: Physical Chemistry / Material Science

Abstract: Cu_2ZnSnS_4 (CZTS) has gained significant interest from scientists and researchers due to its affordability and environmental efficiency as a solar cell material. Using a complexing agent in the deposition process shows promise in enhancing efficiency and reducing detrimental crystalline defects in CZTS thin films. In this study, using the chemical bath deposition (CBD) technique, triethanolamine (TEA) was used as the complexing agent for depositing CZTS thin films on glass substrates. The structural and optical properties of the deposited thin films were analyzed, confirming the synthesis of tetragonal CZTS using X-ray diffraction. The results revealed considerable improvement in the optical properties when using a complexing agent. Optical characteristics were evaluated using UV-visible spectroscopic measurements.

Keywords: Thin film, CZTS, TEA, CBD, Optical properties

Synthesis and Inverse Hierarchy Effect of Nickel Cobaltite and Cobalt Sulfide Hybrid Nanostructures and their Influence on Energy Storage Application

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Subject Category: Inorganic Chemistry / Material Science

Abstract: In this study, Cobalt sulfide (CoS2) and nickel cobaltite (NiCoO2) nanostructures are prepared individually, one over the other and in reverse order to explore their effects on energy storage applications. The prepared nanostructures characterize using techniques involving X-ray diffraction, raman spectroscopy, scanning electron microscopy, energy dispersive X-ray spectroscopy, contact angle analysis and transmission electron microscopy. The electrochemical performance of these nanostructured materials for energy storage is thoroughly investigated, among all the prepared nanostructures, cobalt sulfide grown over nickel cobaltite exhibited the highest specific capacity of 508.6 mAh·g-1 at a specific current of 1 A·g-1, specific energy of 226.4 Wh·kg-1 at specific power of 115.1 W·kg-1 .The electrode retained a specific capacity of up to 90% after 2500 cycles. It is observed that reversing the growth order of nanostructures in an electrode material provides suitable insights to optimize its electrochemical performance for energy storage applications. Moreover, fabricated asymmetric supercapacitors CoS2/NiCoO2//Activated carbon exhibited maximum specific capacity of 49.8 mAh·g-1 at the specific current of 1 A·g-1 and able to retain the specific capacity as 78.9 % after 5000 cycles due to its high specific energy of 93.0 Wh·kg-1. The outcomes suggested that fabricated electrode material has bright future in energy storage applications.

Transformation of Wheat Straw into Bio-Composite

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Subject Category: Material Science (Material Science, Mechanical Engineering)

Abstract: In order to improve cellulose accessibility and enable the production of biodegradable films and products containing polystyrene and stearic acid or Polybutene, this study looks into chemically treating wheat straw. It focuses especially on alkali-treated straw for environmentally friendly uses in packaging and tableware. This study explores how different chemical treatments can improve the surface of wheat straw, which is tough to break down due to its mixture of cellulose, lignin, hemicellulose, and protein. These treatments make the cellulose in wheat straw more accessible for creating environmentally friendly composites. We prepared films using Polystyrene (PS) as a polymer, Stearic Acid or Polybutene as a plasticizer and mixed with untreated wheat straw, and wheat straw treated with alkali (NaOH) and acidic (HCl, and H2SO4). We used a method called solution casting. We studied how these treatments affected the bio-composites, structure, water resistance, and strength. Our findings show that polystyrene (PS) and Stearic Acid mixed with Alkali-treated wheat straw are well-suited for making biodegradable films, which could be useful in green packaging, and dinnerware for different industries.

Synthesis of 4-Aminoquinazoline Derivatives through Sequential *SNAr* and Suzuki Coupling Reactions

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Subject Category: Organic Chemistry

Abstract: Quinazoline derivatives display extensive applications in organic and pharmaceutical chemistry and are pivotal as natural and synthetic materials for medicinal chemistry. This study reports the synthesis of a new series of 6-aryl-4-aminoquinazoline derivatives through sequential Nucleophilic Aromatic Substitution (*SNAr*) and Suzuki Coupling reactions. Starting from anthranilic acid, ten quinazoline derivatives were synthesized, each featuring various amines at the 4-position and an aryl group at the 6-position, achieving yields ranging from 69% to 80%. The structures of these compounds were rigorously characterized using NMR Spectroscopy, including both 1H NMR and 13C NMR. The spectral data obtained provided definitive identification and structural confirmation of the synthesized quinazoline derivatives. The successful synthesis and characterization of these new quinazoline derivatives underscore their potential significance in developing novel therapeutic agents and further highlight the utility of sequential *SNAr* and Suzuki Coupling reactions in complex organic synthesis.

Optimized ZnS:CoS nanohybrid promising materilas for assymetric supercapictors

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Subject Category: Material Science

Abstract: Cobalt and Zinc nano hybrids were successfully synthesized and various factors that may affect the stability and size of synthesized hybrid sulfide nanoparticles were investigated. Factors like pH, temperature, and ratio of precursor metallic salt played a significant role in determining the optimal properties of hybrid metal sulfide nanoparticles. Various techniques were utilized to characterize and confirm the synthesis of pure metal sulfide and hybrid metal sulfide nanoparticles at various reaction conditions. The visible spectroscopic analysis confirmed the successful synthesis of pure cobalt and zinc sulfide nanoparticles along with hybrid sulfide nanoparticles. XRD and SEM FTIR, Fluorescence spectroscopy, and DFT analyses were effectively performed for further characterization of the samples. The hydrodynamic size distribution of NPs and overall stability of the hybrid sulfide nanoparticles were investigated through Dynamic Light Scattering (DLS) and Zeta potential techniques. After successful optimization, their potential for supercapacitor application was analyzed. Cyclic Voltammetry, EIS, and GCD were done that depict that optimized ZnS and CoS hybrid have high discharging capacity, good cycle stability, and have highest specific capacitance.

Green Synthesis of Cobalt Oxide Nanoparticles for Water Splitting to produce Hydrogen for fuel cell

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Subject Category: Material Science

Abstract: This work describes the green synthesis of CoO nanoparticles using plant extracts from the Populus tree. An environmentally friendly route to form materials with potential use in hydrogen generation is explored. Collection of leaves from the Populus tree from a local nursery starts off the project, followed by proper washing in hot distilled water. The leaves were dried at room temperature, crushed into fine particles using an electrical blender, and boiled in distilled water for three hours to create a plant extract. This solution is filtered and stored for further use.A 1M cobalt nitrate solution is prepared as the starting material for synthesizing CoO nanoparticles. A solution of cobalt salt and a plant extract in the ratio of 3:2 is heated at 90°C for a period of 2 hours 30 minutes. The solution on heating is cooled to room temperature, and centrifugation done at rpm 20,000 for 20 minutes isolates CoO nanoparticles. Fourier-Transform Infrared Spectroscopy studies the structure and morphology of these nanoparticles, Scanning Electron Microscopy, X-ray diffraction, and Transmission Electron Microscopy. The prepared CoO nanoparticles are then used in experiments in hydrogen gas production by water splitting. In turn, the techniques of cyclic voltammetry and linear sweep voltammetry are used to further evaluate these CoO nanoparticles. Such results are expected: the green synthesis of CoO nanoparticles and their efficient use in hydrogen generation by splitting water. The work shall add to the development of the sustainable energy solution by providing a green route of synthesizing CoO nanoparticles for manifesting their potential in generating clean hydrogen fuel. Proper documentation of all the results obtained from this in-depth study will be done and published in a good impact factor scientific journal, contributing toward development in the area of green nanotechnology and sustainable energy.

Insights into the Activation Mechanism and Antagonist Binding of the Allergic Mediated Histamine H1 Receptor via Enhanced Sampling Methods

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Subject Category: Organic Chemistry

Abstract: Allergic diseases have surged globally over recent decades. According to the World Allergy Organization, around 10-30% of the world's population now grapples with allergic rhinitis. In Pakistan, a nation of 221 million, asthma affects 4.3%, and allergic rhinitis, the most common allergy, impacts 24.62%, peaking in Sindh at 27.92%. Food allergies, afflicting 1.1-10.8% of the global population, are also on the rise. Central to allergic responses are histamine receptors (H1-H4), key members of the G protein-coupled receptor (GPCR) family, with the H1 receptor (H1R) being a prime target for anti-allergy treatments. Despite extensive studies, the precise mechanism of H1R activation by histamine remains unclear. To unravel these mechanisms and investigate potential antagonist binding, we utilized advanced sampling methods like Replica Exchange Molecular Dynamics (REMD) to explore H1R's conformational dynamics upon ligand binding. Our findings are expected to reveal the H1R's structural and dynamical behavior, offering valuable insights into the future GPCR-targeted drug discovery.

Nanoceramic Based Antimicrobial Composite Membranes: Sustainable Advances in Water Purification and Desalination

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Subject Category: Material Science

Abstract: Pakistan has been undergoing a severe health crisis due to infectious diseases and viral infections. Microbial and chemical pollutants are the primary causes accountable for various public health issues. The geospatial mapping of physicochemical parameters, WQI, and SPI model findings reveal that most of the groundwater is polluted and thus not safe for drinking purposes. In recent years, the development of polymeric nanocomposite membranes has shown greater potential for water treatment and purification. In this work, the use of unique nanoceramic-based inherent antimicrobial nanofiber/membranes is studied, in resolving the issue of scarcity of clean drinking water around the globe. The synergistic effect of ceramic and polymer along with antimicrobial agents (Silver, Magnetite, Copper oxide Nps) not only provides mechanical stability to the membranes but also boosts up resistance towards membrane fouling. The electrospinning method was utilized for the fabrication of nanocomposite membranes. This method is expected to revolutionize the existing water filtration device market. Advanced characterization techniques such as SEM-EDX, XED, FTIR, and DSC/TGA were utilized to study surface morphology, structural properties, functional groups, and thermal stability of the fabricated membrane. The change in the population of the microorganisms in the feed water was examined by the antimicrobial assay using various standard laboratory bacterial strains. The membrane exhibited excellent water flux and superior salt rejection rates in the desalination studies, performing well in both dead-end and cross-flow designs. The results highlighted the potential of these innovative membranes in addressing and mitigating the issues of water contamination on a worldwide scale. Subsequent investigations will further refine these membranes for enhanced desalination effectiveness and robustness in practical uses.

Synthesis, And Therapeutic Potential Of 2-(5-Cyclopropyl-6-Thioxo-1,3,5-Thiadiazinan-3-Yl) Acetic Acid In Animal Models Of Epilepsy Disease.

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Subject Category: Organic Chemistry

Abstract: Oxidative stress has been associated with numerous neurodegenerative diseases, triggering neuroinflammatory responses leading to neuronal apoptosis. The objective of this study was to synthesize and examine the antioxidant capabilities of 2-(5-cyclopropyl-6-thioxo-1, 3, 5-thiadiazinan-3-yl) acetic acid in preventing synaptotoxicity and cognitive impairment caused by oxidative stress induced by pentylenetetrazole (PTZ) in a mouse model. Cyclopropyl amine and carbon disulphide were dissolved in aqueous KOH at an equivalent molarity as a precursor, using a microwave synthesizer. Four research units were randomly assigned to male, fully mature mice, ages 7-8 weeks. Control, PTZ, PTZ + synthesized compound, and synthesized compound-treated groups. Morris and Y-maze tests were used to evaluate memory, western blotting techniques, and antioxidant enzyme studies for antioxidant, anti-neuroinflammatory, pairing protein restoration ability of synthesized compounds against PTZ. The synthetic drug significantly reduced LPO activity in brain homogenates and significantly restored antioxidant enzymes. Additionally, it significantly reduced the memory deficits caused by PTZ by upregulating the expression of pre- and post-synaptic proteins. It successfully inhibited phosphorylated JNK, TNF-α, and COX-2 proteins, reducing PTZ-induced neuroinflammation in adult male mice. In summary, using a model of PTZ-induced mice, this study elucidates the antioxidative, anti-neuroinflammatory, and neuroprotective properties of the synthesized compound.

Design, Synthesis, Computational and Pharmacological Investigation of Novel 2-Mercaptobenzimidazole Derivatives as Anti-Ulcer Agents

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Subject Category: Organic Chemistry

Abstract: Ulcer may develop at any area of elementary canal including esophagus, stomach, duodenum or other. Any imbalance leads to destruction of mucosal lining of gastrointestinal tract resulting in peptic ulcer. The unique structural features of benzimidazole and a wide range of biological activities of its derivatives made it a privileged structure in drug discovery. 20 different chalcones were prepared by reacting substituted aldehydes with ketones, these chalcones were condensed with benzimidazole-pyrazole hybrids to give final products (M3a–M3t). Docking and *In-silico* studies were done by utilizing AutoDockVina and online tools (SWISS ADME). The chemical structures of 2-mercaptobenzimidazole derivatives were confirmed by FTIR, ^{1}HNMR and ^{13}CNMR spectroscopic data. Molecular docking studies were carried out to predict the binding affinities and interactions of the synthesized compounds with target proteins CA-II (PDB ID: 1A42), and H+ /K+ -ATPase (PDB ID: 5YLU). These derivatives were screened for *In-vitro* antioxidant potential (DPPH), CA-II, *ex-vivo* H+ /K+ ATPase assay and *in-vivo* ethanol-induced gastric ulcer in rats among these M3e, M3i and M3m showed promising results (IC_{50} = 17.76 μM, 20.73 μM and 30.69 μM), M3e and M3m showed significant results (IC_{50} = 41.49 μM and 27 μM), M3i and M3m showed enhanced activity 38.82± 3 and 42.45± 2.52. The compound M3e, M3i, M3m exhibited maximum anti-ulcer activity and reduced the ulcer region by, 68 ± 3%, 74 ± 5%, 81 ± 5% respectively. The results obtained from molecular docking studies of synthesized compounds were also in conjunction with their *in-vitro, in-vivo* and *ex-vivo* pharmacological activities. Results support that compounds M3e, M3i and M3m have potent anti-ulcer and antioxidant activities.

Conjugation of Antimicrobial Peptide to Zinc Phthalocyanine for an Efficient Photodynamic Antimicrobial Chemotherapy

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Subject Category: Organic Chemistry

Abstract: Photodynamic antimicrobial chemotherapy is an attractive and novel therapeutic approach to treat microbial infections. Antimicrobial peptides (AMPs) have the potential to specifically target and kill the microorganism while showing no effect toward mammalian cells. In the current study, antimicrobial peptide (GGG(RW)3) was conjugated to zinc phthalocyanine (ZnPc) for photoinactivation assay to enhance the bacterial killing efficacy of the peptide. We observed that the conjugate ZnPc-GGG(RW)3 displayed a photoinactivation effect against the Gram-positive strains of S. aureus. Furthermore, the addition of KI salt to the PS also significantly killed the Gram-negative E. coli strain under red light illumination.

An innovative vapor-phase synthesis approach to obtain MIP-based optical sensors for label free analyte detection

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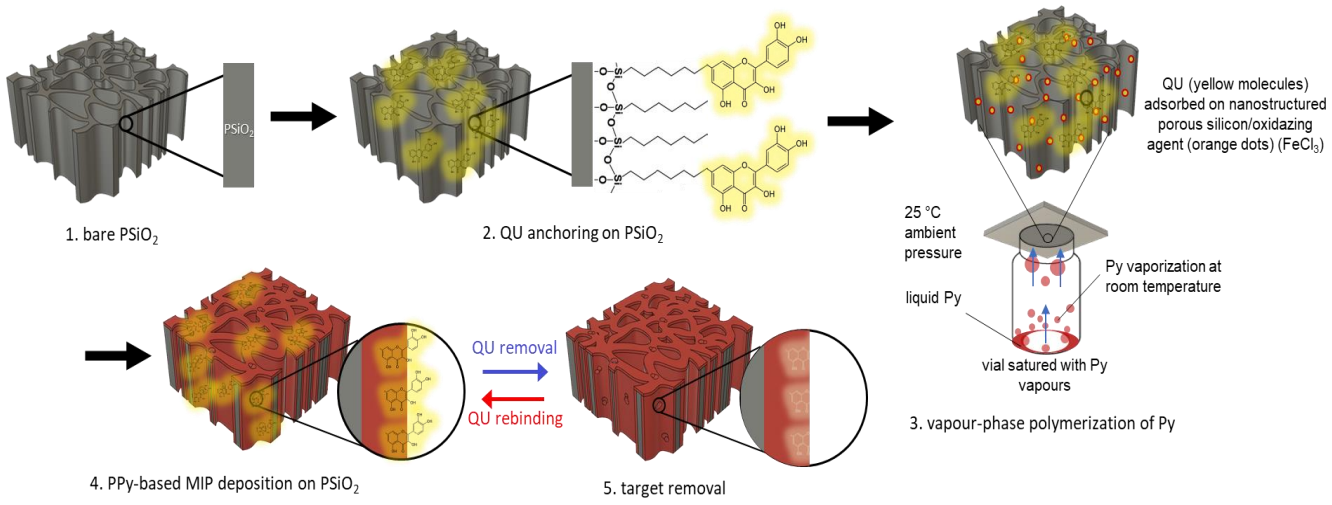
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Subject Category: Analytical Chemistry / Material Science

Abstract: Nanoporous silicon are becoming centrer of attraction for development of label free optical detection because of their huge surface area, lower cost, tunable surface and sensitivity to minute surface changes. But they face serious drawbacks such as lower selectivity that hinders application for real time sensing in complex matrices. Molecularly imprinted polymers (MIPs) are synthetic tailor made receptors renowned for their excellent selectivity but despite enormous research progress and advancements in synthetic methodologies, there is still lack of an easy and versatile approach towards MIP fabrication on complex nanostructured transducing systems. Here, we explore the application of a very simple and effective room temperature vapor phase polymerization technique to obtain MIPs on PSiO2 scaffolds used as interferometer and then optical transducer. This method allows to develop sensitive, selective and robust sensor for label free detection of quercetin (QU) molecule, that represents one of the most important bioflavonoid in human dietary supplements. Finally, the sensor was tested for detection of quercetin in wine samples and the results were validated by HPLC and an agreement in results evidenced the efficiency of sensor for real time detection of quercetin in beverages.

Graphical Abstract:



Doped NMC-811 As Efficient Cathode Material For Lithium Ion Batteries

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Subject Category: Material Science

Abstract: The pristine $LiNi_{0.8}Mn_{0.1}Co_{0.1}O_2$ and the doped (Ti, Nb) NMC (x = 0.01, 0.1and 0.2), as efficient and stable cathode materials for batteries, have been synthesized for the very first-time through CHFS. This work reports a strategy (doping) to stabilize the structure of $LiNi_{0.8}Mn_{0.1}Co_{0.1}O_2$ materials. XRD analysis exhibited that heat treatment at 850 ºC for 12 h using 3X lithium content is the optimized conditions to get the most stable structure. Furthermore, the introduction of dopants into the host structure has notably enhanced the structural and cycling stability of the cathode material. The dopant's effect on the structural and electrochemical properties of the material was studied via a number of techniques, including XRD, EDX, SEM, CV and charge/discharge studies. The pristine NMC material showed very low discharge capacity (~130 $mAhg^{-1}$), which decreased rapidly on cycling owing to structural deterioration, unexpected side reactions and cation mixing. However, increasing the dopant's fraction in the cathode material delivered higher discharge capacities along with enhanced cycling stabilities. The dopants exhibited the discharge capacities in the order: 2 % Nb^{5+} (180 $mAhg^{-1}$) > 0.1 % Ti^{4+} (130 $mAhg^{-1}$). It is highly expected that this work will provide a general approach to improve the electrochemical properties of cathode materials for Li-ion batteries.

POSTER PRESENTATIONS

Poster ID: T- ICRA-251

Kinetics Studies of Removal of Dyes by Using Natural Adsorbent

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Abstract: The removal of methylene blue using natural adsorbents such as banana peels, orange peels, walnut shells and almond shells in aqueous medium at different temperature 298K to 318K was studied spectrphotometrically. The method is safe, non hazardeous and ecofriendly as using raw waste material for removal of dyes from industrial effluents to save environmental pollution. Adsorption data were analyzed using Langmuir and Freundlich isotherm model for adsorption of dye with different natural adsorbents. Adsorption data were fitted well with the Freundlich and Langmuir isotherm model giving the maximum correlation with Freunlich model. This confirmed that multilayer adsorption process is occurred in this adsorption.The kinetics of degradation of methylene blue with natural adsorbents was analyzed by observing change in absorbance with respect to time as function of,concentration of absorbents and temperature. The adsorption process was found to be more favorable at low temperature and high percent removal of dye concentration was observed for banana peels as compare to orange peels, walnut shells and almond shells. The order of percentage degradation is banana peels > walnut shells > Almond shells > orange peels. Results showed that the pseudo-first order adsorption is pre dominant and the overall rate of dye adsorption process is controlled by adsorption reaction.The activation parameters as activation energy (E_A), enthalpy change (ΔH), entropy change (ΔS) and Gibb's free energy (ΔG) are also evaluated.

Poster ID: T- ICRA-252

Developing Eco-friendly Silver Nanoparticles for Enhanced Textile Dye Removal: Optimizing Adsorption and Photocatalysis with Response Surface Methodology

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Subject Category: Physical Chemistry

Abstract: Textile wastewater, contaminated with harmful dyes and chemicals, poses a significant environmental threat. This study explored the potential of silver nanoparticles (Ag-NPs), synthesized using an eco-friendly Azadirachta indica extract, for the removal of Everzol Red 239 dye. Characterized using UV-Vis, XRD, FTIR, AFM, and SEM, the Ag-NPs exhibited effective pollutant removal through a combination of photocatalysis and adsorption. Optimization using Response Surface Methodology (RSM) and Central Composite Design (CCD) revealed that pH and adsorbent dosage were critical factors influencing dye removal efficiency. Optimal conditions (pH 3, adsorbent dosage 0.02 g) led to a 99% removal rate. A decrease in pH enhanced dye removal, likely due to increased electrostatic attraction between the positively charged Ag-NPs and the anionic dye. Conversely, higher adsorbent dosages resulted in increased dye removal capacity. Adsorption isotherm studies confirmed the Freundlich model, indicating heterogeneous adsorption onto the Ag-NP surface. These findings highlight the potential of Ag-NPs as a promising, eco-friendly, and efficient technology for the remediation of textile wastewater

Poster ID: T- ICRA-253

New Insight In The Green Synthesis Of Metallic Nanoparticles, Their Characterization And Application In Environmental Remediation

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Subject Category: Materials Science

Abstract: The present study focuses to resolve the water pollution problems by nanoparticles, adopting adsorption process. The bimetallic Fe-Ni nanoparticles were synthesized by green approach using *MagniferaIndica* leaf extract. The surface morphology and chemical compositions were determined by scanning electron microscopy (SEM), X-ray diffraction (XRD) and Fourier transform infrared spectroscopy (FTIR) techniques. The removal of Everzol Blue (EB) and Congo Red (CR) dye were proceeded by ultrasonicated adsorption process. Response surface methodology (RSM) based on central composite design (CCD) was adopted to study the optimum operating parameters (OOP) for adsorptive removal of EB and CR dyes. The CCD was based on 5 factors, such as time duration, solution pH, amount of Fe-Ni Nanoparticles, concentration of EB and CR dyes was employed.

The present study showed that the nanoparticles efficiently removed the textile dyes. The percent removal efficiency of EB and CR dyes were observed to be 95.65% and 96.73% respectively. Moreover, Adsorption isotherms studied were conducted and EB dye adsorption was observed to follow Freundlich and Harkins Jura model whereas CR dye adsorption was observed to follow Freundlich model. Kinetic studies showed, that it follows pseudo second order kinetics. The pHpzcwas evaluated and it was found to be 6.4. The result of present study show that proposed method is most efficient, environment friendly and cost-effective which efficiently remove EB and CR dyes from the textile effluent.

Keywords: natural adsorbents, dye, Characterization, Adsorption

Poster ID: T- ICRA-254

Spectroscopic Studies of Interactions of Cefuroxime with Cationic Cetrimonium Bromide at Different pH

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Subject Category: Physical Chemistry

Abstract: This study explores the interaction of CTAB with cefuroxime. Cefuroxime is a commonly used second-generation cephalosporin antibiotic which is often used and well-known for its ability to treat a variety of bacterial illnesses. Cetyltrimethylammonium bromide (CATB) is cationic surfactant which has a quaternary ammonium head group ($N(CH_3)_3^+$). The interaction between cefuroxime and CTAB was explored to understand potential changes in the antibiotic's solubility and interactions in the presence of CTAB.The number of drug molecules per micelle, the partition constant, and the binding constant were used to decide cefuroxime's interaction with CTAB. To understand how pH changes impact their interaction and possibly the drug's bioavailability, this study uses UV-visible spectroscopy to investigate the interaction between cefuroxime and CTAB at various pH levels. At pH.1.9 and 12 in the acidic and basic environments respectively, may promotes different types of interactions. Understanding these changes can improve drug delivery systems and are essential for enhancing cefuroxime's efficacy in pharmaceutical formulations.

Poster ID: T- ICRA-255

Gelatin stabilized silver nanoparticles for efficient and selective detection of pioglitazone

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Subject Category: Material Science

Abstract: Metal nanoparticles capped with polymers have been employed for multiple analytical and biomedical applications. In this study, highly stable Gelatin capped silver nanoparticles (G-AgNPs) were synthesized by chemical reduction method. The characterization of G-AgNPs was performed by zeta sizer, UV–visible spectroscopy, Fourier transform infrared spectroscopy (FT-IR), Field emission scanning electron microscopy energy dispersive X-ray (FESEM-EDX), and atomic force microscopy (AFM). Moreover, G-AgNPs were employed as a chemosensor for drug piogligazone. The major strength of the proposed G-AgNPs based assay is its efficiency for on-spot detection of Pioglitazone in ecological, biological, and food samples away from a sophisticated laboratory setup. Moreover, the developed sensor has capacity to be applied as quality control of pharmaceutical formulations containing Pioglitazone.

Poster ID: T- ICRA-257

Improved Delivery of Rifaximin Using Biocompatible Polymer Modified Mesoporous Silica for Combating Gram Positive Infections

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Subject Category: Analytical Chemistry

Abstract: Biopolymer-capped nanoparticles have been employed in number of analytical and biological applications to improve biocompatibility. In this study, the antibacterial efficacy of biopolymer modified mesoporous silica nanocarriers loaded with Rifaximin as a model drug was investigated against various *Staphylococcus Aureus* bacterial strains. The sol-gel approach was employed to synthesize highly stable mesoporous silica nanocarriers (MSNs), which were subsequently amino functionalized with amino silane and cross-linked with biopolymers (CP-MSNs).The functionalization of MSNs with biopolymers (CP-MSNs) provided hydrophilic and biocompatible properties. The characterization techniques including zetasizer, atomic force microscopy (AFM), scanning electron microscopy (SEM), powder X-ray diffraction (P-XRD), Fourier transform infrared spectroscopy (FT-IR) and thermogravimetric analysis (TGA) confirmed the synthesis of CPMS-NPs and drug loaded CPMS-NPs/RI. The uptake and release of Rifaximin were influenced by nanocarriers' properties. In vitro antibacterial investigations and microscopic imaging revealed that Rifaximin administered via CP-MSNs-NPs had a consistently low minimum inhibitory concentration (MIC) compared to free Rifaximin dissolved in water. This study focuses on the potential of mesoporous silica nanocarriers to improve the efficacy of antibiotics against intracellular infections. Hence, it can be concluded that CP-MSNs formulation system was able to prevent the precipitation of Rifaximin for longer period and showed selective antibacterial activity against *S. aureus.*

Poster ID: T- ICRA-258

Synergistic Effects of Mn-Ru Co-Doping in $LaNiO_3$ Perovskite Nanoparticles for Superior Supercapacitor Performance

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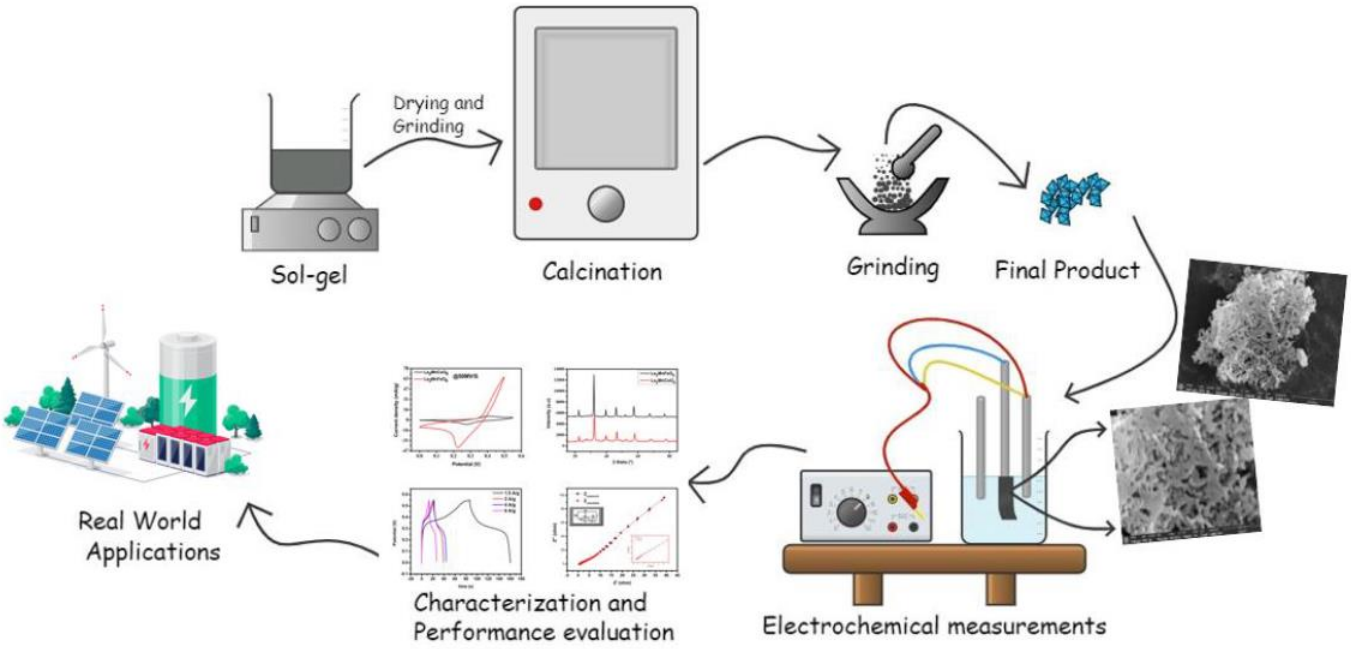
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Subject Category: Material Science

Abstract: Perovskite oxide-(POs) based nanomaterials with the formula ABX3 are gaining much consideration as supercapacitor electrode materials. highly stable structure, high oxygen vacancies, and excellent charge storage capacity make them unique when used as electrode material in supercapacitors. This work presents LaMnxRuyNiO3 (X=0.00, 0.02, 0.05, 0.08 and Y=0.00, 0.08, 0.05, 0.02) perovskite as supercapacitor electrode materials. Nanoparticles were prepared by the Sol-gel method. XRD revealed that LaMnxRuyNiO3 has hexagonal crystal structures with increased lattice constants as we increase the Ru concentration. Field emission Scanning Electron Microscopy (FESEM) confirmed the presence of uniformly distributed nanoparticles. The bandgap, with a value of 1.24 eV for La2Mn0.02Ru0.08NiO3 is determined using the Tauc's plot. The X-ray photoelectron spectroscopy analysis confirmed the presence of La3+, Ni3+, Ni2+, Ru3+, Mn2+, O2- ions on the surface of La2Mn0.02Ru0.08NiO3. The specific capacitance calculated from GCD curves are 349.10, 632.72, 545.53, and 501.18 F/g for LaMnxRuyNi1-(x+y)O3 (x = 0, 0.02, 0.05, 0.08, and y=0, 0.08, 0.05, 0.02) respectively at the current density of 4 A/g. An excellent electrochemical performance in LaMn0.02Ru0.08Ni0.9O3 is explained due to more redox reactions than pure LaNiO3 and more oxygen vacancies are present in the crystal structure tested in a 1M KOH electrolyte. La2Mn0.02Ru0.08NiO3 shows excellent energy and power density of 26.58 Wh/kg and 1139.28 W/kg. The shape of asymmetric CV curves proved that we have battery-type supercapacitors having redox reactions. Moreover, Dunn's method is employed to evaluate the percentage contribution of capacitive and diffusion Behavior.

Graphical Abstract:



Poster ID: T- ICRA-259

Highly efficient donor-acceptor configurated Nicotinaldehyde-based small molecules for NLO materials: Base-free deformylation and DFT study

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Subject Category: Organic Chemistry (Organic synthesis/ Organic Solar Cells)

Abstract: In the current study, 6-methyl nicotinaldehyde derivatives (MNP-BTMP) were synthesized by using the Suzuki cross-coupling method which involved deformylation approach. The FTIR, UV–Visible, 1HNMR and 13CNMR spectroscopic analyses were performed to validate the structures of newly synthesized derivatives. Additionally, the density functional theory (DFT) approach at M06/6–311G (d,p) level was utilized in order to calculate their optoelectronic and nonlinear optical (NLO) properties. Consequently, their HOMO-LUMO band gaps (Egap) were obtained in the range of 4.77–5.55 eV. A significant charge transfer from HOMO towards LUMO was investigated which was further supported by DOS. Moreover, they exhibited the maximum absorption wavelengths (λmax) in the UV- Vis region as 269.989–303.496 nm. Among the derivatives, BTMP was found with the least energy gap (4.77 eV) and the red-shifted absorption spectra (λmax =303.496 nm) as compared to other derivatives. Owing to these unique properties, BTMP was also found with significant linear polarizability ⟨α⟩ and hyperpolarizability (βtot and γtot) values i.e., 2.816×10− 23, 1.855 × 10− 30 and 4.553 × 10− 35 esu., respectively among the titled compounds. These findings indicated the potential use of these organic derivatives in the NLO applications.

Poster ID: T- ICRA-260

Exploring Ag_2S/ZnO Nanocomposites: Synthesis, Structural Analysis, and Multifaceted Applications

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Subject Category: Material Science

Abstract: The innovative development of Ag_2S/ZnO nanocomposite materials holds promising potential for a wide range of applications, leveraging the unique properties of zinc and silver. This study explores the synthesis and characterization of Ag_2S/ZnO nanocomposites and evaluates their effectiveness in various domains such as agriculture, water and air disinfection, antimicrobial agents, and photocatalysis. Ag_2S/ZnO nanocomposites were created using a simple and cost-effective chemical method to ensure sustainability and uniform distribution of Ag_2S and ZnO. Characterization techniques such as Particle Size Analysis (PSA), Scanning Electron Microscopy (SEM), Energy-Dispersive X-ray spectroscopy (EDS), Fourier Transform Infra-Red Spectroscopy (FTIR), X-ray diffraction (XRD), and TGA were employed to analyze the size, morphology, crystalline structure, crystallinity, and stability of the synthesized Ag_2S/ZnO nanocomposite. The results showed a well-defined and stable structure, which opens possibilities for further applications. Spectrophotometric analysis revealed a significant ionic release of zinc, accounting for 47% of the total composition in the Ag_2S/ZnO nanocomposite. In agriculture, Ag_2S/ZnO nanocomposites can enhance crop protection by effectively inhibiting the growth of harmful pathogens, thereby promoting healthier plant growth. In water and air disinfection, these nanocomposites exhibit potent antimicrobial properties, ensuring the eradication of bacteria, viruses, and other pathogens, and contributing to safer and cleaner environments. Moreover, the photocatalytic properties of Ag_2S/ZnO nanocomposites make them highly efficient in degrading organic pollutants under light irradiation, offering a sustainable solution for environmental remediation. The incorporation of silver and zinc not only enhances the functional properties of the nanocomposites but also addresses the increasing demand for these elements in advanced technological applications. This study explores the synthesis, characterization, and potential applications of Ag_2S/ZnO nanocomposites, highlighting their multifunctional capabilities and potential to revolutionize various industrial and environmental processes.

KEYWORDS: Ag_2S/ZnO; nanocomposite; chemical method; Spectrophotometric analysis; antimicrobial; environmental sustainability

Poster ID: T- ICRA-261

Developing MWCNTs/PANI Nanocomposite Electrolytes for Quasi Solid State Dye Sensitized Solar Cell

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Subject Category: Organic Chemistry / Material Science / Polymer Chemistry / Sustainable Energy

Abstract: This study aims to develop efficient DSSC for IoT devices through in-situ polymerization process and produced MWCNTs/PANI nanocomposites. SEM, XRD and FTIR revealed the morphological features, crystal structure, and constituent coordination. Quasi-solid-state electrolytes, comprising MWCNTs/PANI nanocomposites, KI and PC showed varying ionic conductivity based on MWCNTs content and salt concentration. The optimal electrolyte, with 4% MWCNTs/PANI and 0.5 g salt concentration, exhibited a higher ionic conductivity of 197.4 µS/cm and the photovoltaic efficiency of this cell is 2.7%. However, decreased performance was observed at 6% and 8% MWCNTs, potentially due to agglomeration at higher MWCNTs concentrations.

Poster ID: T- ICRA-262

Hydrothermal Synthesis of Bimetallic Nickel Cadmium Sulfides and Their Application as Electrode Material In Electrochemical Water Oxidation

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Subject Category: Electrochemistry

Abstract: Bimetallic transition metal sulfides with nanostructured features have been proposed as efficient electrocatalysts for various energy storage and conversion technologies. Among different energy storage technologies, the electrochemical water splitting is a process, in which water molecule is divided into molecules of oxygen and hydrogen with the help of electricity. Bimetallic nickel cadmium sulfide (NiCdS) was synthesized by hydrothermal method as a potential electrocatalyst for oxygen evolution reaction (OER) in electrochemical water splitting. By varying precursor concentrations, the surface morphology of the composite was tailored. The material's properties were characterized using XRD, SEM, EDX, and FTIR The prepared material was used as anode in electrochemical water splitting. The NiCdS electrode demonstrated exceptional OER performance while operating under ideal conditions. It required an overpotential of 420 mV to oxidise water at a current density of 3 mA cm^{-2} and of 490 mV to oxidise water at a current density of 5 mA cm^{-2} in 1.0 M KOH, with a Tafel slope of 274.47 mV dec^{-1}. The enhanced electrocatalytic activity is attributed to the synergistic effects of the two metals and their modified structure. This work presents a facile synthesis route for highly active NiCdS, suitable for electrochemical water splitting and related applications.

Key words: NiCdS; Hydrothermal method; Electrochemical water splitting; Oxygen evolution reaction

Poster ID: T- ICRA-263

Synthesis and thermoelectric properties of Barium doped Ca3Co4O9

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Subject Category: Material Science

Abstract: The dissipation of energy in the form of heat in automobiles, aircraft turbines, power plants, and factories is hardly recovered since there are several dissipative sources spread in the earth. This dissipation accounts for about 70% of the primary energy produced. To minimize this problem, thermal-electric energy converters can be used to partially recover this energy and increase the efficiency of the system. Thus, it is possible to have a thermoelectric generator (TEG) made up of semiconducting materials, fitting perfectly into this scenario. The classic thermoelectric generators are produced from intermetallic compounds, such as Lead and Bismuth Tellurides, and silicon- germanium. The extent of their application is limited by some factors: decomposition at relatively low temperatures, low availability of precursors, high toxicity, and cost. Therefore, it has been preferred to use thermoelectric oxides to act as charge carriers, because they avoid most of the intermetallic compound limitations. The barium doped thermoelectric bulk samples were prepared via sol-gel method. Thermoelectric properties were discussed through Seebeck coefficient, electrical and thermal conductivity. In addition, X Ray Diffraction (XRD) and Scanning Electron Microscopy (SEM) were carried out to analyze the phase and surface morphology. The analysis of the bulk samples through X-ray diffraction revealed that the samples exhibited single phase. The electrical conductivity, Seebeck coefficient, and thermal conductivity of the materials were systematically measured. Tuning the thermoelectric characteristics of Ba-doped bulk samples resulted in a notable enhancement of the dimensionless figure of merit (zT), reaching $1.35 \times 10{-}12$ at room temperature. This improvement is significant when compared to the un-doped sample, which exhibited a figure of merit of $1.16 \times 10{-}12$. These findings underscore the effectiveness of the sol-gel method in synthesizing Ba-doped thermoelectric materials with enhanced thermoelectric properties, paving the way for improved energy conversion efficiency in diverse applications.

Poster ID: T- ICRA-264

Structure and Thermoelectric Properties of Hydrothermally Grown Bi2Te3-xSex Nanocrystals

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Subject Category: Material Science

Abstract: The large quantity of waste heat generated at intermediate and low temperatures by industry and automotive exhaust requires the advancement of power generation utilising thermoelectric (TE) materials. Bi- and Te-based materials are often utilised in close proximity to ambient temperature. The hydrothermal technique was used to successfully synthesise nanostructured thermoelectric Bi2Te3-xSex (x=0, 0.4, 0.8, 1.2, and 1.6) nanocrystals. By manipulating the concentration of NaOH, which has a significant impact on the synthesis mechanism of Bi2Te3, the particle size and morphology can be regulated. We investigated the structures and morphologies of the synthesized materials by using XRD, SEM, EDX, and Raman spectroscopy. XRD result indicates that the crystal structure exhibits a hexagonal geometry, with a preferred growth along the (015) plane. Additionally, the positions of the diffraction peaks gradually move towards the lower angle side as the Se concentration increased, which is attributed to the substitution of Te atoms by Se atoms in the crystal structure, SEM results indicated that the size of the nanocrystals decreased with an increase in the Se concentration. The power factor of the best performing material, Bi2Te1.4Se1.6, was enhanced to 156 μWm-1°C−2 at 350 °C (the highest reported for Bi2Te3-xSex nanocrystals up to now), compared to the undoped sample (23 μWm-1°C−2), indicating its potential for developing novel materials and devices for TE applications. Bi2Te3-xSex nanocrystals are expected to be a developing material for applications in TE power generation.

Poster ID: T- ICRA-265

Development of Polymer Gel Electrolyte; A Sustainable Material for Third Generation Dye-Sensitized Solar Cells (DSSCs)

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Subject Category: Analytical Chemistry / Material Science / Renewable Energy

Abstract: In DSSC research, employing conductive polymer blend gel electrolytes addresses liquid electrolyte instability, improving electrode adhesion and minimizing leakage for prolonged stability. Our study blended PVDF and PVP, with optimal cohesiveness at PVP concentrations above 40%, and a favorable PVDF-PVP weight ratio of 50%. The resulting electrolyte achieved peak conductivity of 2.33 $mScm^{-1}$, applied in DSSC fabrication. DSSCs with PVDF & PVP polymer blend gel electrolytes demonstrated enhanced durability and 5.43% efficiency, showcasing a promising advancement in solar cell technology. Over two weeks, these DSSCs maintained consistent performance, unlike conventional cells, which experienced a significant decline in both short-circuit current (approx. 36%) and open-circuit voltage (around 33.33%).

Poster ID: T- ICRA-266

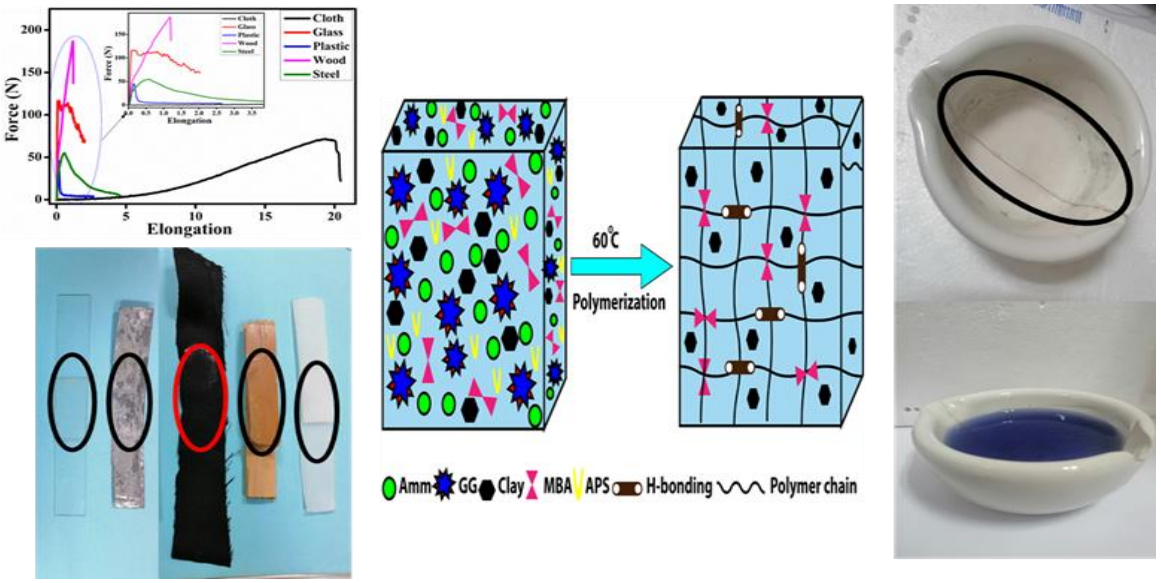
Adhesion tuning by Molsive clay in Guar gum based polymer hydrogels

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Subject Category: Physical Chemistry

Abstract: This study is mainly focused on explaining the effect of the concentration of Molsive clay on the adhesive performance of the synthesized hydrogel. The hydrogels were synthesized by free radical addition polymerization. Adhesive hydrogels have attracted significant interest for various applications such as tissue engineering, wound dressing, crack propagations, and solid surface repairing. The adhering behavior of hydrogels strongly depends on interactions between polymer chains and surfaces. Analysis of rheological data showed an increase in adhesion strength order 4.580>3.885>3.824 kPa with the addition of 0.07, 0.05, and 0.025 g Molsive clay in hydrogels compared to neat sample having 0.286 kPa adhesion strength. Temperature effect on the adhesion performance was also studied and it was found that the prepared hydrogels slightly decrease the adhesion strength at high temperature but still can be utilized for adhesion. For practical application, the hydrogels were used as a binder for solid surfaces made of plastic, wood, glass, cloth, steel, and ceramics. The materials were found effective towards all surfaces but more towards wood surfaces and ceramics. The underwater adhesion also confirms the suitability and practical adaptability of the prepared hydrogels as a binder for different surfaces in the future. Novelty of the present work is that the adhesive hydrogels work best at low as well as high temperature and also works under the water.

Graphical Abstract:



Poster ID: T- ICRA-267

Multifunctional stability of malonic acid bridged conductive hydrogel for durable electronic devices

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Subject Category: Physical Chemistry

Abstract: Flexible and strain-sensing smart materials have significant attention from researchers due to their potential use in human motion detection, soft robotics, epidermis sensors, energy storage devices, etc. However, low mechanical strength, low range sensitivity, high time response, and low anti-fatigue resistance have hampered the application of previously fabricated materials. Herein, malonic acid (MA) reinforced hydrogel was prepared through one pot-free radical polymerization, in which MA makes a bridge by connecting hydrophobically associated polyacrylamide (PAmm) and polydodecyl methacrylate (PDDMA) through physical cross-linking. Ethyl-hexadecyl dimethyl ammonium bromide (EHDDAB), a cationic surfactant is used to ensure the formation of micelles. The micelles and polymer chains are bridged via interactions of electrostatic charge enhanced by the dicarboxylic groups present on MA molecules. Notable mechanical strength was observed for MA4 with 2102% strain, 2.36MPa stress, and excellent cyclic stability. At 500% strain, the hydrogel suggests a significant sensitivity to tensile strain, as indicated by its gauge factor of 6.9 and with fast response recovery time. Meanwhile, an ionic conductivity after the addition of LiCl was calculated as 0.20S/m. Furthermore, practical applications were observed through the detection of different human motions like finger bending, wrist movement, elbow and knee movements. Similarly, small physiological larynx vibration was detected by speaking, coughing, and drinking water. The hydrogel as an electronic pen showed responses to multiple languages both in speaking and writing. The MA-regulated hydrogels show the possibility for use as flexible materials for many different applications, including flexible touch screens, biomedical monitoring, and soft robotic devices.

Poster ID: T- ICRA-268

Synthesis, Characterization and Biological Applications of Novel Uracil Derivatives

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Subject Category: Organic Chemistry

Abstract: Numerous arylidine derivatives of uracil have been synthesized by stirring EAA with urea and aromatic aldehyde at ambient temperature using green approaches by employing biocatalyst i.e., crushed garlic cloves in the presence of water as solvent. Melting points were determined and were used uncorrected. Spectroscopic characterization was done to confirm the structures of the designed series of compounds. Additionally, all the synthesized compounds were screened for their biological activities including antioxidant activity. Among them 6-Methyl-(3-nitrophenylfuran) uracil exhibited the highest antioxidant activity.

Poster ID: T- ICRA-269

Artificial Intelligence for in-silico studies of isolated compounds of *Calotropis procera*

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Subject Category: Natural Organic Chemistry

Abstract: The process of discovering and developing drugs requires the use of artificial intelligence. The traditional drug discovery methods are expensive and time - consuming, whereas the use of software helps in exploring lead compound interactions with their target proteins. The development of drugs from natural sources is frequent. Therefore, the present study was aimed to determine first time the in silico potential of isolated phytochemicals from *Calotropis procera* against different diabetic target enzymes. The lupeol and apigenin 4`-O-α-L- rhamnopyranoside were isolated from *Calotropis procera*. They were investigated against different targeted diabetic enzymes like aldose reductase, α-amylase 1 and α-glucosidase using 1-Click Docking Mcule and UCSF Chimera to investigate their physicochemical properties and binding modes. Both compounds exhibited drug likeness properties, good docking scores and stronger interactions with enzymes through binding pocket residues. First time in silico and docking analysis of compounds suggest that they are good candidate to inhibit diabetes targets enzymes and may be helpful in the treatment of diabetes. They may also be suitable candidates for further formulation studies.

Poster ID: T- ICRA-270

Extraction of raw carbon from sugarcane bagasse for synthesis of graphene based trimetallic nanocomposite: biological study

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Subject Category: Inorganic Chemistry / Analytical Chemistry

Abstract: In this research Hummers method was adopted to make graphene oxide from raw carbon which was obtained from sugarcane bagasse. The morphology of the synthesized samples were examined through Scanning Electron Microscopy (SEM). The Elemental analysis and percentage purity of the samples were checked through Energy Dispersive X-Ray (EDX). The functional groups identification of prepared samples was made through Fourier Transform Infrared Spectroscopy (FTIR). The ABTS assay was followed to check the antioxidant activity of synthesized samples. The antioxidant activity was evaluated at different concentration of the samples in the reaction. Results showed that graphene based trimetallic nano composite when calcined at 400 ˚C showed maximum antioxidant potential as compared to graphene and trimetallic composite.

Poster ID: T- ICRA-271

Preparation of *Ocimum tenuiflorum* essential oil loaded bioactive nano-formulation

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Subject Category: Analytical Chemistry / Material Science

Abstract: Essential oil based nano-formulations are need of the time as the efficacy and stability of such formulations have valuable applications in the field of medicine and cosmetic research. Hence, in present study, the essential oil from *Ocimum tenuiflorum* plant was extracted by hydro-distillation method. The compositional ingredients used to formulate nanoemulsion by low energy spontaneous method. The physical characteristics of emulsion were evaluated by analyzing pH, viscosity, conductivity and surface tension. The droplet size and zeta potential of the nanoemulsion was analyzed by using a Zetasizer. The essential oil was then added in the prepared nanoemulsion in different concentrations, resulting in the formation of 5 samples (F_1 to F_5). The samples showed significant antioxidant activity as tested by DPPH method. Mosquitoes' repellency test of these essential oil loaded nanoemulsions showed that the activity increase with increasing e. oil concentration with 100 % repellency for F5. The results of present study suggested that the prepared formulation can be used as an important ingredient to formulate mosquito repellent products.

Poster ID: T- ICRA-272

Tailored Multilayer Zinc Based Inorganic-Organic Hybrid Material (ZnO@HKUST-1) as an Efficient Anode for High Performance Lithium Ion Batteries

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Subject Category Analytical Chemistry

Abstract: Metal oxide (MO_x) modified with metal organic frameworks (MOFs) are widely used in lithium ion batteries (LIBs) due to their adequacy, low cost, large specific capacity, high theoretical capacity, cyclic stability and superior electrochemical performance. In this work, we synthesized zinc oxide (ZnO) nanoparticles via hydrothermal method and modified with MOF (HKUST-1/MOF-199) using Layer by Layer coating (**LBL-C-10**) method. The synthesized **ZnHK-1** (ZnO@HKUST-1) composite was characterized by EIS and then further utilized as active anodic material for LIBs. The charge transfer resistance (R_{ct}) for ZnHK-1 (1.42kΩ) calculated from Nyquist plot is smaller comparatively to ZnO (43.1KΩ) and HK-1 (14.64 KΩ). The synthesized ZnHK-1 composite exhibit high initial charge (1300 mAh/g) and discharge capacity (1640 mAh/g) with 79.2% initial columbic (C.E%) efficiency and 1158 mAh/g reversible discharge capacity. Galvanostatic charge/discharge capacity 2nd, 5th & 100th cycle is 1160/1420 mAh/g, 1180/1200 mAh/g and 1133/1158 with 81.6 % and 98.33 % and 97.8% columbic efficiency, respectively. Comparatively ZnO and HK-1 show only 72.2 % and 72.7% initial columbic efficiency and 231mAh/g and 692mAh/g reversible capacity at stable current density (100mA/g). The 97.8% C.E and 70.6% retaining capacity of ZnHK-1 at 100 mA/g current density after 100 cycles, indicates that the reported method is efficient way to improve the electrochemical stability of lithium ion batteries.

Poster ID: T- ICRA-273

The most recent developments in nanomaterials-assisted electrochemical sensors for bisphenol A analysis

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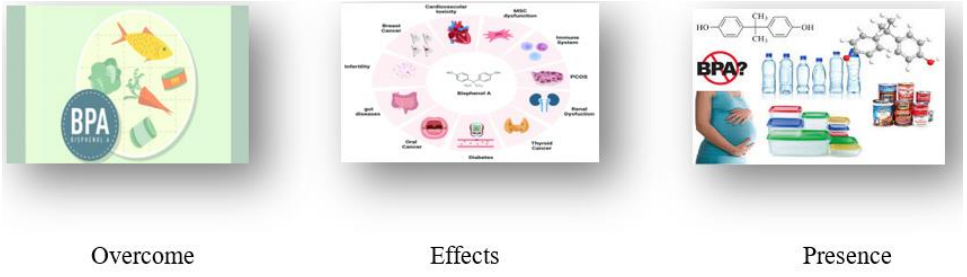
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Subject Category: Organic Chemistry / Physical Chemistry

Abstract: Advances in electrochemical sensor technology have revolutionized the detection of bisphenol A (BPA), a common industrial chemical known to have potential endocrine disrupting effects. This abstract summarizes recent research on the integration of nanomaterials to improve the performance of electrochemical sensors customized for BPA analysis. Materials like metal nanoparticles, carbon nanotubes, and graphene have emerged as critical components in enhancing the stability, sensitivity, and selectivity of sensors. These materials allow for the creation of complex 3D Nano architectures that greatly increase the surface area available for the immobilization of biomolecular recognition elements, thereby augmenting the analytical capabilities of the sensors.

Carbon nanomaterial-based sensors have shown increased sensitivity to BPA. Nanocomposite-based sensors, which combine nanoparticles with polymers, address issues like nanoparticle dispersion and aggregation, improving sensor reliability. Biomolecule-based biosensors, which use particular biorecognition components like aptamers and enzymes, achieve remarkable selectivity in detecting BPA in complex matrices. Case studies highlight the efficacy of novel sensor designs, such as Vertical Mesoporous Silica-Nano channel Film (VMSF) sensors, which are notable for their strong antifouling qualities and high sensitivity, making them ideal for direct BPA detection in difficult environmental samples. Matrix effects and electrode passivation are difficulties in BPA detection, which has led to continued research into more straightforward electrode designs with improved antifouling properties.

Graphical Abstract:



Poster ID: T- ICRA-274

Extraction and biological evaluation of *Citrus sinensis* peels essential oil

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Subject Category: Analytical Chemistry

Abstract: Essential oil is an excellent source of many bioactive compounds due to which it has vide applications such as antimicrobial, antioxidant, antifungal and anti-inflammatory etc. The present research work was aimed to extract essential oil of *Citrus sinensis* peels through the hydrodistillation. For this purpose, the peels collected from local market of Okara, were hydrodistilled to obtain pure essential oil. The yield % w/v obtained was 0.44% which depicted slight variation due to extraction method, climate change, region and soil. Biological activities like antibacterial, hemolytic and antioxidant pure of the pure e. oil were analyzed and it was concluded that e. oil was significantly active. The essential oil was chemically characterized by GC-MS technique. The obtained results have shown that they can be helpful for the preparation of the safe antioxidant agents from the *Citrus sinensis* peel's oil in the food and medicine industries.

Poster ID: T- ICRA-275

Electrochemical sensing of Hazardous Para-Nitrophenol using Modified Ceria based Nanocomposites

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Subject Category: Material Science

Abstract: Hazardous nitrophenols threaten aquatic life, plants, and animals due to NP's poisonous nature. Among them, 4-NP is a top-rated hazardous environmental pollutant and secondary explosive chemical, has been detected by promising Ceria-based nanostructures. CeO_2 nanostructures and modified CeO_2 based nanostructures and reduced graphene oxide (rGO) were synthesized by facile hydrothermal method where an improved Hummer's method was used for graphene oxide synthesis. All synthesized samples were characterized by XRD, SEM, and EDX, which confirmed cubic fluorite structure, phase purity, nanorods morphology, purity of elemental composition and formation of defects of synthesized materials. Electrochemical studies of these materials which include CV, DPV, and EIS were investigated. After deep consideration and with analysis of all these physicochemical and electrochemical characterization techniques, it is confirmed that doped CeO_2/rGO nanocomposite appeared to be the best electrocatalyst for hazardous 4-NP sensing among all the prepared catalysts by showing lowest limit of detection (0.0015 mM), lowest limit of quantification (0.0046 mM) and highest sensitivity (1.33 $mA.mM^{-1}cm^{-2}$) The overall impact of the present study is that the ceria-based electrocatalysts have been prepared by the facile hydrothermal method and are found to be active sensor electrodes for electrochemical sensing of hazardous 4-nitrophenol.

Poster ID: T- ICRA-279

Exploring Electrolyte Additives for the Next Generation Lithium Ion Nanobatteries *via* Molecular Dynamics Simulations

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Subject Category: Material Science

Abstract: Rechargeable lithium-ion batteries (LIBs) have become the cornerstone of modern energy storage solutions, capturing the attention of researchers globally. However, some rooms are still available to improve their cycling and storage capacity. A critical factor influencing LIBs performance is the choice of electrolytes and their additives, which can make a substantial difference. Modern electrolyte additives have captivated the significant attention from researchers due to their role in forming an effective solid electrolyte interphase (SEI) when silicon is used as the anode material. A well-chosen solvent with these additives can greatly enhance the solvation and diffusivity of lithium ions. Fluoroethylene carbonate (FEC) and Vinylene carbonate (VC) are extensively used additives for an increased lifespan of Li-ion batteries (LIBs). This research employs molecular dynamics simulations to explore the effects of these additives on a silicon anode within a nanobattery model. We investigated the physicochemical, structural, and dynamical properties of six distinct electrolyte systems. Additionally, we examined the transport properties of LiPF6 in these systems to determine the role of FEC and VC. Utilizing our simulation findings, we developed a comprehensive nanobattery model incorporating the most effective electrolyte system, subsequently evaluating its properties to showcase the improved performance of the LIBs model.

Poster ID: T- ICRA-280

Development of Hydrophobic Nanoparticles Coating for Improved Self Cleaning Function of Photovoltaic Applications

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Subject Category: Material Science

Abstract: Silica nanoparticles have unique optical and electronic properties due to which they are used in wide range of applications. They allow precise control over size and have refractive index of 1.4 which enables high light transmission. They can easily be functionalized to change properties. In this research we synthesized SNPs of different sizes by Stöber process using TEOS as a precursor, coated them on FTO glass substrate by nebulizer spray coating technique and treated with HMDS to increase their hydrophobicity; for the purpose of self-cleaning coating for photovoltaic applications. We performed different characterization techniques including, scanning electron microscopy(SEM), particles size analysis (PSA), Fourier transformation infrared spectroscopy(FTIR), UV-Visible Spectroscopy and contact angle measurement at different stages of our research. The coating became hydrophobic by exposing silica-coated glass to an HMDS solution. The contact angle of average 140˚ of treated SNPs was achieved while also increasing the transmittance of the film. The hydrophobic antireflective film we prepared can restrain the adsorption of the moisture from the air and will improve the optical performance of the optical system efficiently.

Poster ID: T- ICRA-281

Response surface methodological study for removal of Methylene Blue Dye from water using Iron oxide-based *Acacia nilotica* wood bio composite to investigate equilibrium parameters

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Subject Category: Analytical Chemistry

Abstract: Heamatite based nanocomposites have gained importance due to their low cost utilization in water pollutants treatment. Anthropogenic sources have increased water pollution; to combat this low cost and feasible green approach is necessary. Composites in combination with *Acacia nilotica* wood biomass using iron chloride tri hydrate solution,for the withdrawl of methylene blue dye from the aqueous solution enhanced pollutants uplift efficiency in less time and cost. Nanomaterial was characterized using SEM, FTIR, and XRD techniques. The impacts of dye concentration, contact time, photocatalytic degradation, adsorbent dose, and temperature were explored using batch adsorption. It was unearthed that the percentage removal of the dye was expanded up to 96% as the initial dye concentration was changed from 20mg/L to 100mg/L. The investigation of equilibrium data showed that the results best fitted Freundlich isotherm which proposed that multi-layered adsorption occurred on adsorbent's surface. The contact time study revealed that maximum dye removal of 97% was achieved at a contact time of 120 minutes. The kinetic study of experimental data revealed that adsorption process was following pseudo second order kinetics which means chemisorption occurred. The experimental results also showed good agreement with intra-particle diffusion model suggesting the diffusion of adsorbate in inner layers of adsorbent's surface. With change in adsorbent's dose from 0.1g to 0.5g, percentage removal was increased up to 96%. The photocatalytic activity of adsorbent at 120 minutes showed maximum dye removal with a percentage of 97. The thermodynamics study of experimental data disclosed that process was endothermic, spontaneous, and feasible in nature. Response surface methodology and central composite design were also applied to experimental data to predict the effect of multiple parameters revealing good correspondence between these parameters.

Poster ID: T- ICRA-282

Laser-Induced Graphene (LIG) based Efficient Electrodes for Electrochemical Energy Storage

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Subject Category: Material Science

Abstract: Clean energy is efficiently replacing fossil fuel energy for future electrochemical applications. Research in developing and fabricating low-cost materials providing high output and efficiency has been a main consideration hitherto. Polyimide's (PI) conversion to laser-induced graphene (LIG) using laser scribing has attracted much attention due to the exceptional properties of LIG like high porosity, high surface area, high thermal stability, and excellent conductivity, and LIG has replaced many materials in electrochemical applications. In this research, we have fabricated graphene-based nanocomposites LIG/TiO2 through the deposition of Titanium dioxide (TiO2) powder on LIG using the Electrophoretic deposition (EPD) method having 12% Titanium (Ti) deposited and through Aerosol Method having 77% Ti deposited. An increase in the capacitive response of LIG has been observed after deposition of high concentration of TiO2 since it has high dielectric constant. Field Emission Scanning Electron Microscopy (FESEM) has been performed to study morphology, size of particles, and indicatory peaks of all desired elements are observed by Energy Dispersive Spectroscopy (EDS). Furthermore, this nanocomposite has been investigated to study different electrochemical properties for energy storage purpose. Various characterization techniques like Electrochemical Impedance Spectroscopy (EIS), Cyclic Voltammetry (CV), and Linear Sweep Voltammetry (LSV) etc. have also been used to study a significant difference in the properties of PI, LIG, and LIG/TiO2 nanocomposite. The LIG showed an imaginary impedance of 410 -Z" /W cm while the nanocomposite showed an imaginary impedance of 650 -Z" /W cm which shows an increase in the capacitive behavior of nanocomposite. Study of this property difference has suggested LIG/TiO2 to be a promising candidate for various electrochemical applications since it has shown a better electrochemical response for energy storage as compared to LIG.

Keywords: Polyimide (PI), Laser-Induced Graphene (LIG), LIG/TiO2, Efficient Electrodes, Electrochemical Applications

Poster ID: T- ICRA-283

Novel amide comprising dinitro and diamines for potential DNA binding and antioxidant evaluation

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Subject Category: Analytical Chemistry

Abstract: Novel dinitro compound, namely, A1 [N,N′-{Oxo bis(4,1-phenylene)}bis(4-nitrobenzamide)] and A2 [N,N′-{methylene bis(4,1-phenylene)}bis(4-nitrobenzamide)] were synthesized and reduced to their respective diamines B1 [N, N′- [oxo bis(4,1-phenylene] bis(4-aminobenzamide)] and B2 [N,N′- [methylene bis(4,1-phenylene)]bis(4-aminobenzamide)]in the presence of 10% Pd charcoal under inert conditions. The synthesized compounds were characterized by Fourier transform (FT-IR) and Nuclear magnetic resonance (NMR), and the solid state structure of dinitro compound was elucidated by single-crystal X-ray crystallography. A1 was crystallized in the triclinic system with P -1 space group, and A2 was crystallized in monoclinic system with space group C2. DFT calculations were performed for structure-property relationship information of synthesized compounds. The mode of interaction of these synthesized compounds with SS-DNA was examined using standard Ultra-Violet spectroscopy, cyclic voltammetry and molecular docking techniques. These studies confirmed the groove binding mode of examined molecules with DNA. Additionally, the DPPH (2,2-diphenyl-1-picrylhydrazyl) scavenging activity of synthesized compounds was checked to evaluate their antioxidant behaviour. Synthesized diamines showed more DPPH scavenging activity than their respective dinitro compounds.

Poster ID: T- ICRA-285

Dynamics and Kinetics of Ligand Binding to New Delhi Metallo-β-Lactamase (NDM-1) – A Molecular Dynamics Simulation Study

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Subject Category: Organic Chemistry

Abstract: New Delhi Metallo-β-lactamase (NDM-1) belongs to an emergent class of β-lactamases used by bacteria for resisting the activity of antibiotics. Major producers of NDM-1 are K. pneumoniae and E. coli which are well known for their pathogenicity in causing most nosocomial infections. As of now, there is no FDA-approved inhibitor against NDM-1, therefore, the cure and treatment of Hospital-acquired infections (HAIs) are becoming a global challenge. Co-administration of antibiotics with inhibitors of β-lactamases has been attempted which demonstrated promising results for instance a few FDA-approved, commercially available drugs such as clavulanic plus amoxicillin and sulbactam plus ampicillin etc. Nevertheless, they showed potential inhibitory effects only against Serine-based β-lactamases. We have performed an all-atom molecular dynamics (MD) simulation to understand the mechanism of inhibition of different inhibitors evaluated experimentally but lack an atomic-level description under dynamical conditions. Moreover, as kinetic parameters are considered to be the best indicators of a drug's in-vivo efficacy, we have evaluated the kinetic parameters of the inhibitors by employing biased MD simulations to understand the binding and unbinding mechanisms. Our study aims to possibly assist the mechanism-based development of such inhibitors for their enhanced in-vitro and in-vivo activities against NDM-1.

Poster ID: T- ICRA-286

Remediation Of Lead-Contaminated Water Using Iron-Zinc Bimetallic Nanoparticles Synthesized By Azadirachta Indica Leaf Extract

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Subject Category: Material Science (Environmental Chemistry)

Abstract: Advances in nanotechnology have created effective nanoparticle-based technologies for environmental remediation. The present study is aimed at synthesizing *Azadirachta indica*-mediated Iron Zinc nanoparticles (FeZn NPs). 0.1M ferrous sulfate heptahydrate and 0.1M zinc sulfate heptahydrate were used as precursor salts. The morphological and optical features were evaluated by using various approaches such as UV-Vis, XRD, and FTIR. The effect of two parameters (initial lead concentration and adsorbent dosage) was also evaluated on lead removing ability of FeZn-NPs. The results showed that 0.06 g of FeZn NPs showed the highest adsorption efficiency of 80.3%, at the initial lead concentration of 150 mg/L. Further, the adsorption efficiency decreased as the concentration of lead increased. The optimal dose of FeZn NPs for removal of lead from 40 mL of 100 mg/L lead ion concentrated water, under pH 6, was 0.06 g. Beyond this value, even an increased adsorbent amount did not produce significant results. In conclusion, green-mediated FeZn-NPs are safer and effective in environmental remediation applications.

Poster ID: T- ICRA-287

Thermodynamic and Kinetic Studies on the Adsorption of Amoxicillin from Aqueous Solution using Graphene Oxide

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Subject Category: Physical Chemistry

Abstract: The focus of the present study was to develop an efficient adsorbent for the removal of Amoxicillin, which is a mircopollutant, from aqueous solutions. For this purpose, Graphene Oxide, derived from Graphite Powder, using modified Hummer's method was used which showed favorable adsorption capacity. Different analytical techniques such as XRD, SEGM, FTIR, TGA and Point of Zero Charge (PZC) were used for the characterization of the adsorbent. The PZC of Graphene Oxide was determined by salt addition method in the presence of background electrolyte (NaNO3) at 298±1K. The PZC values were found to be 2±0.1 for Graphene Oxide. A comprehensive kinetic study of Amoxicillin adsorption onto Graphene Oxide was conducted at 278,298 and 308K. The pH was maintained at 2±0.1. The different kinetic models were applied to the experimental data to probe into the mechanism of amoxicillin adsorption. The present experimental data was best described by pseudo second and pseudo first order kinetics. Adsorption experiments were carried out using batch adsorption techniques. The removal of amoxicillin from aqueous solution was carried out as a function of pH, concentration and temperature. The effect of pH showed maximum adsorption at pH 2. The range of pH investigated was between 2 and 11. The adsorption was found to decrease with increase in pH. The adsorption capacities of Graphene Oxide were substantially increased by increasing concentration while a decrease in Amoxicillin adsorption was noted at lower temperatures. The Langmuir, Freundlich and DR models were used to interpret the experimental data. The Langmuir parameters i.e binding energy and adsorption maxima were found to be temperature dependent. The mean free energy was calculated from DR model confirmed, that the process of adsorption is physisorption. The thermodynamic parameters for the adsorption process were calculated using well-known Vant Hoff equation. The ΔH values suggested that the overall process is endothermic in nature. The negative values of ΔG and positive values of ΔS proposed that the adsorption is spontaneous and thermodynamically favored.

Poster ID: T- ICRA-288

Synthesis of thieno [3, 2-b] thiophene-based donor-acceptor conjugated copolymers for the fabrication of low-voltage organic field-effect transistors

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Subject Category: Material Science

Abstract: To fabricate organic field-effect transistors (OFETs) with high electrical performance and stability by low-cost processing, it is still an important challenge to develop organic semiconductor materials that are solution-processable and environmentally stable. Similarly, dielectric engineering is a vital tool to tackle the challenge of high energy consumption by developing OFETs with low operational voltage. A new series of donor-acceptor co-polymers was synthesized, incorporating 2,5-di(thiophen-2-yl)thieno[3,2-b]thiophene (TT) as donor blocks with different substituents. These donor units were coupled with 4,7-diethynylbenzo[2,1,3]thiadiazole (BTD) acceptor unit via Sonogashira cross-coupling polymerization. These co-polymers were employed as solution-processable semiconductor layer in OFETs. Khaya gum was used as water-processable, biodegradable and environment friendly natural dielectric layer to investigate the charge transport characteristics of OFET devices at low voltage. All the devices displayed a p-type field-effect behavior with lowest average threshold voltage of -0.5 V. Among these, alkyl chain (C9H19) substituted polymer exhibited highest average saturated hole mobility (μsat) of 0.086 cm2 V-1 s-1, on/off current ratio (Ion/Ioff) of 1.0 × 103, and subthreshold swing (SS) of 425 mV dec-1. These results corroborate that the novel TTBTD polymers have promising potential for electronic applications, in particular, where tunability of the field-effect behavior is essential for performance.

Poster ID: T- ICRA-290

In vitro and in vivo antimicrobial efficacy of lithium complex against multi-drug resistant Acinetobacter baumannii

*Sidrah Asghar, Moatter Zehra and Ayaz Ahmed**

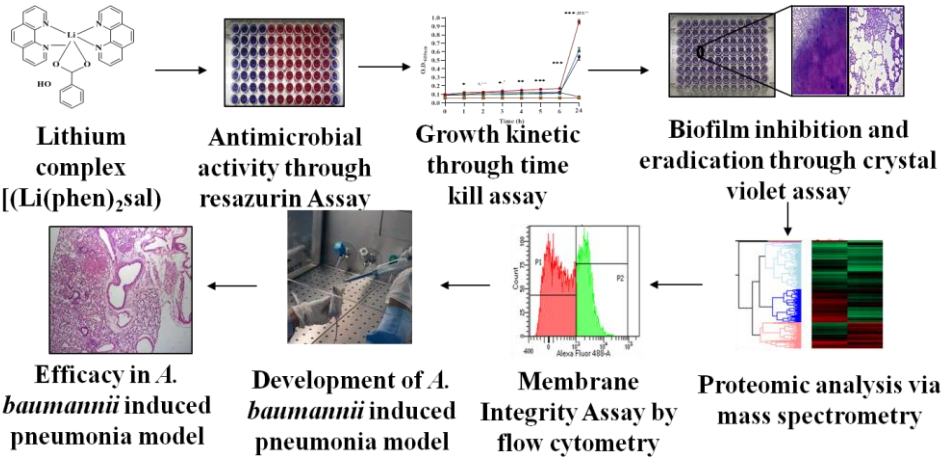
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Subject Category: Organic Chemistry

Abstract: *Acinetobacter baumannii* is a Gram-negative ESKAPE pathogen, accountable for lethal nosocomial infections, owing to high pathogenicity and multi-drug resistance (MDR). Hence, the World Health Organization (WHO) has prioritized this bacterium against which new pharmacophores are required urgently. A heterocyclic organic compound 1,10-phenanthroline is known for its ability to chelate metal ions. When it complexed with lithium ions, it forms a stable coordination complex and the presence of salicylic acid enhance this chelation or stabilize the complex further. In this study, the antimicrobial potential of lithium complex based on salicylic acid, and 1,10-phenanthroline ([Li(phen)$_2$sal]) against MDR *A. baumannii* (MDRAB) was evaluated. Moreover, *A. baumannii*-induced pneumonia model was developed and characterized to assess *in-vivo* efficacy. Lithium complex significantly inhibited the bacterial growth at 16-32 µg/ml. Proteomic analysis revealed that the compound exerted oxidative stress on *A. baumannii* which may contribute to cell rupturing. Flow cytometry confirmed its role in cytoplasmic membrane disintegration. It was observed that lithium complex ameliorated the pneumonia symptoms at non-toxic doses of 20 and 40 mg/kg as evident by the significant reduction in bacterial burden and parenchymal inflammation. These findings suggest that the lithium complex can be a potential drug candidate against MDR *A. baumannii.*

Graphical abstract:


Lithium complex [(Li(phen)2sal)
Antimicrobial activity through resazurin Assay
Growth kinetic through time kill assay
Biofilm inhibition and eradication through crystal violet assay
Proteomic analysis via mass spectrometry
Membrane Integrity Assay by flow cytometry
Development of A. baumannii induced pneumonia model
Efficacy in A. baumannii induced pneumonia model

Poster ID: T- ICRA-291

Characterization of Wax Produced from Pyrolysis of Waste Baby Diapers

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Subject Category: Organic Chemistry

Abstract: Waste baby diapers are a significant source of waste with millions of tons generated annually worldwide. This study aimed to produce wax from diapers. First pulp of waste diapers were collected from manufacturing industries then a pyrolytic reactor was designed composed a container of iron sheets with a condenser, outlet and inlet then 5 Kg feedstock was introduced in reactor. The process of pyrolysis was conducted by using wood as fuel that provided 300 °C to 350 °C temperature. The pyrolysis process yielded gases, wax (11%) and water content (10.62%). Various analytical techniques were employed including GC-MS, FTIR, and elemental analysis to investigate the composition of wax. The FTIR results showed peaks at 3300 cm^{-1} represents O-H stretching vibration, indicating the presence of -OH group. Peak at 2913 cm^{-1} was observed indicating C-H stretching vibration, typical for saturated hydrocarbons. At 1704 cm^{-1} C=O stretching vibration, characteristic of carbonyl groups. A peak at 1464 cm^{-1} indicating C-H bending vibration. A C-O stretching vibration of was observed at 1170 cm^{-1} and 1245 cm^{-1}. The presence of hydroxyl and carbonyl groups suggests that the wax may have some polar functionality, which could impact its properties and potential applications in cosmetics and shoe polish.

Poster ID: T- ICRA-292

Thermal decomposition of cotton seeds press cake over Zeolite Y, Hydrogen into useful products and its kinetic studies

Marrij Afraza

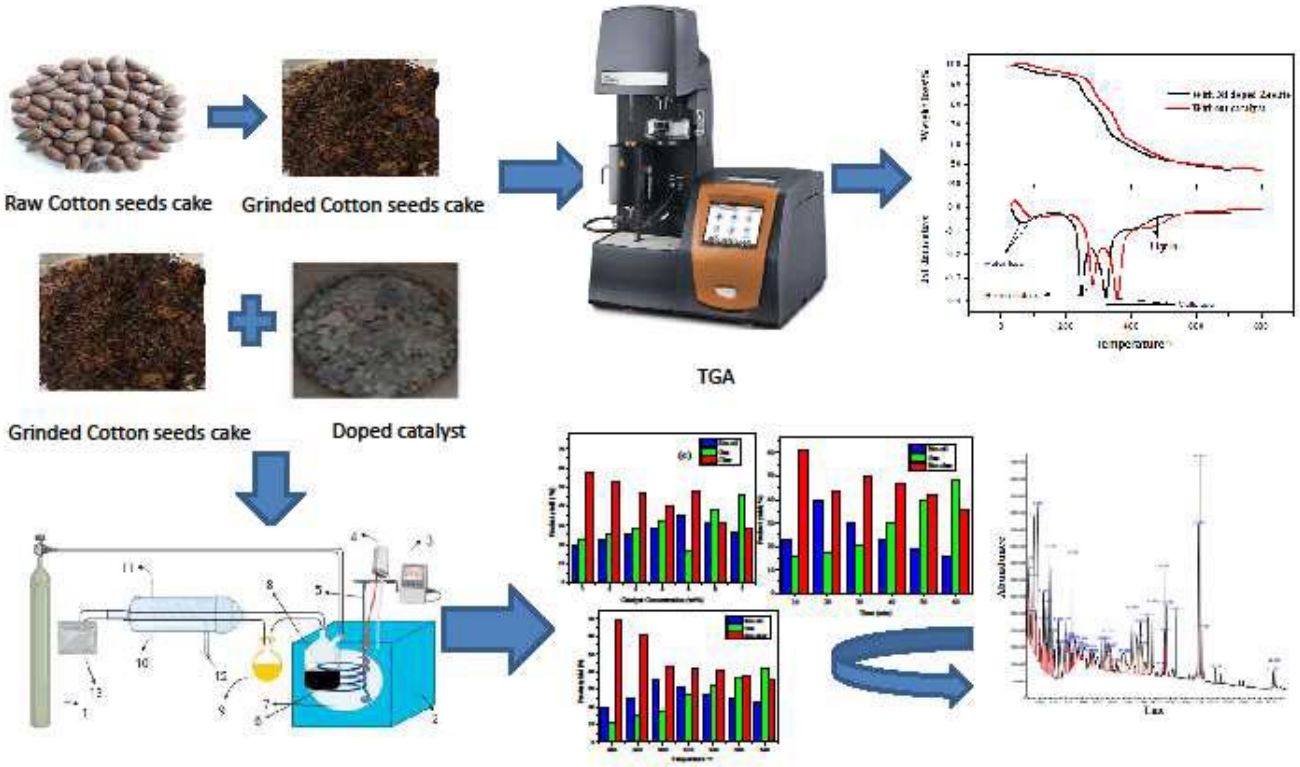
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Subject Category: Physical Chemistry

Abstract: In the recent study, our focus was the production of value added compounds from utilizing waste cotton seeds press cake through an environmental friendly approach. Initially, zeolite was loaded with nickel using wet impregnation method and characterized. The de-oiled cake was exposed to thermo gravimetric analysis (TGA) with and without nickel doped zeolite for the determination of maximum degradation point which exhibited a four-step weight loss, attributed to the evaporation of water and the degradation of hemicellulose, cellulose, and lignin. To further investigate the effects of catalysts, pyrolysis experiments followed with TGA was conducted at 25 to 800°C at different heating rates min under a nitrogen atmosphere. Kinetic parameters were determined using the Kissinger equation which indicated the lowering of activation energy and pre exponential factor with the use of nickel impregnated Zeolite. Pyrolysis experiments in the presence and absence of Ni-Zeolite were carried out to optimize reaction conditions such as time, temperature, catalyst in order to produce maximum bio-oil yield and pyrolysates were analyzed using SEM, XRD, EDX, and GC-MS. The oils obtained from the catalyzed and uncatalyzed reactions were subsequently analyzed using gas chromatography mass spectrometry (GCMS). Moreover, results indicated that catalyst enhance the quality of bio-oil and with further upgrading, the obtained oil holds promising potential for applications across various industries.

Graphical abstract:


Raw Cotton seeds cake
Grinded Cotton seeds cake
TGA
Grinded Cotton seeds cake
Doped catalyst

Poster ID: T- ICRA-293

Isolation of a New Anti-cancer Clerodane Furanoditerpenoid Glucoside and Potent Anti-Glycating Compounds from *Tinospora bakis* (A. Rich.) Miers

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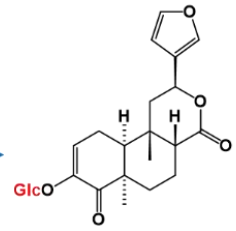
Subject Category: Organic Chemistry

Abstract: *Tinospora bakis* (A. Rich.) Miers, belonging to the family Menispermaceae, has been the central focus of this research due to its profound medicinal importance. Despite its extensive use, the active components of the plant have never been investigated. In this work, a series of furanoditerpenoids (**1–10**), one flavonoid (**11**), and one alkaloid (**12**) were isolated from *T. bakis*. Notably, one new compound was discovered and named: tinobakiside (**1**). The structural elucidation was performed using NMR, UV-Vis, FT-IR, CD spectroscopy, and mass spectrometry. Remarkably, compounds **4** and **7** exhibited a superior anti-glycation effect with IC_{50} values of 37 ± 0.48, and 25 ± 0.25 µM, outperforming established agents like rutin (IC_{50} = 69 ± 0.12 µM) and quercetin (IC_{50} = 104 ± 1.75 µM) in a lab model of protein glycation with glucose while compound **6** showed moderate activity with an IC_{50} value of 909 ± 5.86 µM. Moreover, compounds **1** and **7** displayed moderate anti-cancer activity against the MCF-7 cell line with IC_{50} values of 31.0 ± 7.70 and 36.0 ± 8.93 µM, respectively, while Doxorubicin (IC_{50} = 2.3 ± 0.2 µM) was used as standard. Cytotoxicity against the BJ-cell line revealed no cytotoxic effects for any compound. This study highlights the diverse therapeutic potential of *T. bakis* compounds as promising anti-glycating and anti-cancer drug candidates.

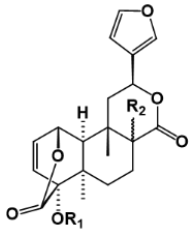
Graphical Abstract:



Tinospora bakis



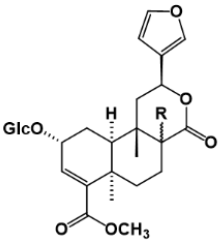
1. IC_{50} = 31.0 ± 7.70 μM$^{\Delta}$



2. R_1 = H, R_2 = β-H
3. R_1 = H, R_2 = α-H

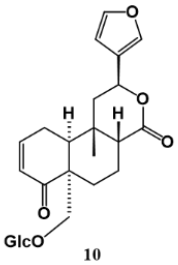
6. R_1 = β-D-glc, R_2 = β-H
IC_{50} = 909 ± 5.86 μM*

7. R_1 = β-D-glc, R_2 = α-H
IC_{50} = 36.0 ± 8.93 μM$^{\Delta}$
IC_{50} = 25 ± 0.25 μM*



4. R = α-H
IC_{50} = 37 ± 0.48 μM*

5. R = β-H
8. R = β-OH
9. R = α-OH



10

Δ Anti-cancer Activity

* Anti-glycation Activity

Poster ID: T- ICRA-294

Green synthesis of metal nanoparticles by using Mangifera indica kernel extract, their characterization and biological evaluation

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Subject Category: Analytical Chemistry.

Abstract: Nanotechnology is one of the emerging and fast growing concepts in the field of science and technology. Nanomaterials are unidimensional or multidimensional particles of nanoscale size usually 1-100 nm (1) and owing to the distinctive nature of nanomaterials as they possess unique physicochemical and biological properties, nanoparticles have been the focus of extensive research in recent years. It has been discovered that they show tremendous promise for use in pharmaceutical drug delivery systems, biosensor devices, organic catalysis along with several other chemical and medicinal applications. Nanoparticles are generally synthesized by physical or chemical methods usually by utilizing top-down and bottom-up approaches. Although effective, these approaches have several disadvantages such as use of potentially hazardous chemicals, environmental toxicity, expensive equipment, lengthy reaction cycles, high energy requirements and production of non-eco-friendly by products. In view of the disadvantages of the conventional approaches, a green synthesis methodology is becoming famous for the synthesis of nanoparticles because of its eco-friendly and cost effective nature. Green synthesis involves the use of living organisms and living systems for the synthesis of nanoparticles with defined particle size and morphology. A review of literature revealed that the use of plant materials such as extracts provide an excellent mean of synthesizing nanoparticles due to easy availability and easier handling of plant materials when compared with the other biological organisms and systems.

Several studies have reported the synthesis of metal nanoparticles by using different plant extracts such as silver (Ag-NP) nanoparticles by *Vitex negundo* L., Cobalt nanoparticles (Co-NP) by *Celosia argentea*, copper nanoparticles (Cu-NP) by *Azadirachta indica* and zinc oxide nanoparticles (ZnO-NP) by *Laurus nobilis* etc.

Mangifera indica is a plant belongs to the Ancardiaceae family, is commonly known for its very sweet fruit. Traditionally, mango is used for the treatment of several ailments including diarrhea, diabetes, cancer, toothache, urinary tract infection and some chemical constituents also exhibit hepatoprotective and anti-inflammatory activities (11). Mango kernels (seeds) also provides a plethora of chemical constituents including polyphenols which possess the potential for use in diabetes. Usually the kernels go to waste after

the utilization of fruit pulp and by utilizing the kernel for extraction of phytochemicals will provide a cost effective way getting the plant constituents.

The primary objective of the present study is to utilize the mango kernel as a cost effective source of extract for the green synthesis of metal nanoparticles of silver (Ag), copper (Cu), lead (Pb), zinc (Zn) and iron (Fe) their characterization and evaluation of anti-diabetic potential.

Poster ID: T- ICRA-295

Untargeted metabolomics analysis of gentamicin-induced tolerant colonies of Klebsiella pneumoniae

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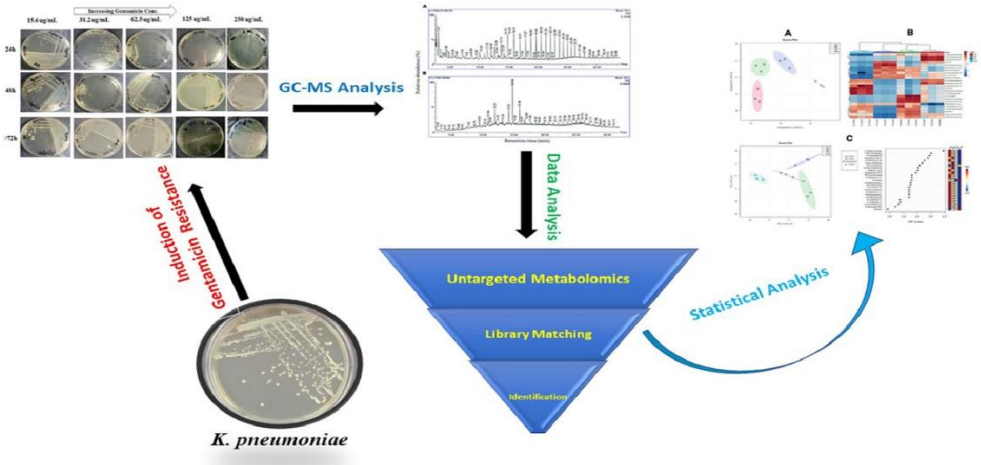
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Subject Category: Analytical Chemistry

Abstract: Antibiotic resistance in pathogenic bacteria like *Klebsiella pneumoniae* poses a severe threat to humankind. Understanding bacterial interactions with antibiotics and how they develop resistance is crucial. This study analyzes metabolomics in *K. pneumoniae*-sensitive and gentamicin-tolerant (resistant) strains. Strains were treated with gentamicin at five concentrations (16.2 µg/mL to 250 µg/mL) over three periods (24 h, 48 h, 72 h). Colonies from these conditions underwent metabolomics analysis using GC–MS. Morphological changes were notable with increasing gentamicin. Strains grown at 250 µg/mL tolerated 1 mg/mL gentamicin, showing resistance. A total of 459 metabolites were identified. Sequential down/up-regulation in 4, 3, and 4 metabolites correlated with rising gentamicin concentrations at 24 h, 48 h, and 72 h. Comparative analysis showed seven down- and sixteen up-regulated metabolites between resistant and sensitive strains. Some fatty acids and sugars increased in concentration, while metabolites like inosine, tyrosine, 1-propionylproline, and 2-hydroxyacetic acid decreased in resistant samples. These regulatory metabolites may be linked to resistance development in *K. pneumoniae* against gentamicin and could aid in the rapid detection of gentamicin-resistant clinical strains.

Graphical Abstract:



Poster ID: T- ICRA-296

Removal of arsenic from water through modified chitosan based biocompatible hybrid beads

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Subject Category: Physical Chemistry

Abstract: This research is centered on the synthesis of biocompatible crosslinked chitosan/PVA beads with improved physicochemical properties for application in adsorption of arsenic (III). Arsenic (III) being carcinogenic poses a serious threat to life. The crosslinking agent employed has not been reported previously with the aforementioned blend. The modified chitosan beads were synthesized by phase-inversion method and characterized using FT-IR spectroscopy and XRD which confirmed the crosslinking of chitosan/PVA blend. Along with SEM and EDX analyses, the degree of swelling and the degree of porosity were also determined. The experimental was performed using cyclic voltammetry and gravimetry. The crosslinked chitosan/PVA beads demonstrated superior adsorption capacity for arsenic (III) as compared to chitosan beads, chitosan/PVA beads and crosslinked chitosan beads. Adsorption studies at different pH showed best performance of the adsorbent at pH 8 which was further confirmed by the pHpzc. The experimental data satisfied different monolayer and multilayer adsorption models. The adsorbent's maximum sorption capacity for arsenic (III) was 329 mg/g. Future prospects include the adsorption of other heavy metals and oil-water separ.

Poster ID: T- ICRA-297

Synthesis, characterization, and in vitro anticholinesterase screening of novel indole amines

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Subject Category: Organic Chemistry

Abstract: The present study involved the targeted synthesis and characterization of novel indole amines with antiacetylcholinesterase profiling. A series of proposed indole amines was virtually screened against human acetylcholinesterase. A few indole amines (**23**, **24**, and **25**) showing strong enzyme binding in the in silico studies were synthesized in the laboratory and characterized using spectroscopic (IR, UV, NMR, single crystal XRD) and spectrometric (EIMS, HR-EIMS) methods. The indole amine **23** was crystallized from EtOH and analyzed with single crystal XRD. These ligands interacted with the PAS site in the enzyme, and their binding may disrupt the activity. The in vitro acetylcholinesterase inhibition studies revealed that the IC_{50} values for indole amines **25** and **24** (4.28 and 4.66 µM, respectively) were comparable to that of galantamine (4.15 µM) and may be studied further as cost-effective acetylcholinesterase inhibitors.

Poster ID: T- ICRA-298

Imidazolium ionic liquid modified cellulose based antimicrobial membranes for desalination of sea water

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Subject Category: Organic Chemistry

Abstract: Freshwater scarcity is a global problem and 6 billion people are expected to experience water shortage by 2050. Membrane based water purification is employed due to its high selectivity, cost effectiveness, energy efficiency, simple operation and environmental friendliness. Polyamide, polyether sulfone and cellulose based membranes are prevalent in the field of water treatment owing to their good separation efficiency and robust mechanical properties. But these membranes tolerate a narrow range of pH and are prone to fouling, resulting in a decrease in both filtration efficiency and water flux. This project aims the modification of naturally abundant cellulose with imidazole based ionic liquids that are well known for their antibacterial, antiviral, antifungal and anti-inflammatory properties. This modified novel material is blended with cellulose acetate in different weight percentages. Cellulose nanocrystals are incorporated in it for enhanced hydrophilicity, improved mechanical strength, high salt rejection and elevated water flux. The physiochemical properties of fabricated membranes are evaluated through FTIR, SEM, XRD, contact angle, tensile testing, water flux and salt rejection capacity. The results show that Imidazolium Ionic liquid modified cellulose membrane showed high salt rejection i.e. 85.03% for , 76.1% for NaCl and 77.9% for sea water composition. Membrane 10 showed highest water flux of 1627.68 L/h with minimal fouling and good mechanical strength. The membranes possess high hydrophilicity and porosity, and exhibit remarkable antibacterial and antifungal properties which prevent them from bio-fouling and hence expand their durability.

Keywords: Desalination, Antimicrobial activity, hydrophilicity, water flux, fouling

Poster ID: T- ICRA-300

Synthesis and Derivatization of 11H-indeno[1,2-b]quinoxaline

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Subject Category: Organic Chemistry

Abstract: (E)-11-((4-acetylphenyl)imino)-11H-indeno[1,2-b]quinoxaline-7-carboxylic acid is an excellent starting materials for the synthesis of 11H-indeno[1,2-b]quinoxaline derivatives. It was synthesized by firstly reacting ninhydrin with 3,4-diaminobenzoic acid in methanol and then the product obtained further with 4-aminoacetophenone in methanol. A novel series of compounds was obtained by reacting (E)-11-((4-acetylphenyl)imino)-11H-indeno[1,2-b]quinoxaline-7-carboxylic acid with furfuraldehydes in water/ethanol solvent using NaOH as catalyst. The synthesized compounds were analyzed by FTIR. Their melting points were also determined. In short, we developed an efficient and cheap strategy for synthesizing a series of complex indenoquinoxaline compounds which can be utilized in almost every field of science, especially in pharmacology.

Poster ID: T- ICRA-302

Extract of Tamarix Aphylla Bark: An Efficient Reducing and Capping Agent for Ag, Au and Ag/Au Nanoparticles with their In-Vitro Biological Activities

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Subject Category: Organic Chemistry

Abstract: This study reports the biogenic synthesis of Ag, Au and Ag/Au bimetallic nanoparticles using bark extract of *Tamarix aphylla*. Silver nitrate ($AgNO_3$) and hydrogen tetra chloroaurate trihydrate ($HAuCl_4.3H_2O$) were used as precursors while deionised water and methanol (CH_3OH) were used as solvents. Synthesized nanoparticles were characterized through UV-visible spectroscopy, SEM (scanning electron microscopy), TEM (transmission electron microscopy) and FTIR (Fourier transform infrared) for their morphology, structure, and identification of different functional groups. The UV-visible spectra of AgNPs, AuNPs and Ag/Au bimetallic nanoparticles showed peaks at 436, 532 and 527 nm respectively due to the excitation of Surface Plasmon Resonance. SEM and TEM images showed spherical and well distributed nanoparticles with particle size as 29 nm (AgNPs), 13 nm (AuNPs) and 26 nm (Ag/Au NPs). The synthesized nanoparticles are significantly active against inhibition of free radicals, α-amylase, α-glucosidase and have anti inflammatory potential with AgNPs having the highest percent activity at 400 µg/ml, followed by Ag/Au bimetallic NPs. The same trend (AgNPs>Ag/AuNPs>AuNPs) has been observed at all concentrations i.e. 100 µg/ml, 200 µg/ml and 400 µg/ml. AuNPs have shown lowest activity at all concentrations. So the current study opens up a new possibility of synthesizing Ag and Au nanoparticles using bark extract of *Tamarix aphylla* with promising biological activities.

Poster ID: T ICRA-303

Rational Design of metal dopping in AlCo layer double hydroxide electrode for electrochemical hydrogen generation

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Abstract: Hydrogen fuel is considered future energy to overcome the high energy crisis in coming future. The electrochemical water splitting is a promising technique to produce hydrogen evolution reaction (HER). However, it still suffers from many challenges, including sluggish alkaline HER kinetics, insufficient activity and limited lifetime of noble metals electrocatalysts for enhanced alkaline HER. Here, we report an efficient alkaline HER electrocatalyst wherein a 3D flowered shaped superstructure of AlCo-layer double hydroxide (AlCo-LDH) is designed to serve as a bifunctional electrocatalyst for HER and oxygen evolution reaction (OER). The nonprecious zinc is introduced into the self-supported AlCo-LDHs 3D-superstructure synthesized by hydrothermal method and exhibits a preeminent electrocatalytic activity toward the HER with an ultralow overpotential of 146 mV to obtain 10 mA cm^{-2} in alkaline solution (1.0 M KOH). Zn doped AlCo-LDHs can effectively facilitates the mass transfer, enhance the charge transfer kinetics and endowing excellent long-term durability under industrial scale application. This work provides a strategy to develop efficient metal layered double hydroxide as HER catalysts.

Poster ID: T- ICRA-304

Synthesis of nanoemulsions based on natural products for neurogenesis

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Subject Category: Organic Chemistry

Abstract: Nanoemulsions are finely dispersed systems consisting of nanometer-sized droplets, which offer improved stability, bioavailability, and targeted delivery of therapeutic agents. In nanotechnology, they play a crucial role in enhancing the efficacy of natural products for various biomedical applications. One significant advantage of nanoemulsions is their ability to cross the blood-brain barrier, a major challenge in delivering therapeutic agents to the brain. This study explored the neurogenic potential of Quinic acid-loaded nanoemulsions prepared using a green, scalable method. Quinic acid, dissolved in olive oil, was emulsified with PEG-400 and Tween-20 using probe sonication. The resulting nanoemulsions exhibited a mean particle size of 124.9 nm with a polydispersity index of 0.232. The nanoformulation of Quinic acid showed significant growth-supporting activity and neuronal differentiation of primary cortical cultures of neonatal rats at 24, 48, and 72-hour time points, with doses ranging from 10 to 100 µM. This research highlights the potential of nanoemulsions in advancing neurological therapies.

Poster ID: T- ICRA-305

Self-assembled cyclic peptide nanotubes enhancing drug delivery efficiency and neuroregenerative potential.

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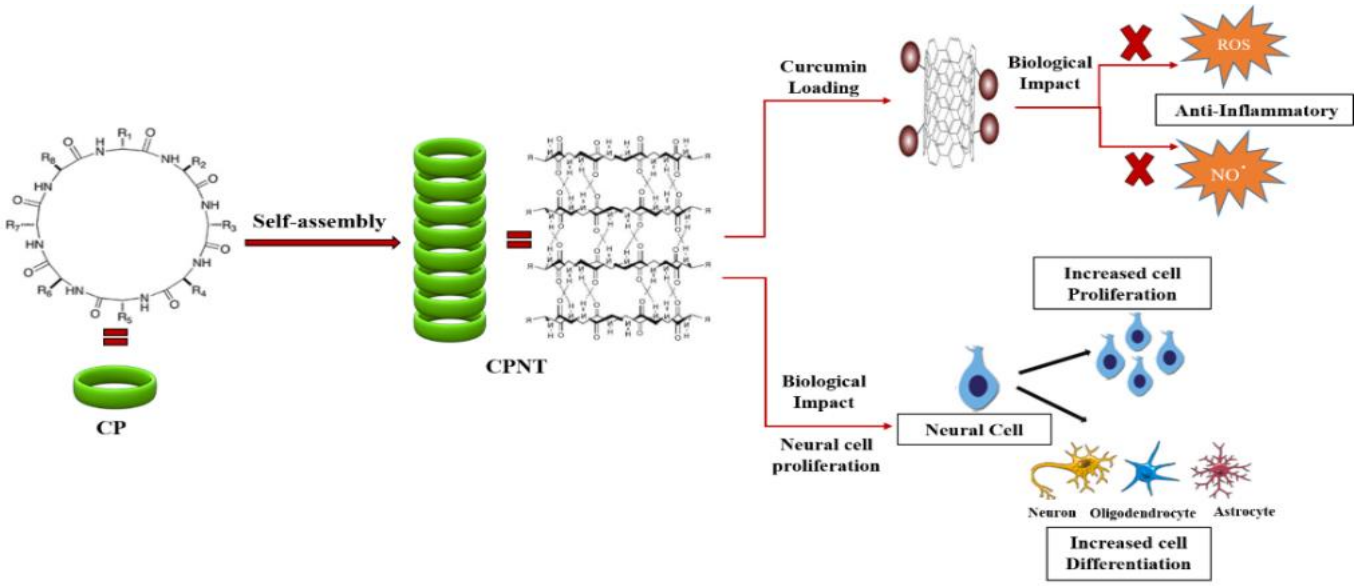
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Subject Category: Organic Chemistry

Abstract: Self-assembled Cyclic Peptide Nanotubes (CPNT) are unique supramolecular nanobiomaterials with a tubular structure. These nanotubes offer a novel approach to enhance drug delivery efficiency. Cyclic peptide was synthesized by using a Solid Phase Peptide Synthesis (SPPS) strategy and self-assembled in the form of nanotubes by using the Phase Equilibration method. CPNT were loaded with curcumin, overcoming its solubility issues and resulting in enhanced biological activity and non-toxicity on normal mammalian cell lines BJ and NIH-3T3. In vitro, release studies revealed a controlled and sustained release profile, which increases the bioavailability and therapeutic efficacy of curcumin. Biological evaluations indicate that the nanotubes notably enhanced curcumin's anti-inflammatory properties by inhibiting intracellular reactive oxygen species (ROS) from human whole blood phagocytes and nitric oxide ($NO^{\cdot}$) with IC_{50} values of (1.8 ± 0.1 and 44.4 ± 1.4 µg/mL respectively) as compared to curcumin which was insoluble in PBS and showed IC50 value of >100 µg/mL on both parameters. Additionally, the MTT assay indicates that CPNT at the concentration of 50 and 100µM displayed the growth-supporting activity of primary cortical cultures obtained from the brains of rat pups. This study emphasizes the transformative potential of self-assembled cyclic peptide nanotubes in enhancing therapeutic outcomes and advancing biomedical research.

Graphical Abstract:


Self-assembly
CP
CPNT
Curcumin Loading
Biological Impact
ROS
Anti-Inflammatory
NO˙
Increased cell Proliferation
Biological Impact
Neural Cell
Neural cell proliferation
Neuron
Oligodendrocyte
Astrocyte
Increased cell Differentiation

Poster ID: T- ICRA-306

Synthesis and Electrochemical study of Hybrid for supercapattery application.

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Subject Category: Inorganic Chemistry/ Material Science

Abstract: The supercapattery is an innovative energy storage device that merges the long-lasting cycle stability, high power density, and rapid charge/discharge capabilities of supercapacitors with the higher energy density of batteries. This study involved synthesizing a hybrid material composed of transition metal oxides, carbon material and polymer using a straightforward hydrothermal method and in-situ polymerization. The synthesized material's structure, composition, and morphology were analyzed through various techniques such as X-ray Diffraction (XRD), Fourier Transform Infrared Spectroscopy (FTIR), Scanning Electron Microscopy (SEM), Energy-Dispersive X-ray Spectroscopy (EDS) and elemental mapping, Thermogravimetric Analysis (TGA), and X-ray Photoelectron Spectroscopy (XPS). The electrode material demonstrated impressive capacitive performance, achieving a specific capacity, power density and an energy density, remarkable cyclic stability. This outstanding electrochemical performance is attributed to the material's structural stability and the synergistic effects of its components.

Poster ID: T- ICRA-307

Bimetallic Fe/Ni-BTC MOF Decorated MXene Hybrid for Improved Oxidation of Water

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Subject Category: Material Science

Abstract:

Electrochemical production of green hydrogen via water-splitting is an emerging technology to generate a substitute source of energy. However, due to the slow kinetics of oxygen evolution reaction (OER), high cost, lesser availability and easy oxidation of noble metal-based electrocatalysts led the explorers to find an efficient and low-cost electrocatalysts. In the current communication, we have developed a synthesis strategy for the preparation of hybrid electrocatalyst composed of bimetallic (iron, nickel) based metal organic framework (FeNiBTC MOF) and MXene ($Ti_3C_2T_x$) via solvothermal reaction. Thanks to the ultrathin heterostructure with high electrical conductivity of MXene, with abundant active sites of FeNiBTC MOF, the as-prepared hybrid electrocatalyst FeNiBTC@MXene, leads the high efficiency OER in an alkaline environment. FeNiBTC MOF was impeccably decorated on the surfaces of MXene nanosheets with different FeNiBTC to MXene ratios and was characterized via pXRD, FESEM/EDS, XPS and BET. The optimized hybrid structured catalyst (FeNiBTC@Mx-3) revealed the best performance for OER with a low overpotential of 210 mV vs. RHE at a current density of 10 mA/cm^2 and a Tafel plot value of 38.4 mV/dec and stable up to 1000th CV cycles.

Poster ID: T- ICRA-308

Enhanced Ibuprofen Encapsulation and Controlled Release at different pH and Concentration Using Zr-MOF Nano carrier

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Subject Category: Organic Chemistry

Abstract: This study investigates the loading and release of Ibuprofen using Zirconium-based Metal-Organic Frameworks (Zr-MOFs) as nano-carriers. Ibuprofen, a widely utilized non-steroidal anti-inflammatory drug, was efficiently loaded into Zr-MOFs, characterized by high surface area and tunable pore sizes. Zr-MOF was characterized by FTIR, PXRD, BET and SEM techniques.The drug loading process was analyzed by using techniques such as UV-Vis spectroscopy. Controlled release studies revealed pH and temperature-dependent release profiles, highlighting the potential of Zr-MOFs as effective nano-carriers for controlled drug delivery applications. This research provides insights into optimizing drug encapsulation and release mechanisms using Zr-MOFs, aiming to enhance therapeutic outcomes in pharmaceutical formulations.

Poster ID: T- ICRA-309

Fe, Mn and Co based efficient bifunctional electrocatalyst derived from MOF/ZIF for oxygen electrode in zinc-air batteries

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Subject Category: Material Science.

Abstract: Zinc-air batteries (ZABs) are gaining significant attention in renewable energy due to their high safety, low cost, high theoretical energy density (1084 $Wh \cdot kg^{-1}$), and environmental friendliness. However, the slow kinetics of oxygen reactions necessitate noble metal-based electrocatalysts for rechargeable zinc-air batteries (RZABs), which are expensive and have stability issues. This calls for the development of cost-effective, ultra-stable, and efficient non-noble metal-based bifunctional electrocatalysts for oxygen reduction (ORR) and oxygen evolution (OER) reactions. We present a simple approach to create a trimetal-based nitrogen-doped hierarchically porous carbon (FeMnCo/N-C) bifunctional electrocatalyst. By mixing Fe/Mn-MOF and ZIF-67, followed by pyrolysis, we obtained FeMnCo/N-C. This catalyst exhibits excellent bifunctional activity, with half-wave potential ($E_{1/2}$) = 0.84V for ORR and overpotential to achieve 10mA/cm^2 current density (η_{10}=390mV) for OER, and a full zinc-air cell displayed a potential gap of 0.78V, outperforming Co/N-C and Fe/Mn-C. It also shows superior stability compared to Pt/C and RuO_2.The electrocatalyst's dual functionality is attributed to the synergistic effects of the tri-metals, nitrogen-doped carbon, and its porous structure, making FeMnCo/N-C an excellent candidate for advancing efficient and sustainable zinc-air battery and green energy technology.

Poster ID: T- ICRA-310

MOF Derived Co-Mn based Bi-functional Electrocatalyst for Oxygen Electrode In Zinc-Air Batteries.

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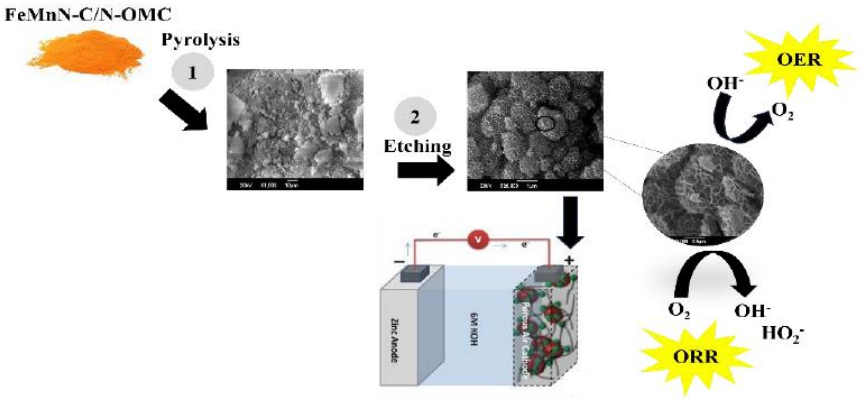
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Subject Category: Material Science

Abstract: As energy demand grows and fossil fuel supplies decline, the development of innovative and efficient electrochemical storage systems becomes crucial. Zinc-air batteries (ZABs) are highly promising for energy storage in stationary applications, portable electronics, and electric vehicles due to their high energy density, eco-friendliness, safety, and affordability. Despite these advantages, they face substantial challenges such as sluggish oxygen reduction and evolution processes, which limit their commercial application. This study presents the synthesis of a bimetallic cobalt manganese-based nitrogen doped mesoporous carbon structure (Co-Mn/N-MC) as bifunctional electrocatalyst using a template casting method. The electrocatalyst exhibits a half-wave potential (E1/2) of 0.82 V for the oxygen reduction reaction (ORR) and demonstrates superior oxygen evolution reaction (OER) performance with a lower overpotential of 360 mV at 10 mA/cm2 and a Tafel slope of 65 mV dec^{-1}. This work addresses the existing research gaps and offers a scalable, cost-effective solution, potentially accelerating the commercialization and widespread application of -air batteries.

Graphical Abstract



Poster ID: T- ICRA-311

Phytochemical Characterizations of *Maranthes polyandra* (Benth.) Prance

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Subject Category: Organic Chemistry

Abstract: *Maranthes polyandra* (Benth.) Prance yielded two new ursane-type triterpenoids, called Polyanside A (**1**) and B (**2**), and eleven known compounds (**3–13**), all of which were isolated and classified. These compounds' structures were determined using a combination of numerous spectroscopic collections of data and chemical evidence. The anti-inflammatory, anti-cancer, and cytotoxic properties of the isolated compounds were assessed using a normal human cell line (BJ). They were all non-cytotoxic and inactive. Through GC-MS analysis, 41 chemicals were identified from the hexane fraction. This is the first thorough investigation of *M. polyandra's* phytochemistry. The cytotoxicity and anti-inflammatory and anti-cancer properties of compounds **1, 2, 5, 6, 8, 11**, and **13** were assessed. They were all inactive. This plant may have anti-inflammatory components based on its historic applications. It might take more research to find strong anti-inflammatory compounds in this species

Poster ID: T- ICRA-312

Role of silver nanoparticles loaded polyurethane based nanocomposites in preparation of durable inks

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Subject Category: Organic Chemistry

Abstract: Polyurethane (PU) based composite polymers were fabricated with silver (Ag) metallic nanoparticles via chemical reduction method while using $NaBH_4$ as reductant. Different properties of polyurethane (PU) and PU-Ag nanocomposites were illustrated by various analytical techniques such as UV-visible, FTIR, XRD, TGA and DLS, etc. Surface Plasmon resonance band appeared in UV-visible spectra indicates the presence of Ag metallic nanoparticles in polymer sieves. FE-SEM analysis confirms the spherical shaped morphology of Ag nanoparticles. EDX analysis depicts the presence of Ag in composite particles. XRD spectra shows the crystalline nature of metal nanoparticles in form of sharp, narrow and low intensity peaks. The conductive inks from different biological sources such as turmeric, turmeric & slaked lime and caramelized sugar have been synthesized and then PU-Ag nanocomposites have been added into inks to get synergistic characteristics of silver nanoparticles and polyurethane. The synthesized inks were characterized via different techniques such as UV-visible, FTIR, etc. The prepared inks showed exceptional rheological properties and stability due to PU-Ag nanocomposites. PU-Ag nanocomposites and synthesized inks were also proved excellent antibacterial agents against gram-positive and gram-negative bacteria.

Poster ID: T- ICRA-313

Analysis of XDR Salmonella typhi from Whole Blood Samples

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Subject Category: Analytical Chemistry / Biochemistry

Abstract: *Salmonella enterica* serovar typhi (*S. typhi)* is a Gram-negative bacteria that causes typhoid fever. Food or water contaminated with infectious dose of *S. typhi* is a common cause of typhoid fever. Transmission may occur through poor hygiene and sewage contamination of the water supply. The source of infection may may disseminate from an infected person (carrier) to healthy individuals by fecal sample contamination to drinking water or food consumbles. Typhoid fever can be fatal in children and aged persons if remined undiagnosed and untreated. The convential diagnosis test for Salmonella infections involves use of IgG an IgM antibodies but now this test is considered obseletes due to false positive Salmonella dectection. Blood culturing is an only option available for the diagnosis of *S. typhi* infections but it takes a long reporting time (3-8 days). For the early detection of typhoid, certain molecular tests of S. typhi could be used incclduing PCR-based detection methods. Compared to other blood culturing and microbiological testing, the overall diagnosis took less time, greater sensitivity and precise results. PCR allows selective amplification to distinguish S. typhi from the other closely related species. Use of multiplex PCR can better help to attain more sensitivity and specificity to detect microbial pathogens in clinical specimens. This study help to provide a way towards *S. typhi* diagnosis using PCR in whole blood samples.

Poster ID: T- ICRA-314

Seasonal profile of physicochemical and microbial water parameters of Rawal Lake, Pakistan

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Subject Category: Environmental Chemistry / Inorganic Chemistry

Abstract: Rawal Lake plays vital role for its surrounding communities, related biodiversity and Pakistan's two largest cities, i.e. Rawalpindi and Islamabad. Thus, water quality of Rawal Lake is imperative. The study presented seasonal variations in water parameters and impact of meteorological factors on water quality. Water samples were taken from twelve different sites in Rawal Lake in pre-monsoon, monsoon and post-monsoon seasons. Alkalinity, hardness, phosphate (PO_4^{-3}), sulfate (SO_4^{-2}), turbidity, electrical conductivity (EC), total dissolved solids (TDS), pH, dissolved oxygen (DO), temperature and total fecal coliforms (TFC) were among the parameters measured in the samples. Except for SO_4^{-2}, statistically significant seasonal variations were found for every parameter; which showed sulfur-containing water bed of Rawal Lake. Mean levels of EC, PO_4^{-3}, and TFC were found higher than allowed limits at each sampling month, indicating that water was not fit for human consumption. The monsoon season raised concerns as most of the water parameters exceeded from the standard limits in this season. On the other hand, this season correspondingly carried Rawal Lake's DO and alkalinity levels back within acceptable ranges. The impact of meteorological variables on these parameters and the relationships among water parameters were also strongly associated, according to statistical analysis.

Poster ID: T- ICRA-315

EpoR/βcR Heterocomplex Modeling and Epo Mimetic Screening for Tissue-Protective Therapeutics

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Subject Category: Organic Chemistry, Computational Chemistry

Abstract: Erythropoietin (Epo), a glycosylated cytokine, modulates erythropoiesis and exhibits tissue-protective activity through an anti-apoptotic pathway in its heterodimeric form, EpoR/βcR. These receptors consist of three domains: Extracellular, Transmembrane (TM), and Cytoplasmic (Cyt). The TM and Cyt domains are crucial for dimerization and tissue protection, yet their tertiary structures are undocumented. To explore EpoR/βcR interactions, we used in silico tools to model and refine the cytoplasmic domains, verifying their quality with Rosetta. These domains were then merged with their extracellular and TM counterparts. The complete receptor structures underwent geometry optimization and energy minimization via molecular dynamic simulations, followed by protein-protein docking to form the heterocomplex. We report optimized structure coordinates of EpoR/βcR and the electrostatic interactions between interfacial residues. Using the STRAPEP Database, we screened for Epo mimetic agents, identifying four hits (4HQX, 4XO8, 1TPO, 3EZM) with strong binding affinity and molecular interactions. These peptides demonstrated stability in the EpoR/βcR binding site, suggesting potential for future characterization studies.

Poster ID: T- ICRA-316

Synthesis Of Phakellistatin 18 And Its Analog

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Subject Category: Organic Chemistry

Abstract: The cyclic peptides of phakellistatin class of natural product are reported as potent anticancer compounds. Phakellistatin are the proline-rich cyclic peptides, which are isolated from marine organism. So far, twenty-two members of phakellistatin were isolated from marine sponge *Phakellia*. Phakellistatin 18 (1) isolated from marine sponge *phakellia fusca* in which all three proline residues were reported in *cis* conformation. In current studies, Solution-Phase macrocylization route was adopted for the total synthesis of natural product 1. As a result, all three proline-linkages were found in *trans* conformation, cytotoxicity, structural and synthetic studies of phakellistatin 18 (2) and its new analog [Ala 1,3,6] Phakellistatin 18 (3) were prepared by substituting all three proline residues by Alanine residue. These peptides were purified by using RP-HPLC and their structures were confirmed by various spectroscopic techniques such as ESIMS, ^{1}H-NMR, ^{13}C-NMR and HR-ESI-MS. In vitro anticancer screening of peptides were performed against different cancer cell lines, trans-phakellistatin 18 (2) (IC_{50}=67.5±2.938 µM) showed comparable cytotoxicity against HepG2 cancer cell line with standard drug doxorubicin (IC_{50}=63.88±6.48 µM). we are first to reported the structural and synthetic studies on *trans*-phakellistatin 18 (2) and its anticancer screening against HepG2 cell lines. Furthermore detail of current studies will be describe during presentation.

Poster ID: T- ICRA-317

Exploring the coordination chemistry and biological efficacy of hydrazone ligands and their metal (II) complexes

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Subject Category: Inorganic Chemistry / Material Science

Abstract: This study investigated the formation of metal (II) complexes with hydrazone derivatives to understand their role in biological systems and to highlight their bioactivity and the stability of metal complexes, which are important in coordination chemistry. We synthesized a series of hydrazone ligands from the condensation of substituted isatin and aromatic aldehydes and then examined the coordination complexes formed with these ligands and Co(II), Ni(II), Cu(II), and Zn(II) ions. The newly synthesized compounds underwent thorough characterization using a range of analytical techniques, including FT-IR, EI-mass spectrometry, elemental analysis, UV-Visible spectroscopy, magnetic susceptibility, and conductivity studies. Elemental analysis revealed a 1:1 metal-to-ligand ratio for Cu(II) and Zn(II) complexes and a 1:2 ratio for Co(II) and Ni(II) complexes. Magnetic susceptibility data indicated a square planar geometry for Cu(II) complexes, a distorted tetrahedral geometry for Zn(II) complexes, and octahedral geometries for Co(II) and Ni(II) complexes. The primary aim was to assess the antimicrobial and antioxidant properties of the hydrazones and their complexes. The results showed that the metal complexes were significantly more effective as antibacterial and antifungal agents compared to the ligands. Conversely, the ligands exhibited better DPPH radical scavenging activity than the metal complexes.

Poster ID: T- ICRA-318

Role of Co and Ni ferrites in the fabrication of Saccharum officinarum bioadsorbents for removing As(III)

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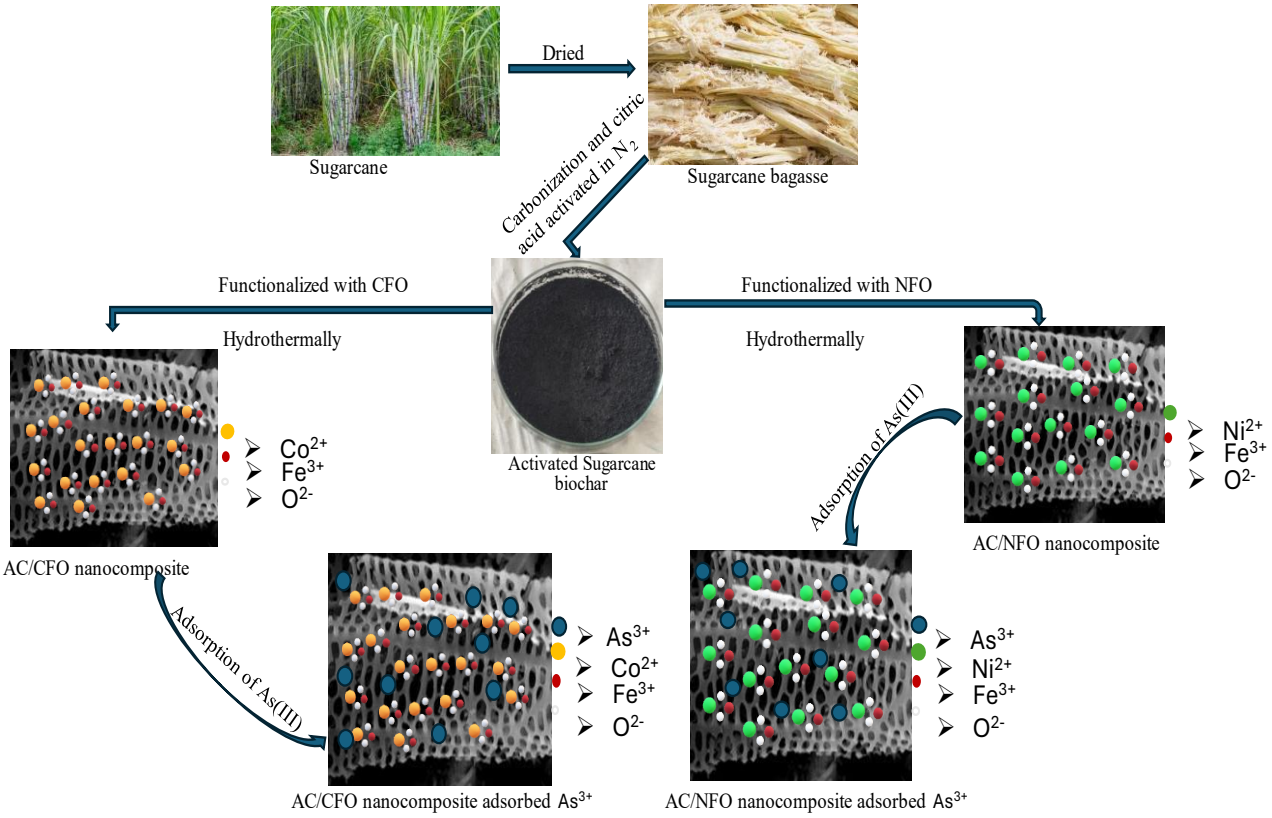
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Subject Category: Physical Chemistry / Material Science

Abstract: Removal of arsenic from drinking water is one of the most significant worldwide concerns. Adsorptive removal of arsenic is highly practical and efficient among the various methods. The green synthesis using sugarcane extract was eco-friendly, and cost-effective and resulted in adsorbents having magnetic behavior and a high capacity for arsenic. In the present study, activated carbon (ACs) of sugarcane and its surface, functionalized with nickel ferrite (AC/NFO) and cobalt ferrite (AC/CFO) were used as biosorbents for the removal of As(III) from aqueous medium. The functionalization was undertaken in a controlled hydrothermal method using Saccharum officinarum extract as a green source. The morphology, functionality, thermal stability, structure, magnetic behavior, surface area and porosity of adsorbents were confirmed by SEM, FTIR, TGA, XRD, VSM, BJH and AFM. The sorption behavior was determined with the effect of pH, time, concentration, temperature and dose. Pseudo-second order and Langmuir models were found suitable to validate the sorption data. Thermodynamics of the sorption process and the desorption studies suggested that the governing mechanism is chemisorption. The adsorption capacities for AC2, AC2C2 and AC2N2 were 961 µg/g, 1103 µg/g and 1000 µg/g respectively.

Graphical Abstract:


Sugarcane
Dried
Sugarcane bagasse
Carbonization and citric acid activated in N_2
Activated Sugarcane biochar
Functionalized with CFO
Hydrothermally
AC/CFO nanocomposite
Co^{2+}
Fe^{3+}
O^{2-}
Functionalized with NFO
Hydrothermally
AC/NFO nanocomposite
Ni^{2+}
Fe^{3+}
O^{2-}
Adsorption of As(III)
Adsorption of As(III)
As^{3+}
Co^{2+}
Fe^{3+}
O^{2-}
AC/CFO nanocomposite adsorbed As^{3+}
As^{3+}
Ni^{2+}
Fe^{3+}
O^{2-}
AC/NFO nanocomposite adsorbed As^{3+}

Poster ID: T- ICRA-320

Design and Synthesis of novel indazole-based thiadiazole compounds containing thiazolidinone moieties: in-vitro Acetylcholinesterase/Butyrylcholinesterase inhibition studies along with molecular modelling protocols

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Subject Category: Organic Chemistry

Abstract: An indazole-based thiadiazole containing thiazolidinone moieties **(1-17**) was synthesized and its inhibition of acetylcholinesterase (AChE) and butyrylcholinesterase (BuChE) activities was evaluated. The IC50 values of the synthesized analogues ranging between 0.86 ± 0.33 μM to 26.73 ± 0.84 μM, in comparison to Donepezil (IC50 = 1.26 ± 0.18 μM) against (AChE) and 0.89 ± 0.12 μM to 27.08 ± 0.19 μM in comparison to Donepezil (1.35 ± 0.37 μM) against (BuChE) enzymes. Among the series the following 9, 14 & 1 scaffolds with IC50 values for these scaffolds were determined to be 9 (0.86 ± 0.30 & 0.89 ± 0.12 μM), 14 (0.92 ± 0.10 & 0.98 ± 0.48 μM), 1 (1.10 ± 0.37 & 1.19 ± 0.42 μM). These scaffolds were found to be the most potent analogues bearing para-CF3, ortho-OH, and para-F substitutions on the phenyl ring attached to the thiadiazole skeleton, as well as meta-Cl, -NO2, and para-chloro substitutions on the phenyl ring. The synthesized scaffolds have been further characterized by using ^{1}H-NMR, ^{13}C-NMR and (HR-MS) to bring out their structural characteristics. To gain a better understanding of how binding takes place in the most potent synthesized derivatives, an in-depth study of molecular docking has been carried out in order to determine the mode of binding for the most potent synthesized derivatives.

Poster ID: T- ICRA-321

Synthesis and characterization of Biomedical Hydrogel by using Chemically Modified Biopolymer for Drug delivery application

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Subject Category: Material Science / Analytical Chemistry

Abstract: PVA mixed with Alginate and chemically cross-linked with tetraethyl orthosilicate (TEOS). Alginate/PVA based hydrogel were synthesized for controlled drug delivery systems. The characterizations of hydrogel were done by different techniques FTIR, SEM, TGA, and UV-VIS Spectroscopy. Surface morphology of hydrogel was checked by Scanning electron microscopy (SEM), thermal stability of hydrogel was checked by Thermal gravimetric Analysis (TGA). Linkages between polymers of hydrogel were detected by Fourier transform infrared spectroscopy (FT-IR). It was found that hydrogel is porous which play a key role in swelling of hydrogel. In different solutions such as pH (buffer as well as non-buffer), electrolytic and aqueous solution the swelling of hydrogel is checked. At low pH the swelling of hydrogel is maximum and minimum at high pH. The hydrogel was loaded with two different drugs, Sitagliptin and Ciprofloxacin HCl. For In-vitro release study both the drugs were selected as modal drug and studied in simulated gastric fluid (SGF) and simulated intestinal fluid (SIF). In SIF Sitagliptin release 99% in 180 minutes. The Ciprofloxacin HCl shows 98.8% release in SIF for 5 hours. So, Ciprofloxacin HCl and Sitagliptin both the drugs are compatible with Alginate/PVA hydrogel which is determined from drug release study of drugs.

Poster ID: T- ICRA-322

Synthesis, Characterization and Biological Applications of Novel Metallo-phthalocyanine Derivatives

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Subject Category: Organic Chemistry

Abstract: Different Metallo-phthalocyanines were synthesized in this research work. These Metallo-phthalocyanines were synthesized by adopting Multi-Component Multi-Step chemical reactions synthetic scheme under specific reaction conditions. Chalcone and Chalcone-substituted phthalonitrile which in turn prepared from aldehydes, were considered as precursor for the synthesis of different Metallo-phthalocyanines. Different solvents were used like DMF, DMSO, EtOH, Water etc. After the synthesis of these Novel products, these were characterized using different techniques such as FTIR and TLC. The obtained products have different industrial and pharmaceutical applications. Metallo-phthalocyanines are applicable biological activities such as anti-cancer, antimicrobial, anti-oxidant properties. %Yield of all obtained products was calculated when actual yield calculated. All of the obtained 7 different products have moderate yield.

Poster ID: T- ICRA-324

Targeted Analysis of Veterinary Drugs in Food Samples by Developing a High-Resolution Tandem Mass Spectral Library

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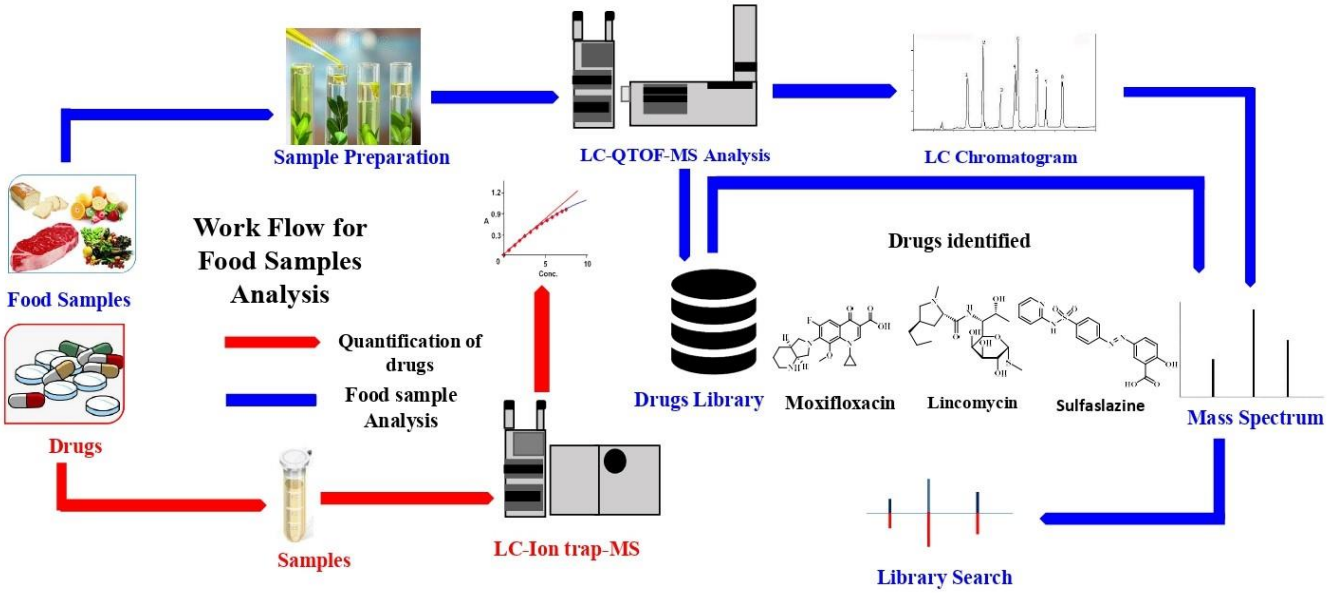
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Subject Category: Analytical Chemistry

Abstract: Veterinary drug residues present in foods can pose severe health threats to the population. The present study aims to develop a high-resolution mass spectral library of 158 veterinary drugs of 16 different classes for their rapid identification in food samples through Liquid Chromatography−High-Resolution Electrospray Ionization-Tandem Mass Spectrometry. Standard drugs were pooled according to their log P values and exact masses before analysis. Spectra were collected at system automated collision energy, i.e., of 25−60 eV and four predetermined collision energies (10, 20, 30, and 40 eV) for each compound using a schedule precursor list of $[M+H]^+$, $[M+Na]^+$, and $[M+NH_4]^+$ ions. The utility of the developed database was checked by analyzing food samples. A total of 17 veterinary drugs based on the reference Standard Retention Times, HR-MS spectra, and MS/MS spectra were identified in the analyzed samples. Moreover, five veterinary drugs were selected for quantitative analysis, including doxycyclinehyclate, lincomycin, sulfasalazine, moxifloxacin, and diphenoxylate, using Liquid Chromatography−Ion Trap Mass-Spectrometry. Concentrations of the drug were obtained to vary from 0.0805 to 0.9731 mg/kg in food samples and were found to be exceeded in most of the cases as per the maximum residue levels described by FAO/WHO. The MS data were submitted to the MetaboLights online database (MTBLS2914). This study will help in the high-throughput screening of multiclass veterinary drugs in foodstuffs.

Graphical Abstract:


Work Flow for Food Samples Analysis
Food Samples
Drugs
Sample Preparation
Samples
LC-QTOF-MS Analysis
LC-Ion trap-MS
LC Chromatogram
Drugs Library
Drugs identified
Moxifloxacin
Lincomycin
Sulfaslazine
Mass Spectrum
Library Search
Quantification of drugs
Food sample Analysis

Poster ID: T- ICRA-325

α-MnO2@ZIF-67 as Bifunctional Electrocatalyst for Air Cathode in High Performance Rechargeable Zinc-Air Batteries

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Subject Category: Material Science.

Abstract: The development of a cost-effective and highly efficient electrocatalysts that can perform both oxygen reduction reactions (ORRs) and oxygen evolution reactions (OERs) are of immediate necessity of Zn-air batteries (ZAB). Driven by the advancement of infrastructure generated from metal organic frameworks, a unique composite structure derived from ZIF-67 and MnO2 nanowires is designed. The thermal treatment at high temperature results in the formation of ZIF-67 based nanoparticles evenly distributed and enclosed with the MnO2 nanowires. The MnO2@ZIF-67 structure exhibits a half-wave potential of 0.85 V for the oxygen reduction reaction (ORR) and a low onset potential of 1.67 V for the oxygen evolution reaction (OER) at a current density of 10 mA/cm2. MnO2@ZIF-67 based primary and rechargeable ZAB was assembled and cycled to demonstrate superior performance as bifunctional electrocatalyst for zinc air battery device which showed excellent performance in terms of power density of 203.4 mW/cm2, discharge specific capacity of 821.7 mAh/g, an open circuit voltage of 1.28 V, and high electrochemical stability of up to 200 hrs with comparatively smaller elevation in the potential gap. Thus, this study offers a simple, inexpensive, and eco-friendly approach to systematically design microporous bifunctional electrocatalysts used in zinc-air batteries and energy storage devices.

Poster ID: T- ICRA-326

Effect of Marble Waste/Silica on Physical Properties of Epdm/Nbr Composite

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Subject Category: Physical Chemistry

Abstract: This study aims to develop a cost-effective rubber composite with excellent physical and mechanical properties by compounding and blending techniques. By combining NBR and EPDM rubbers, we leveraged the strengths of each component to achieve improved properties. NBR offers high resistance to oil and solvent swelling but lacks ozone resistance and heat aging properties, while EPDM excels in heat aging and ozone resistance but has poor solvent resistance. The NBR/EPDM blend exhibits good solvent and heat aging properties but requires enhancement of its mechanical properties, which we achieved by incorporating MS and Silica as fillers. This research investigates the optimal ratio of silica to marble waste to achieve superior physical and mechanical properties in the blend. The results show that the mechanical and swelling properties of EPDM/NBR blends are significantly influenced by the marble waste/silica ratio and CR content. The addition of silica significantly enhances tensile strength, tear strength, and hardness due to reinforcement, while incorporating CR in the marble waste/silica-filled composite further improves these properties. However, elongation at break, abrasion resistance, compression resistance, and rebound resilience decrease due to crosslinking. The study finds that the polarity of CR plays a crucial role in enhancing interactions within the blend, leading to improved properties."

Keywords: Marble Sludge (MS), Chloroprene Rubber (CR), Ethylene Propylene Diene Monomer (EPDM),Nitrile Butadiene Rubber (NBR).

Poster ID: T- ICRA-327

Computational study on enhanced CO sensing properties of boron-doped $Zn_{12}O_{12}$ nanocluster

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Subject Category: Material Science and Computational Chemistry

Abstract: CO gas is a threat to human life and environment. It causes CO poisoning and atmospheric pollution. CO sensing has become a huge challenge because of its colorless and odorless nature. To protect the environment from the toxicity of CO gas, different devices have been identified for its detection and capturing. Among them, metal-oxide semiconductors (SnO2, WO3, TiO2 etc.) used as sensing materials for various chemical gases. However, boron-doped Zn12O12 nanocluster is not explored for adsorption of CO. It is a metal oxide semiconductor that rationalizes the resistance behavior upon CO adsorption. Doping manipulated the nanocluster geometry making it highly sensitive to CO gas based on change in conductivity due to presence of CO molecules. Thus, we offer novel kind of boron-doped nanostructured Zn12O12 signaling CO hence reducing the critical risk factors related to CO gas. Herein; through DFT calculations, we investigated that the doped cluster is efficient to CO sensing as compared to pristine Zn12O12, dealing with its electronic, thermodynamic and spectroscopic properties. Electronic (EHOMO, ELUMO, Eg, DOS, QT) and thermodynamic (ΔG, ΔH, S) parameters were calculated to measure pristine and doped Zn12O12 nanocluster sensing potential for CO. Introduction of boron in Zn12O12 nanocluster, reduced the HOMO-LUMO gap i.e. 1.04 eV, and resistance hence enhancing the adsorption energy (i.e. 2.0), the stability and the electrical conductivity of pristine nanocluster towards CO. The increasing entropy and negative value of ΔH and ΔG indicated that adsorption of CO over doped cluster is energetically more favorable. Spectroscopic analysis revealed that the doping of boron resulted in the shifting of spectral range to fluorescent region depicting radiation emission by doped cluster upon CO detecting and capturing.

Graphical Abstract:

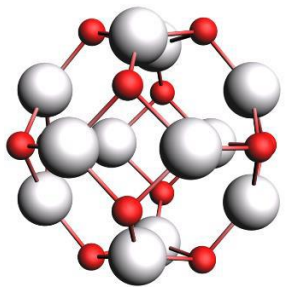


Figure: Optimized geometry of Zn12O12 nanocluster calculated at DZ basis set at GGA: BLYP/DFT performed at ADF modelling suite.

Poster ID: T- ICRA-331

Synthesis of Aryl-substituted Thio/Carbohydrazides based Chromoflourescent Sensor for detection of Heavy metal ions

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Subject Category: Organic Chemistry

Abstract: Molecular sensors which are based on organic molecules, employed as optical probes played crucial role in sensing and detecting toxic metal ions in water, food, air and biological environment. A series of Aryl-substituted Thio/Carbohydrazide based chromofluorescent sensor molecules derived from Thio/carbohydrazide, Ar-aldehydes were successfully designed and synthesized. The synthesized sensor molecules exhibit significant shift in their absorption spectra with visual color changes on interaction with heavy metal ions. In addition, the optimized sensor showed excellent selectivity towards Cu(II) ions with fluorescence "turn off" and "turn on" modes respectively. Along with this, prepared sensor-ion complex solution exhibited colorimetric "On-Off" sensing of Cu(II)ions, which was confirmed through UV-visible titration experiments.

Poster ID: T- ICRA-332

Total Synthesis and Biological Studies of Cyclic Peptide Stylissamide G

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Subject Category: Organic Chemistry

Abstract: Cyclic peptides containing one or more proline residues provide greater rigidity to the cyclic structure and enhance its biological potential. Proline containing cyclic peptides are abundant in nature. According to reports, various proline-rich cyclic peptides have been isolated from various marine organisms. Stylissaamide G is a proline rich natural cyclic heptapeptide isolated and identified from the Bahamian sponge Stylissa caribica. The synthesis of these proline-containing peptides becomes complicated due to the multiple possibilities of conformational isomers at proline residues. The focus of this study was to confirm the structure of natural product peptide by total synthesis. The synthesis of the natural product stylisamide G was achieved by solid-phase peptide synthesis of linear precursor, followed by liquid phase cyclization to obtain the desired peptide. Its structure was confirmed by HRESIMS and NMR spectroscopy data. In this study, by changing the cyclization site of linear precursor, we obtained the same cyclic peptide having different conformations as well. We have further synthesized analogues of stylissamide G just by replacing alanine with proline. Structural and biological studies of cyclic Peptide conformers and alanine-substituted analogues were also performed.

Poster ID: T- ICRA-333

Synthesis of New Bioactive Organic Molecules from Pyridine Carboxylic Acids and Their Arene Ruthenium Complexes

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Subject Category: Organic Chemistry

Abstract: Our molecular design is based on the documented activity of isonicotinic acid as a pharmacophoric moiety in order to unleash the new bioactive entities achieved through multistep synthesis. The creation of new structures based on arene ruthenium complexes of this moiety has garnered considerable attention in the field of medicinal chemistry research. Pyridine containing molecules like isonicotinic acid (pyridine-4-carboxylic acid) and nicotinic acid (pyridine-3-carboxylic acid) are small bioactive molecule, which have long been used for designing antibiotics and antituberculosis preparations. In multi-step organic synthesis approach to prepare nicotinoyl and isonicotinoyl containing compounds used 4-benzyloxy,4'-hydroxybiphenyl ligands as linker. Divergent synthetic strategy was employed to prepare the target compounds, which involved multitudinous of reactions ranging from the classical Williamson's reaction to the esoteric steglich esterification, Boc- Protection, Benzylation , Boc- deprotection, Griess diazotization, Sandmeyer reaction and arene ruthenium complexation. The lipophilicity of these macromolecules helped them traverse the cell membrane via EPR effect, subsequently triggering the activation of cellular receptors eliciting a biological response.

Poster ID: T- ICRA-334

Sustainable Dyeing of Chemically Modified Cotton Fabric with Reactive Dyes in Acidic Condition

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Subject Category: Organic Chemistry

Abstract: This study focuses on the improvement of the dyeability of cellulosic fibre with reactive dyes using a sustainable approach. A previously reported organic compound containing two epoxide rings, N^1, N1'-(ethane-1,2-diyl) bis(N^2-(oxiran-2-ylmethyl)ethane-1,2-diamine) (AXL-2) was employed as a cross-linker to the cotton fabric. AXL-2 was synthesized by reaction of epichlorohydrin (ECH) with triethylenetetramine (TETA). Reactive dyeing generates the most problematic effluent containing 20–50% dye in hydrolyzed form (due to the use of alkali in the dyeing process) along with large amounts of inorganic salts. Cotton fabric makes covalent interactions with AXL-2 that subsequently binds with the dye molecule containing free NH2 or OH groups in acidic conditions rendering efficient dyeing in the absence of alkali and salts. ESI-mass spectrometry and ATR–FTIR spectroscopy were used to characterize the synthesized compound, while the treated cotton fabric was characterized by ATR–FTIR and SEM–EDX. Dyeing properties of AXL-2 treated cotton fabrics were studied by measuring their K/S values, % exhaustion, % fixation, and % total fixation. Color difference between treated and untreated fabrics (DE) was evaluated using CIE lab system that exhibited excellent color depth on the AXL-2 treated cotton fabric compared to pristine cotton fabric. Color fastness properties of the dyed fabrics were also studied. No significant change in the absorbency and thermal stability of the fabric was noticed after treatment. The tensile strength of the treated cotton fabric before dyeing (TC) was reduced by 7.6% in warp and 3.5% in weft directions as compared to untreated cotton fabric (CC). However, the treated cotton fabric had lesser reduction in the strength warp-wise compared to untreated cotton fabric after dyeing along with an increase in the strength weft-wise. The novel dyeing method of treated cellulosic fibre by reactive dyes renders a sustainable and eco-friendly approach.

Poster ID: T- ICRA-335

DABCO-F Based Ionic Liquids as Sustainable Approach for the Syntheses of Heterocycles

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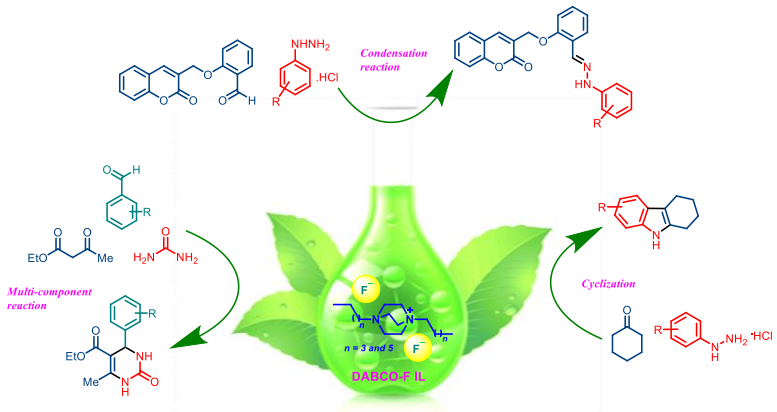
[b]*Third World Center for Science and Technology, International Center for Chemical and Biological Sciences, University of Karachi, Karachi and Pakistan*

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Subject Category: Organic Chemistry

Abstract: Green chemistry refers to a fundamental approach for developing chemical products in an environment-friendly way, which helps us to explore efficient methods. Though synthesis is the core for preparing materials in industries, including pharmaceuticals, supramolecular, *etc*. However, most of the conventional methods utilize catalyst or solvents that are hazardous to the environment. On contrary, green methods offers sustainable approach for preparation of compounds efficiently. In particular, ionic liquids (ILs) are considered as green solvents that act as catalyst as well as reaction media. They serve as an alternate to traditional solvents and have gained immense attention of chemists since their discovery. They offer several advantages such as ease of handling, recyclability, storage, thermal stability, non-volatility, and their dissolving ability. Thus, the use of ILs in organic syntheses is one of the sustainable methods. Herein, two new ILs are prepared *via* alkylation of DABCO followed by halide-exchange. Their decomposition temperatures were measured *via* TGA/DSC; while quantum mechanical calculations were performed to obtain their configuration and stability. Also, cytotoxicity studies of these ILs also showed them as safe reagents. Thus, efficiency of these ILs was explored *via* preparing different libraries of heterocycles utilizing condensation, cyclization, and multi-component reactions. Results, when compared with conventional methods, revealed high yields for products in ILs; highlighting their importance as efficient media.

Graphical Abstract:



Poster ID: T- ICRA-336

Chemical Modification of Cellulose and Polyester/Cellulose Blended Fabric to Make It Disperse Dyeable; A Sustainable Approach to Achieve Dyeing of P/C Blend with Disperse Dyes

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[b]Textile Institute of Pakistan, Karachi-75020, Pakistan

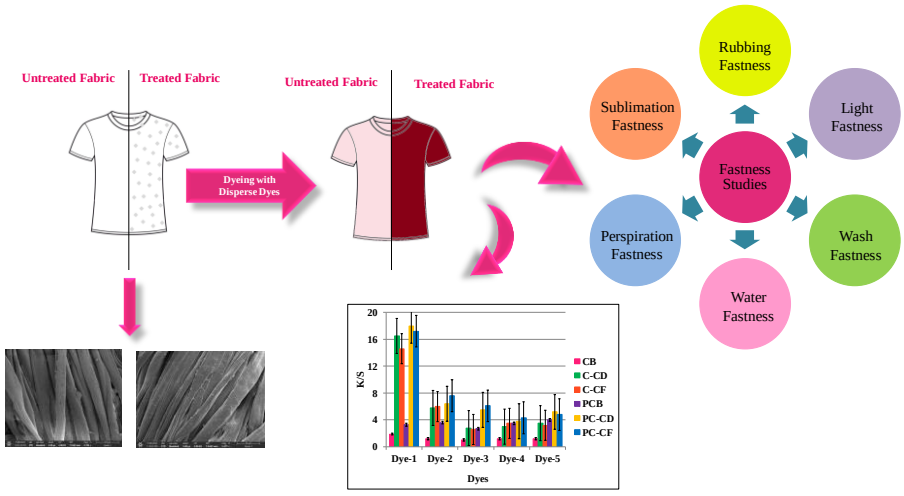
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Subject Category: Materials Chemistry

Abstract: An attempt was made to modify the cotton and polyester/cotton (P/C) fabric with a bi-functional crosslinking compound, hexamethylene diisocyanate, to covalently link the cellulosic part of the fabric with the selected disperse dyes. The chemical composition, morphology, and strength of the modified fabric were examined by ATR-FTIR, SEM-EDX, tensile strength testing, water absorbency test, and thermogravimetric analysis (TGA). The results verified the binding of the crosslinker with the cellulose fabric under neutral conditions. The crosslinked fibres were dyed with commercially available disperse dyes bearing amino and/or hydroxyl groups by using pad-dry-cure dyeing method. The K/S values were measured to analyze the depth of shade on the modified fibers in comparison with the respective untreated dyed samples. The fastness properties of the dyed fabrics were studied by employing ISO standard protocols. It was observed that the modified fabrics have a good overall fastness.

GRAPHICAL ABSTRACT



Poster ID: T- ICRA-339

Carbazole-Triphenylamine based Polyimides for Batteries and Supercapacitors

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Subject Category: Organic Chemistry / Material Science

Abstract: The burgeoning energy demands set of by human population is causing serious depletion of fossil fuels due of their limited reserves along with increasing environmental concerns that sparked intensive investigations on sustainable and efficient energy conservation or power generation by electrochemical organic reactions, use of metal-ion batteries, metal-air batteries, fuel cells and supercapacitors. Polyimide (PI) with dense aromatic structure has gained considerable attention due to its facile and economical synthesis, as well as good chemical and thermal stability. Particularly, aromatic polyimides with carbonyl groups and reasonably high thermal/mechanical stability, are becoming significantly suitable for supercapacitor and battery applications. Carbazole linked triphenylamine based diamine (DA) was successfully synthesized in high purity and good yield and further employed for the preparation of series of neat and co-polymides. The prepared polyimides are used as substrates for the CO_2 Laser scribing and further employed to develop bendable supercapacitors. In general, abundant N atoms exist in the molecular chain of PI, thus ensuring the formation of in situ N-doped carbon skeletons after pyrolysis. Different molecular structures of the polyimides led to the development of different porous graphene formation hence different capacitance was observed. Furthermore, we have also investigated these prepared polyimide materials for their applications as fire retardant materials in Zinc/Lithium / Sodium-ion batteries.

Poster ID: T- ICRA-340

One Pot Solvothermal Synthesis of Zr-MOF and DNA@MOF Composite for improved Current Density Towards Oxygen Evolution

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Subject Category: Organic Chemistry / Physical Chemistry

Abstract: The development of efficient and sustainable electrocatalysts is crucial for advancing renewable energy. To address the sluggish Oxygen Evolution Reaction (OER), this study presents a novel method for synthesizing a Metal-Organic Framework (MOF) using zirconium (Zr) and fumaric acid, integrating DNA for enhanced stability and catalytic activity. Several characterizations confirmed the structural, compositional, morphological, textural, and electrochemical properties of the composite material. The DNA@MOF exhibited exceptional OER performance, with an onset potential of 1.49 V, an overpotential of 294 mV, and a low Tafel slope value of 59 mV/dec. Electrochemical impedance spectroscopy (EIS) showed a low charge transfer resistance of 1.2 Ω and the composite material demonstrated long-term stability over 50 hours and a high electrochemical active surface area of 411.25 cm^2. This research highlights the potential of Zr-MOF-based composites as efficient, durable electrocatalysts for water splitting, offering a promising alternative to noble metal-based catalysts for energy conversion and storage.

Poster ID: T-ICRA-343

Electrochemical Performance of Carbon Modified LiNiPO4 as Li-Ion Battery Cathode: A Combined Experimental and Theoretical Study

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Subject Category: Physical Chemistry Material Science

Abstract: This study demonstrates the synthesis of olivine $LiNiPO_4$, and carbon modified $LiNiPO_4$ (LNP/C-composites) cathode materials for use in lithium-ion batteries (LIBs) synthesized via non aqueous sol-gel process. The LNP/C-composites were fabricated through high energy ball-milling of LiNiPO4 with different weight ratios of conductive carbon black. The electrochemical performance of LiNiPO4 has been considerably improved by modifying the material with conductive carbon black which enhanced cathode performance as thoroughly studied by electrochemical analysis. Discharge capacities of LNP/C-composite cathodes with 25 wt% carbon were 175 mAh g^{-1}, 150 mAh g^{-1} and 125 mAh g^{-1} with corresponding capacity retention of 82.7%, 84.1% and 82.2% after 100 cycles at 0.05C, 0.1C and 1C rates, respectively. High-temperature electrochemical impedance spectra correspond to decreased charge transfer resistance with increased electronic conductivity and minimum cell polarization for the LNP/C powders. Additionally, the inflow of lithium-ion flux in cathode particle was simulated by using phase-field modeling indicating the coexistence of Li-poor and Li-rich phases during charging and discharging processes. The findings are significant for the development of optimal battery electrode materials as the methodology and insights used are readily transferable to other ion insertion-based electrodes.

Poster ID: T- ICRA-345

Ultrasonic Nanoconfinement-Mediated Highly Dispersed Ni/Co- Nanoparticles for Enhanced Water Electrolysis

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Subject Category: Material Science

Abstract: Developing a sustainable electrochemical water splitting (EWS) system requires a catalyst that is synthesized from cost-effective and abundant sources. This study demonstrates a facile ultrasound nanopore-confinement method for growing well-dispersed Ni/Co nanoparticles inside pores of a high surface area porous carbon nitride (PCN) network. Incorporating subnanmt Ni or Co into PCN significantly reduces the onset potential for the OER and HER compared to pure PCN, showing it to be an efficient bifunctional electrocatalyst for overall water splitting. Co-f-PCN and Ni-f-PCN electrocatalysts exhibit improved onset potentials for OER at 1.38 and 1.40 V (vs. RHE) respectively, compared to PCN at 1.43 V (vs. RHE). Co-f-PCN demonstrates high efficiency as a bifunctional electrocatalyst, requiring very small overpotentials of 210 and 70 mV vs. RHE to drive the OER and HER at 10 mA cm-2, respectively. Co-f-PCN an efficient electrocatalyst for overall water splitting, achieving a current density of 10 mA cm-2 at a cell voltage of 1.5 V, in a two-electrode system with a stable response for up to 24 h. Furthermore, PCN, Co-f-PCN, and Ni-f-PCN catalysts all show good long-term durability.. This method effectively integrates Ni or Co, especially Co into high-N content PCN, significantly enhancing electrocatalytic OER and HER performance.

Poster ID: T- ICRA-346

EDTA Functionalized Wheat Straw Based Adsorbents For The Removal Of Crystal Violet Dye Kinetic Isotherm And Mechanistic Studies

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Subject Category: Physical Chemistry

Abstract: .Wheatstraw based adsorbent functionalized with EDTA (MWS) was prepared and characterized by different spectroscopic, microscopic and analytical methods including Fourier transform infrared (FTIR) spectroscopy,Scanning electron microscopy (SEM), and X-ray powder diffraction (XRD), thermogravimetric analysis (TGA),Energy Dispersive X ray (EDX) and point of zero charge (PZC) .The adsorption of crystal violet (CV) dye on MWS was optimized for several parameters, like pH of medium, time of equilibrium, amount of adsorbents, concentrations and solution temperature. To investigate the rate of adsorption and reaction mechanism, kinetic experiments were carried out at 298, 308 and 318K and 328 K ±1K .The results of kinetic modeling, adsorption isotherms, and thermodynamic parameters showed that pseudo-second-order and Langmuir isotherm gave a better fit to the kinetics and adsorption data . The sorption process was spontaneous and endothermic.

Poster ID: T- ICRA-347

Tannic acid strengthen adhesion of poly(AAm-co-GG) hydrogels for multiple solid surfaces repairing

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Abstract: In this study Copper-nickel mixed oxide (CuO-NiO) was prepared by co-precipitation method. CuO-NiO was modified by Amberlite IR120 (R) resin to prepare CuO-NiO Amberlite IR120 resin modified (CuO-NiO-R) composites. The synthesized material was used as adsorbents for Basic Blue 41 (BB41) removal from aqueous medium. The adsorbents were characterized by FT-IR, SEM, EDX, TGA, and XRD. Different experimental parameters like contact time, adsorbates concentration, dosage of adsorbent, pH effect, and temperature were investigated. Salt addition method was employed to determine the point of zero charge (pH_{PZC}) of CuO-NiO-R composites at room temperature. A significant effect of pH was observed on the adsorption of Basic Blue 41 . Different kinetics and isotherm models were applied. The best fitting models were pseudo second order and Langmuir with correlation coefficient R^2 >0.99. The calculated maximum adsorption capacity (qmax) obtained from Langmuir model was 124 mg g^{-1}. Negative values of ΔG° showed that the adsorption process is spontaneous in nature. The positive values of ΔH° showed the endothermic nature of adsorption process onto CuO-NiO-R. The used adsorbent were regenerated and used for five consecutive adsorption-desorption cycles.

Poster ID: T- ICRA-348

Extraction and Purification of Value-Added Chemical from Apricot Seed

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Subject Category: Organic Chemistry

Abstract: In recent years, environmental concerns and resource depletion have underscored the need for innovative waste management and resource recovery approaches. Chemical recycling and up-cycling processes have emerged as promising strategies for converting organic waste materials into valuable chemicals, promoting a sustainable and circular economy. This thesis focuses on the extraction of value-added chemicals from waste sources, specifically apricot kernels, highlighting their dual benefits of waste reduction and high-quality chemical production. Leveraging various pretreatment technologies and catalytic systems, this study was aimed to optimize conversion processes and enhance the yield of target chemicals such as levulinic acid from renewable feedstock. Furthermore, the extracted chemical was utilized to synthesize a biofuel precursor i.e. Valeric acid. The nature of the synthesis was electrochemical reduction. This process not only enhances the value of the extracted chemical but also contributes to the development of sustainable biofuels. The electrochemical reduction method offers a controlled and efficient pathway for the synthesis of valeric acid, showcasing the potential of integrating waste-derived chemicals into renewable energy applications

Poster ID: T- ICRA-349

MnSe/MnO Heterostructure Electrocatalyst: Emphasizing the Exceptional OER performance and Stability

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Subject Category: Materials Chemistry

Abstract: The development of advanced catalytic materials is indispensable for effective energy conversion through water splitting. The first of its kind MnSe/MnO heterostructure is presented in manuscript that is synthesized using a hydrothermal technique. The heterostructure demonstrates remarkable performance by accomplishing an overpotential of just 250 mV at high value of current density (40 mA/cm^2) for oxygen evolution reaction (OER), together with a small value of Tafel slope, which is only 27.97 mV/dec (even lower compared to benchmark catalysts). The extraordinary catalytic accomplishment and stability of the heterostructure are largely due to the additive relationship between MnSe and MnO. The MnSe, a semiconductor, provides abundant active sites for OER, while MnO acts as an insulating stabilizer, protecting MnSe from dissolution and degradation during the OER process. The Mn d5 electronic configuration is instrumental in enabling smooth transitions between various oxidation states, thus facilitating electron transfer necessary for water oxidation. Additionally, the high spin state associated with Mn d5 system, characterized by five unpaired d-electrons, significantly enhances the electron transfer efficiency, promoting oxygen gas evolution. This work offers a novel Mn-based catalyst with remarkable OER properties and sheds light on the development and application of cutting-edge catalysts for water oxidation reactions.

Poster ID: T- ICRA-350

Investigating Interface Inhibitory Mechanisms of Anti-Inflammatory Agents Targeting Interleukin-1β to Alleviate Depression

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Subject Category: Organic Chemistry

Abstract: Interleukin-1β (IL-1β), a major inflammatory mediator affecting memory and mood, is linked to depression. Despite many inhibitors, no specific drug has been effective, signaling a need for new treatments. A multifaceted virtual screening strategy including ligand-based pharmacophore modeling, virtual screening, and molecular docking was used to find novel compounds inhibiting interaction of IL-1β with its cognate receptor IL-1RI. This led to the identification of six promising compounds, further analyzed through molecular dynamics simulations to understand their behavior. These simulations showed that the compounds stabilized within IL-1β targeted pocket, potentially blocking the formation of an active heterocomplex and disrupting associated signaling by interacting with key residues such as Gln15, Asp120, Gly140, Glu128, Asp142, Leu62, Val141, and Val140. Compound-283 and Compound-157 exhibited the highest binding energies at −174.04±14.3 and −162.34±11.9 kcal/mol, respectively, indicating strong interactions with IL-1β. Molecular simulations confirmed these stable interactions, primarily driven by polar solvation and Van der Waals forces. In vitro tests further validated these results, with Compound-157 and Compound-283 showing IC_{50} values of 1.6±0.1 and 9.1±1.7 µg/mL, respectively. As IL-1β is a key mediator in an array of behavioral actions, the blockade of IL-1β could provide a viable therapeutic option for stress-related mood disorders including depression.

Poster ID: T- ICRA-351

Preparation of Silk Protein Sericin-based Wound Dressing Matrix and Evaluation of its Antimicrobial Activity

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Subject Category: Organic Chemistry

Abstract: The present research work is a green approach for the utilization of waste silk protein sericin blended with plant extracts and silver nanoparticles to produce wound dressings. According to different surveys about 50,000 tons of silk sericin is disposed of as wastewater every year worldwide. Silk sericin plays a promising role in wound healing by enhancing the growth of human skin cells. In this work, silk sericin protein was successfully extracted from the waste water of sericulture industry and was lyophilized to get the silk protein sericin. Silk sericin was characterized using UV-visible spectroscopy, TGA and FT-IR to know the thermal behavior as well as structure of silk sericin. Then 5% w/v of sericin was blended with 10% w/v of PVA, and 2% w/v of chitosan in order to improve the mechanical properties of the silk sericin protein so it can be used to prepare wound healing film. Azadirachta indica modified silver nanoparticles were synthesized and characterized using UV-visible spectroscopy, FT-IR, and zeta potential analyzer. Results suggested the successful synthesis of nanoparticles giving the zeta potential of -34 Mv. 0.5 mL of NPs were added into the PVA, sericin and chitosan blended mixture making a ratio 8:1:1:0.5. 0.1 mL extract of marigold, honey, and aloe vera was added into each separate mixture and termed as M1, H1, and A1 to improve the antibacterial activity of the wound healing film. The prepared films M1, A1, and H1 were again analyzed by the FT-IR to observe the structural modification. Nutrient broth assay was carried out and 2% and 5% of M1 and H1 exhibit excellent antibacterial activity, whereas, only 5% A1 in the broth inhibited the growth of bacteria. So, this film A1, M1, and H1 can be used as a wound healing matrix.

Poster ID: T- ICRA-352

Facile synthesis of Carbon Supported Cu_3P nano-catalyst for Selective Hydrogenation of nitroarenes to amines

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Subject Category: Inorganic Chemistry

Abstract: In several facets of the hydrogen economy, from the evolution of hydrogen to reactions of selective hydrogenation, transition metal phosphides have recently been recognized as potential substitutes for rare noble metal catalysts. But the need for dangerous and costly phosphorus precursors and limited support dispersion of the metal phosphide substantially lowers the appeal and performance of metal phosphides in catalysis. Here in, we demonstrate the use of a supported copper phosphide nanostructure (Cu_3P/C@800) as a highly effective, selective and reusable heterogeneous catalyst for selective hydrogenation of nitroarenes. A facile and green approach (i.e., single source precursor) was used to synthesize Cu_3P/C@800. In order to promote highly exposed active reaction sites, the Cu_3P was supported on activated carbon (AC). In this study, the Cu_3P/C@800 catalyst achieves superior cyclic stability, 100% conversion, and >98% selectivity of substituted nitroarenes. The as-synthesized catalyst also sustains its catalytic efficiency during reusability tests.

Poster ID: T- ICRA-353

Nanotech Solutions: Adsorption and Sensing of Biomolecules and Pollutants with Biomass Derived Green and Cost-effective Composites

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Subject Category: Material Science/ Analytical Chemistry

Abstract: Green chemistry aims to develop sustainable technologies with minimal environmental disruption. The present study explored the prospects of synthesis of magnetically modified nano-biosorbents derived from biomass like Banana peels and Pinus needles for pollutant removal and chemical/antioxidants sensing. These biosorbents were prepared using a simple and sustainable co-precipitation method integrated with magnetite (Fe_3O_4) to enhance their functional properties. The nanocomposites showed peroxidase-like activity for hydrogen peroxide detection in environmental samples and efficient adsorption of methylene blue dye from wastewater. The optical sensing platform based on magnetized pine needles and banana peels provided rapid and selective H_2O_2 detection with a limit of detection as low as 0.007μM and 0.008μM, respectively. Adsorption experiments displayed high dye removal efficiencies, with an excellent 1193 mg/g for pine needle-based and 296.4 mg/g for banana peel-based composites. This research underscores the potential of biomass based green nanomaterials in dual environmental as well as biological applications, offering a cost effective and sustainable approach for pollutant monitoring and wastewater treatment.

Poster ID: T- ICRA-354

Identification of Mutant Anaplastic Lymphoma Kinase Inhibitor by Utilizing Integrated *In-silico* Approaches

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Subject Category: Computational Chemistry

Abstract: Non-small-cell lung cancer (NSCLC), driven by the ALK mutation, is a leading global cause of death. Despite the development of ALK inhibitors, the emergence of the I1171N + L1198H mutation confers resistance led to limited therapeutic options. The emergence of drug resistance and the need for extremely selective inhibitors have prompted the exploration of novel therapeutic strategies. To address this, we developed a pharmacophore based on first, second, and third-generation ALK inhibitors to screen chemical data base for potential ALK inhibitors. Compounds matching the pharmacophoric features have been assessed against the I1171N + L1198H ALK mutant. Molecular Dynamics simulations were used to evaluate dynamic stability and Density Functional Theory was utilized to examine chemical reactivity of promising hits. The MM-PBSA method determined the binding free energy of the protein-ligand complexes. With reasonable pharmacokinetic profiles, steady dynamic behavior, attractive interactions, and promising affinities, five chemical scaffolds were found to have the potential to be further optimized as ALK inhibitors. This study provides a foundational framework for developing novel inhibitors targeting ALK resistance mechanisms in NSCLC, thereby contributing to the advancement of therapeutic strategies.

Poster ID: T- ICRA-355

Integrating Dynamic Molecular Insights and Inhibitory Strategies for Interleukin-17A Interface: A Computational Approach

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Subject Category: Organic Chemistry

Abstract: IL-17A, a key pro-inflammatory cytokine, implicated in a broad spectrum of immune responses. Dysregulation of IL-17A contributes to multiple pathological conditions, from mild inflammation to cancer. When IL-17A binds to its receptor IL-17RA, it forms a complex that triggers molecular signaling cascades linked to inflammation. Therefore, targeting IL-17A and blocking its interaction with IL-17RA presents a unique opportunity for developing new therapeutic strategies. Currently, there are no oral-specific drugs available, emphasizing the urgent need for new anti-inflammatory treatments to combat autoimmune and inflammatory diseases. This research employs ligand-based pharmacophore modeling, virtual screening, and molecular docking simulations, which resulted in identification of six promising compounds targeting IL-17A/IL-17RA interface. Molecular dynamics simulations demonstrated the stability and inhibitory potential of these compounds by revealing interactions within the IL-17A binding site, effectively locking the associated signaling cascade via key interfacial residues (Arg39, Trp51, Trp67, Gln94, Glu95, Leu97, Leu99, Lys114, and Ser118). Binding free energy calculations further confirmed that the identified hits, particularly compounds **1, 3, 4,** and **6,** exhibit higher affinities than the native inhibitor (7AMA). These potent IL-17A inhibitors show significant promise as potential drug candidates for treating inflammatory diseases.

Poster ID: T- ICRA-357

Synthesis and Characterization of Metal-Doped Brushite Cement for Bone Regeneration

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Subject Category: Material Science

Abstract: The major goal of this research is the synthesis and characterization of metal-doped brushite cement for bone regeneration, introducing a novel class of biomaterials with significant biomedical applications. Brushite, or dicalcium phosphate dihydrate, is widely used in bone cement for tissue regeneration. The study synthesizes brushite doped with zinc (Zn^{2+}) and silver (Ag^{+}) ions using standard precipitation and continuous flow methods. Characterization was performed using powder X-ray diffraction (PXRD), Fourier transform infrared spectroscopy (FTIR), and scanning electron microscopy (SEM), along with bioassays to assess antibacterial and antifungal properties. Initial testing indicated non-toxicity and potential for mineralized tissue engineering. PXRD and FTIR confirmed the incorporation of Zn and Ag ions, significantly altering the structural properties of brushite. SEM imaging showed nanocrystalline structures with random flake-like and plate-like forms prone to agglomeration, and particle size analysis revealed an approximate size of 90 nm. Ag-doped brushite exhibited strong antimicrobial activities, suppressing fungal growth and showing considerable antibacterial effects on Bacillus thuringiensis and moderate effects on Escherichia coli. Zn-doped brushite also demonstrated notable antifungal properties. These findings suggest that metal-doped brushite compounds, particularly those doped with Ag^{+} and Zn^{2+}, hold promise as active materials for bone regeneration.

Poster ID: T- ICRA-359

Synthesis and characterization of Ecofriendly synthesized EDTA-coated Iron oxide Nanoparticles for removal of heavy metals from Wastewater

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Subject Category: Material Science

Abstract: Globally, water pollution is increasing due to the rapid expansion of industrial activity and human population. The major contribution of heavy metals (Pb, Cd, and Cr) can be extremely toxic and contribute significantly. Nanotechnology has provided potential options for efficient and cost-effective heavy metal extraction from water. In this study, bared iron oxide nanoparticles (IONPs) and EDTA-coated IONPs were synthesized by aqueous guava leaf extract using the green co-precipitation method to remove heavy metals (Cr6+ and Pb2+) from wastewater. Several analytical approaches were used to characterize the nanoparticles, including UV-Vis, FTIR, PSA, Zeta potential, VSM, SEM, and XRD. Characterization techniques confirmed the crystalline nature of functionalized IONPs with spherical morphologies. FTIR analysis indicated biomolecule involvement in nanoparticle formation. Batch adsorption experiments systematically investigated the influence of pH, contact time, initial metal ion concentrations, and adsorbent concentration on Cr6+ and Pb2+ adsorption. Maximum Cr6+ removal occurred at pH 2.0 (84.3%) and Pb2+ at 4.0 (99.1%), respectively. Adsorption kinetics followed a pseudo-second-order model while equilibrium data fit the Langmuir isotherm. The results of this work indicate that plant-mediated EDTA-coated IONPs are promising potential adsorbents and exhibited remarkable reusability for metal ions removal in wastewater treatment.

Poster ID: T- ICRA-360

Nanofabrication of Metal-Based Electrospun Nanomembranes for Sustainable Water Purification

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Subject Category: Material Science

Abstract: Contaminated water is one of the major issues around the world. Urbanization and industrialization are causing water contamination, threatening livelihoods and health. This paper uses the coprecipitation method to explore the synthesis of **(metal Oxide)** like AgO, CuO, ZnO, and MnO nanoparticles using a green source (Eucalyptus camaldulensis, Phyllanthus emblica) as a reducing agent. Incorporating these nanoparticles into the electrospun nanomembrane is a modified membrane version for enhancing water purification. Electrospun nanofibrous membranes (ENMs) are a novel approach that outperforms traditional membranes in terms of flow and rejection rates. The sewage and water purification processes are being revolutionized by these ENMs because of their energy-efficient, lightweight, and economical characteristics. These nano-membranes have a great deal of potential for use in water treatment applications. Characterization of the membrane and nanoparticles was conducted using XRD, SEM+EDX, FTIR, PSA, and UV-visible, confirming the successful fabrication and uniform distribution of nanoparticles within the membrane. These metal oxides used in nanomembrane exhibit remarkable antibacterial and antifungal activity against microorganisms. The aim was to create environmentally friendly and effective nanomembranes for water purification. The progress of electrospun nanomembranes is expected to promote the prosperity and utilization of various industries for water treatment.

Poster ID: T- ICRA-361

Synthesis and Characterization of Copper Nanoparticles By Using Punica Granatum, Musa acuminata, and Mangifera Indica Peel Extracts for Biomedical Applications

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Subject Category: Material Science

Abstract: The ongoing progress in nanotechnology has highlighted the significance of creating metallic nanoparticles by utilizing their physicochemical characteristics. Copper nanoparticles, or CuNPs, are especially significant, but have drawbacks like toxicity, agglomeration, and instability. The majority of CuNPs are currently manufactured in laboratories due to high costs and contamination concerns, which hinder industrial production. Overcoming these obstacles requires a sustainable method of producing pure CuNPs. This study uses extracts from banana (Musa acuminata), mango (Mangifera indica), and pomegranate (Punica granatum) peels as reducing and stabilizing agents in the green synthesis of copper oxide nanoparticles (CuO NPs). This approach is inexpensive, dependable, and eco friendly. It does, however, confront obstacles such as microbiological contamination, seasonality, cost, and environmental considerations about fruit peels. CuO NPs are being synthesized for sustainable photocatalytic technology, and their potential biomedical applications—such as medication delivery, tissue engineering, antimicrobial, anti-cancer, and biosensors—are being investigated. Fruit peel extracts offer a green synthesis process that lowers manufacturing costs and has a less negative impact on the environment. The difficulties associated with extracting and storing nanoparticles will be examined in this work, which will ultimately aid in the creation of efficient and sustainable biomedical applications. This research intends to manufacture high-quality CuO NPs that are sustainable and appropriate for a variety of cutting-edge biomedical applications by utilizing the special qualities of peel extracts from bananas, mangoes, and pomegranates.

Poster ID: T- ICRA-362

Nucleoside derivatives as new antibiofilm agents against *Staphylococcus aureus*

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Subject Category: Biochemistry

Abstract: *Staphylococcus aureus* is a major human pathogen, responsible for a number of diseases. *S. aureus* associated infections are usually acquired from community as well as from hospital settings, and are difficult to treat because of the emerging resistance against available antibiotics. One of the key reasons for their resistance is the biofilm formation. Biofilm is a microbial community composed of cells irreversibly bound to a surface, which are capable of producing an extracellular polymer matrix, and thus, display an altered phenotype. This biofilm prevents antibiotics to physically reach the bacterial cells. Therefore, targeting biofilm formation is a promising approach for the development of novel drugs. In this study, a number of diverse classes of natural products and their synthetic analogues were systematically screened against *Staphylococcus aureus* biofilm by using 96-well plate crystal violet assay. Among them, some candidates from nucleoside class were found to be active against biofilm formation. Compound **1, 2** and **3** significantly inhibited biofilm formation showing ≅90% inhibition. The effect of these compounds was also evaluated at molecular level by using Real Time-Polymerase Chain Reaction (RT-PCR) to see the change in expression of *ica A, ica D,* and *fnb* genes which are involved in biofilm formation. Moreover, microscopic techniques were also used to see the changes in biofilm morphology.

Poster ID: T- ICRA-363

Synthesis and Characterization of Analogs of Natural Product Chartergellus- CP1 Peptide and their Anti-Proliferation Effect on Cancer Cell lines

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Subject Category: Biochemistry

Abstract: Cancer is a life-threatening disease that occurs as a result of the uncontrolled proliferation of cells. Prostate, cervical and breast cancers are the most prevalent issues in females and males, respectively. The side effects and the ineffectiveness of traditional therapies have drawn the attention of researchers to new cancer therapeutic strategies. Peptides are emerging as an alternative potential therapeutic candidate against cancer due to various advantages over the traditional therapies particularly due to non-specificity of current treatments. Chartergellus-CP1 peptide was isolated from the social wasp (Chartergellus communis) venom and reported as potential anti-seizure peptide. The sequence was confirmed by Edman degradation and molecular weight was confirmed by mass spectrometry. In the current study, the Chartergellus-CP1 peptide and its analogs were synthesized on a solid support using rink amide resin, and their masses were confirmed by LRFAB-MS, and LRESI-MS, the structure of the synthesized peptides was elucidated by NMR studies. The anti-proliferation potential of the Chartergellus-CP1 and its analogue was evaluated by MTT assay on HeLa, PC3 and MCF-7 cancer cell line in-vitro. The Chartergellus-CP1 peptide and its analogs were found to possess anti-proliferative activity specifically on cancer cell lines, while the synthesized peptides were non-cytotoxic on normal skin fibroblast cell line (BJ cells) with an effective dose. Moreover, anti-proliferation activity of most potent Chartergellus-CP1 was confirmed by microscopy, ROS measurement, and FACS technique. The synthesized peptide CP1 and CP1-LA found to have potential for future anticancer drug development.

Poster ID: T- ICRA-365

Design and Evaluation of Propanehydrazide Derivatives as Lipoxygenase Inhibitors for Anti-Inflammatory and Anticancer Therapy

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Subject Category: Organic Chemistry, Analytical Chemistry

Abstract: In inflammation and cancer, lipoxygenase (LOX) is the major enzymes. The lipoxygenases are found within a human body and have a significant impact on triggering inflammatory responses. When there is an excessive presence of the reactive oxygen species, it can lead to an inflammation, which triggers the liberation of cytokines and so activates lipoxygenases. LOXs play a role in producing leukotrienes and prostaglandins, compounds associated with the progression of the disease. Thus, inhibiting the activity of LOXs is considered a crucial step in preventing the development of such diseases. Several inhibitors of LOX have been synthesized to treat these complications, but many of these inhibitors have failed to pass the clinical trials due to its poor toxicity and pharmacokinetics properties. In this case, this research has performed virtual screening to design the propane hydrazide derivatives. Pharmacophore-based modeling has been utilized to synthesize the compounds with hydrazide ring and functional groups. In addition, LOX was extracted from different sources to determine the activity of these derivatives **1(a-q)** against this enzyme. The observed lowest IC50 values for LOX was 4.05 ± 0.30 µM and 3.65 ± 0.71 µM. The results for molecular docking showed good interactions of compounds with LOX. It was found that these compounds have the potential to interact with different amino acid residues. In addition, molecular simulations suggested that the lead compounds **1i** and **1n** are stable compounds. Therefore, it was concluded that **1i** and **1n** can be used as a potent compound against LOX. These two derivatives can act as a drug candidate for the treatment of inflammation as well as cancer.

Poster ID: T-ICRA-366

Discovery of Hydrazone Derivatives as Potent Inhibitors of Beta Secretase 1 for the Management of Alzheimer's Disease

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Subject Category: Organic Chemistry, Analytical Chemistry

Abstract: Alzheimer's disease, a neurodegenerative disorder, is a major healthcare concern due to its profound effects on brain function, associated with the build-up of amyloid-β (Aβ) peptide as amyloid plaques in the brain. This excessive buildup of Aβ peptides is attributed to the uncontrolled activity of a rate limiting enzyme namely beta secretase 1 (BACE1). By inhibiting this enzyme, the formation of Aβ can be limited; therefore, the disease can be prevented. In this regard, novel beta secretase 1 inhibitors (5a-o) have been synthesized and evaluated. Among these inhibitors, 5c, 5d, 5g and 5i presented good inhibitory potential; however, 5d was the most potent of all with an IC50 value of 2.81 ± 0.02 µM. Molecular docking evaluation via BioSolveIT suit predicted that the binding energy of 5d in complex with beta secretase 1 was -7.4 kcal/mol and its binding affinity was in micromolar range indicated by HYDE, SeeSAR. Moreover, pre-clinical investigation via SwissADME proved its druggable properties and rendered it specific with no PAINS alerts. At last, it was observed that 5d obstructs the binding site of beta secretase by interacting with residues among which Tyr71 is of prime importance because it is the part of beta secretase 1 flap. This flap has role in promoting the closed confirmation of beta secretase 1 upon interaction with compound. Consequently, 5d is the most potential inhibitor of beta secretase 1 in Alzheimer's disease but in vivo analysis is required to further validate the results.

Poster ID: T- ICRA-367

Evaluation of antifungal potentials of Albizia kalkora extract as a natural fungicide: In vitro and computational studies

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Subject Category: Analytical Chemistry

Abstract: The antifungal bioactivity potential of the organic extract of silk tree (Albizia kalkora) was investigated in the current study. The crude extracts of A. kalkora and methanol, n-hexane, chloroform, and ethyl acetate fractions were prepared. The antifungal activity of obtained fractions of A. kalkora was studied at different concentrations ranging from 0.39-50 µg/mL. Dimethyl sulfoxide (DMSO) was taken as a toxicity control, whereas thiophanate methyl (TM) as a positive control. All the fractions significantly reduced the FOL growth (methanolic: 9.49–94.93 %, n-hexane: 11.12–100 %, chloroform: 20.96–91.41 %, and ethyl acetate: 18.75–96.70 %). The n-hexane fraction showed 6.25 µg/mL MIC as compared to TM with 64 µg/mL MIC. The non-polar (n-hexane) fraction showed maximum antifungal bioactivity against FOL in comparison with chloroform, methanol, and ethyl acetate fractions. GC/MS analysis exhibited that the n-hexane fraction contained hexadecanoic acid, 9,12,15-octadecatrienoic acid, 9,12-octadecadienoic acid, bis(2-ethylhexyl) phthalate, methyl stearate, and [1,2,4]triazolo[1,5-a]pyrimidine-6-carboxylic acid. The results of in vitro antifungal inhibition were further reinforced by molecular docking analysis. Five virulence proteins of FOL i.e., pH-responsive PacC transcription factor (PACC), MeaB, TOR; target of rapamycin (FMK1), Signal transducing MAP kinase kinase (STE-STE7), and High Osmolarity Glycerol 1(HOG1) were docked with identified phytocompounds in the n-hexane fraction by GC/MS analysis. MEAB showed maximum binding affinities with zinnimide (−12.03 kcal/mol), HOG1 and FMK1with α-Tocospiro-B (−11.51 kcal/mol) and (−10.55 kcal/mol) respectively, STE-STE7 with docosanoic acid (−11.31 kcal/mol), and PACC with heptadecanoic acid (−9.88 kcal/mol) respectively with strong hydrophobic or hydrophilic interactions with active pocket residues. In conclusion, the n-hexane fraction of the A. kalkora can be used to manage FOL.

Poster ID: T- ICRA-368

Fabrication of antimicrobial cloth by in-situ synthesis of Cu-MOF on Ti/CNT composite and its deposition on NaOH treated cotton fabric

Misbah Iqbal, Fatima Rana, Tanzeela Nazar, Manzar Sohail

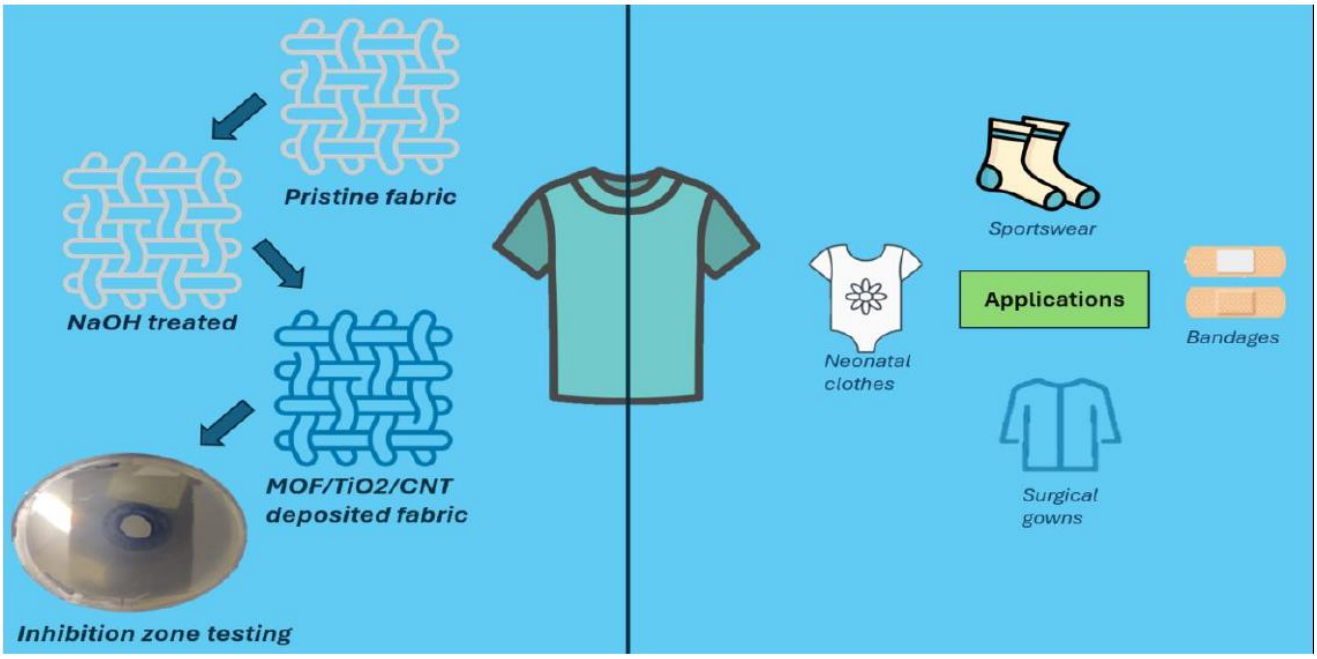
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Subject Category: Organic Chemistry

Abstract: Fifteen out of every hundred patients will contract at least one health-acquired infection (HAI) during their hospital stay and on average one in every ten affected patients will die. Introducing antibacterial fabrics into healthcare settings effectively reduces the transmission of HAIs thereby improving patient's health and well-being. Additionally, sustainable, self-cleaning textiles will conserve the environment by minimizing chemical usage and waste generation. The final composite deposited on the fabric is a Cu-MOF and Ti based hybrid using carbon nanotubes as a support material. It displayed best of both precursors; efficient antimicrobial properties and recyclability with an inhibition zone of 7.5-8 mm and 6-6.5 mm radius against *Staphylococcus* and *E. coli* respectively. Our product is a recyclable, non-cytotoxic and effective antimicrobial fabric that may help health care professionals eliminate the risk of hospital acquired infections by preventing surface and aerial transfer of bacterial and fungal infections from one patient to the other.

Graphical Abstratct:



Poster ID: T-369

Green synthesis of antioxidant, polymeric, rod-shaped lanthanum oxide nanomaterials from *Polygonum minus* leaf extract

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Subject Category: Inorganic Chemistry and Material Science

Abstract: Metal nanoparticles synthesis via green techniques is an innovative research field. This research focus on utilizing a simple and speedy green technique for producing biogenic lanthanum nanoparticles from *Polygonum minus* leaf extract as a reducing agent. The reduction of La^{+3} to elemental La was done at room temperature. The size of nanoparticles was determined primarily by leaf extract concentration and pH. The characterization of the synthesized nanoparticles was accomplished through FT-IR, EDX, FE-SEM, PXRD, and UV-Vis spectroscopy. The FTIR of the La_2O_3 NPs revealed characteristic La-O band at 614 cm^{-1}. The successful synthesis of La_2O_3 NPs was further confirmed by Uv-Vis band that ranges between 300 to 400 nm. The EDX analysis indicated the presence of La in the nanoparticles. The FESEM demonstrated nanoparticles as an elongated rod-like structure with a consistent particle size of around 343 nm (image J software), confirming their polymeric nature. The XRD spectrum showed practically broad band which means that the nanomaterials generated are totally amorphous and lack periodic crystal structure. The TG-DTA results established thermal stability. The DPPH assay investigations for antioxidant activity demonstrated that La_2O_3 NPs have a 28.3% inhibition percentage. The synthesized La_2O_3 NPs have also potential for other significant biological applications.